DEVELOPMENT OF A WEB-BASED THESIS AND CAPSTONE PROJECT MANAGEMENT AND WORKFLOW AUTOMATION SYSTEM FOR HAU-SOC

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Area of Specialization in Web Development

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APPROVAL SHEET

This capstone entitled **"DEVELOPMENT OF A WEB-BASED THESIS AND CAPSTONE PROJECT MANAGEMENT AND WORKFLOW AUTOMATION SYSTEM FOR HAU-SOC"**, prepared and submitted in partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology with Area of Specialization in Web Development has been examined and is recommended for acceptance and oral examination.

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Acknowledgment

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Development of a Web-Based Thesis and Capstone Project Management and Workflow Automation System for HAU-SOC

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Introduction

Managing thesis/capstone projects is a core part of the educational process, but the workflow remains highly fragmented in many universities. Currently, manual procedures require a diverse group of stakeholders, including students, supervising faculty, panel evaluators, and academic administrators, to perform interconnected processes over distributed systems. The students are required to upload different versions of their documents through different websites, while the advisors and the panel members give their important feedback through different means such as emails, messaging applications, and written notes. At the same time, the program coordinators and the administrative staff have to arrange the assignment of the project roles, defense, and clearance after the defense manually through the use of forms. This creates situations where the student, faculty member, and administrator run into administrative barriers, and it becomes difficult to keep a centralized, audited record of academic progress, hindering the essential reviewing, defending, and archiving processes.

Most institutions are actively pursuing digital transformation of higher education, but existing EDMSs are often not designed to accommodate the highly dynamic, overlapping responsibilities inherent in academic supervision. In general, workflow digitalization increases the effectiveness of institutional processes (Aliazas et al., 2024; Benavides et al., 2020), but traditional EDMSs are constrained by the complex, overlapping responsibilities of university personnel (Tsupari et al., 2024). The above limitation is clearly visible, especially within the Holy Angel University – School of Computing (HAU-SOC), where management

of several computing courses necessitates highly coordinated supervision for numerous undergraduate research students. Traditional RBAC grants permissions according to predefined roles of organizational units, but it does not fit well in environments where users perform multiple responsibilities. Kayes et al. (2017) defined CAAC as an expansion of traditional authorization mechanisms that takes into account the context of the operation and gives permissions based on the user's responsibilities. Thus, the academic supervision environment at HAU-SOC fits well with the principle of CAAC, as a faculty member performs different responsibilities at the same time: he/she acts as an Adviser for one group of students, as a Panel Member for another, and as a Coordinator in the department. This means there is a considerable research gap in developing a system that extends simple file management with end-to-end workflow automation using CAAC, distinguishing between global institutional roles and particular roles in each project (Sandhu et al., 2000; Leite et al., 2023).

Thus, this capstone project aims to propose the Development of a Web-Based Thesis and Capstone Project Management and Workflow Automation System for Holy Angel University – School of Computing (HAU-SOC), which will address the aforementioned operational drawbacks. In order to prove empirically that the proposed system is beneficial, the research will first start with the analysis of the inefficiencies in the current process and thereby show how the new system can effectively address these problems. In order to address these issues directly and convert these scattered manual procedures into an automated workflow process, the system will have some core functionalities. First, there will be a centralized submission, version control, and a context-aware access control model (Global Roles vs. Project-Based Roles) for automating the entire document routing procedure from

submission to archiving. Moreover, in order to enable real-time monitoring of workflow processes, the system will issue email notifications to inform the users of their pending actions (Adegoke & Olalekan, 2026). The process of evaluation of the documents will further include Google Gemini Flash AI-generated summaries of the manuscript parts generated by external AI services (Zhang et al., 2024) at the proposal stage and at the time of final defense. These summaries will assist in preparing the panel, and non-destructive annotations will be provided through which the adviser, panel, and assigned instructor can provide suggestions for the further improvement of the manuscript on the upload document itself (Sun et al., 2023). Finally, to preserve the integrity of the data, the domain-based restricted registration and authentication process will be used for allowing entry into the system only by the registered university users only using Firebase (Grassi et al., 2017; Mohammed et al., 2023).

Literature Review

Web-Based Information Systems and Workflow Automation

One of the trends that have been recently observed in the digitalization of education is that while there is much digitalization taking place in the management of the students in general, some of the end-to-end processes like thesis supervision are done outside these systems and through different decentralized platforms (Tsupari et al.). In order to modernize the mentioned infrastructure, the solution utilizes web frameworks to turn the process of thesis supervision into a workflow and thus digitize it. By using React, the software provides users with a responsive SPA that guarantees transparency of the project progress at all times.

Process automation is ensured by Node.js using a RESTful API via Firebase Cloud Functions. It enables the asynchronous processing of workflow transitions from the systemic process of writing the manuscript from the Adviser's review queue to the Panel's evaluation queue. This process of digitalization is in line with research findings that technology should be integrated into the workflow and not seen as merely a data repository (Leite et al.).

Electronic Document Management and Version Control

While regular Electronic Document Management Systems (EDMS) solutions serve well as a basis for storage, they need some enhancement in accommodating the inherently iterative process of document routing in academia (Aliazas et al.). Since existing academic systems such as HAU-SOC have difficulties with version control as the number of manuscript revisions increases and managing them using decentralized mediums, it becomes necessary to develop a specific document structure that would help streamline the process. As a result, the document infrastructure that employs Firebase Firestore for real-time NoSQL data synchronization and Firebase Cloud Storage for secure object storage will be developed. While Firestore tracks the timestamp, ID of the submitter, and the workflow phase, Cloud Storage provides secure storage of the physical binary files of the PDFs. The combined solution allows for tight version control where panel members are able to review only the latest version of a manuscript, and the older versions are stored securely.

Context-Aware Access Control (Global Roles and Project-Based Roles)

Role-Based Access Control (RBAC) is the common mechanism used in traditional access management in portal applications for academics; here, static access permissions are

defined based on global roles from the organizational structure (Sandhu et al.). Nevertheless, the above mechanism entails some structural limitations in a dynamic academic environment because, in such an environment, the role of a particular faculty member changes depending on the specific thesis or capstone group to be evaluated. To optimize the structure, Context-Aware Access Control (CAAC) mechanisms are applied through the use of dynamic and contextual variables when making authorizations (Kayes et al.). The Context-Aware Access Control architecture is implemented in the current system across two enforcement points corresponding to the two categories of database operations it governs. For write operations, record creation, updates, and deletions, the corresponding Cloud Function evaluates the requesting user's Global Role (such as Student, Dean, Associate Dean, Program Chair/Coordinator, System Administrator, URO) and Project-Based Role (such as Instructor 1, Instructor 2, Adviser, Panel Chair, Panel Member) against the CAAC_permissions object before the change is committed. For read operations, this same evaluation is performed directly by Firestore Security Rules, which consult the requesting user's CAAC_permissions object to determine which documents they may access without the request ever reaching a Cloud Function. In both cases, the system evaluates not only the institutional identity of the user but also his/her context in relation to the manuscript to be accessed. Consequently, a faculty member will have annotating and approving roles within one group he/she advises, while in another group where he/she acts as a panel member, he/she issues defense verdicts.

Artificial Intelligence Summarization and PDF Annotation

The cognitive burden involved in evaluating long thesis papers for academics is bound to cause administrative friction in the evaluation stage, especially with the increase in

the number of students. However, recent developments in natural language processing reveal that large language models (LLMs) can assist in text summarization through quick recognition of research methods and key findings (Zhang et al.). To assist in the review process, the proposed system integrates an external AI module powered by Google's Gemini Flash via its application programming interface (API). Gemini Flash is used due to its large context window, able to analyze multi-chapter manuscripts in a single pass without lossy text segmentation, and its fast inference speed in serverless functions. The module is programmed according to strict prompting guidelines instructing it to analyze the user-uploaded PDF text only and to provide structured contextual summaries in the React application without hallucinations or using external information. To maintain the security and academic integrity of the data, the system works with the API through an active paid service plan, meaning that Google does not use uploaded manuscript text or the prompts to train its base models and does not involve any human annotators in reviewing the content (Google, n.d.). Apart from the automation of the process, Sun et al. provided the definition of digital annotation as a collaboration tool allowing users to annotate content and add notes to the documents. In this regard, the proposed system introduces the built-in PDF annotation tool, allowing advisers and members of the panel to annotate the manuscripts.

Automated Email Notification Systems

The follow-up process through manual means is not sustainable with an increasing number of thesis groups working at the same time, since the follow-up of submissions and approval processes of the groups becomes impossible through ad-hoc e-mails alone for a supervisor or an adviser. In the design of a workflow-based student clearance system by

Adegoke and Olalekan (2026), it was discovered that automatic e-mail alerts were instrumental to the success of the system. Their result supports treating notification as part of the workflow logic itself rather than as a separate, manually-triggered task. Applying this principle, the proposed system automatically sends an email whenever a workflow milestone is reached — for instance, once a student submits a new manuscript version, the assigned adviser receives an email prompting review; once the adviser completes that review, the student is notified of the outcome; and once a panel chair approves a defense schedule, both the student and panel members are notified of the confirmed date. This ensures that status communication is bound to the same event that updates the workflow record, thus addressing the corresponding gap highlighted by Aliazas et al., according to which document management systems are tracking documents but not status communication.

Domain-Restricted Access Authentication

The preservation of the authenticity of institutional academic records necessitates effective authentication processes in order to prevent any form of access by unauthorized persons. The existing digital identity strategy emphasizes the significance of having efficient ways of authenticating identities since the existing method requires the user's identity to be authenticated using a strict authentication system instead of traditional open registration (Grassi et al.; Mohammed et al.). The suggested system incorporates restricted access within a specific domain where accounts cannot be created without having a verified @hau.edu.ph or @student.hau.edu.ph institutional email. The system uses Firebase Authentication to handle all the authentication processes and manage system access through university credentials that generate JSON Web Tokens for the user session.

Digital Transformation in Higher Education

Digitalization in universities has shown that it is vital to consider the integration of technology into the already present processes rather than the use of separate digital applications (Tiwari, 2024). Although many processes in the administration of universities have been digitized through different technologies, processes like the supervision of theses may still employ various technologies and means of communication, causing fragmentation of processes (Tsupari et al.). The above situation makes it clear that there is a need for process-based systems that will help in facilitating collaboration of processes between the students, professors, and administration. This paper aims to develop a system that will facilitate the digitalization of thesis and capstone project management at HAU-SOC.

Digitalization of Thesis and Capstone Management Processes

The administration of thesis and capstone projects is a highly intricate process complicated by traditional decentralized models, especially as student numbers grow. The conventional supervision process uses a decentralized model, which complicates the work of supervisors and coordinators when trying to control student work in real time (Leite et al.). Transitioning from conventional to centralized models significantly increases administrative and educational quality by allowing process-based discussions and creating tight time structures (Leite et al.). In order to achieve these advantages in HAU-SOC, the new system uses a centralized web portal instead of fragmented communication channels by applying role-based dashboards and workflow routing to facilitate tracking student progress in real time for advisers and coordinators.

Workflow Automation and Document Management in Academic Institutions

In order to increase the efficiency of conventional submissions, the contemporary education system should be based on EDMS and workflow automation as a way of reducing dependence on manual archiving and enhancing governance (Gökmen, 2025). In universities, the electronic document management system will automate administrative functions and increase accessibility of information (Aliazas et al.), while the paperless system will help to increase the performance of the institutions (Ifeoma & Hillary, 2025). Applying the concept to HAU-SOC, instead of using the conventional EDMS, the proposed EDMS will have its own electronic document management features via Firebase Firestore and Cloud Storage as a way of centralizing manuscript versioning through an automated workflow engine.

Context-Aware Security and Access Control in Information Systems

Access control standards often require supplementary approaches due to the multi-layered, dynamic nature of academic collaborations. Kayes et al. described Context-Aware Access Control (CAAC) as an authorization model that takes into consideration dynamic contextual attributes like relationships to resources in order to grant access to more permissions than the static authorization roles usually do. Distinguishing global roles from situational roles is essential when dealing with complicated collaboration processes since this helps to adjust decision-making processes to the present context of the user (Zheng et al., 2019). As such, following the above guidelines, the system will implement CAAC through the application logic so as to differentiate the global roles of the user from his situational project-based roles.

Artificial Intelligence for Academic Document Processing

The implementation of generative AI (GenAI) in document management helps enhance the summarization of lengthy scholarly articles. Zaman et al. (2024) state that the application of AI for the summarization of long scientific papers produces a concise version of long scientific articles without omitting any critical information. Even though AI summarization is helpful in identifying important information, Corbin & Walton (2025) advise that its use should be restricted so as not to interrupt the deeper analysis of the source documents. AI summarization techniques should thus be used at the beginning of the review process, where information is presented in a concise manner (Schott et al., 2026). In line with this, the proposed system uses AI summarization as an auxiliary tool for reviewers at major defense steps only.

Online Thesis Management and Supervision

Doni et al. (2026) investigated the process of modernizing thesis supervision in an Indonesian health polytechnic. The purpose of the research was to move from the traditional method of document routing to a web-based system due to the increasing number of students. Using the Waterfall software development methodology, the authors developed the system facilitating the submission of documents online and scheduling the defense. The key results of the research demonstrated that the introduction of the system allowed for considerable optimization of the interaction between students and advisers and increased the capability of monitoring the progress of projects. At the same time, the main limitation of the system included its inability to differentiate between the roles of advisers and panel members due to

the use of the RBAC model. The system is highly relevant to the current project, proving the necessity of centralizing thesis supervision processes and providing the justification for introducing Context-Aware Access Control (CAAC). Distinguishing Global Roles from Project-Based Roles is one of the major features of the HAU-SOC system.

Electronic Document Management in Philippine Higher Education

Within the local setting, Aliazas et al. focused on the institutional impact of the process of digitalization of administrative procedures. The main objective of the research was to investigate the changes in the process of switching from using paper documents to tracking documents in an Electronic Document Management System (EDMS). Using a descriptive-correlational approach, the authors studied the perception of the efficiency of operations conducted by university employees prior to and after the implementation of digitalization. The main results showed that EDMS has successfully helped to centralize the process of administration, making data easily accessible and keeping institutional records in place. However, one of the main drawbacks of the use of the EDMS is that it functions only as a storage place; all of the stakeholders have to resort to some other form of communication (such as an e-mail chain or a messenger app) to indicate that a certain document needs to be reviewed. This research is highly relevant to the proposed project. Though it shows that Philippine institutions are prepared to embrace digital repositories, it also emphasizes the importance of the implementation of the automated Node.js-based email notification system that will help to manage the workflow in terms of tasks and notifications.

Information Security and Access Control in Educational Systems

The problem of security of academic documents in an electronic environment was considered in Bumalod and Velasco (2024) research. The key aim of the study was to create an information security system for handling confidential educational documents. The authors employed a research method combining traditional RBAC and cryptographic encryption for the regulation of document access. The most important findings were related to the idea that structured access control essentially improves the data integrity of the institution and helps to keep sensitive documents visible only to authorized persons. However, one of the weaknesses of this research was the lack of perimeter security, which means that the user's identity cannot be established during remote access to the system. The current study is highly relevant to the proposed capstone because it proves the importance of protecting academic manuscripts. Nevertheless, the research shows the necessity of having perimeter security. Thus, the HAU-SOC system incorporates the use of domain-restricted access (@hau.edu.ph).

EasyChair

EasyChair is a common Web-based conference management system that serves as a means of managing the process of academic paper submissions, reviewing, and conference organization. In particular, EasyChair enables the process of document submission, assignment of reviewers, their work with submitted documents, and management of the whole review process for the conference chair using a role-based access and workflow management approach (EasyChair, n.d.). One of the major advantages of the system is the possibility of managing document routing through the management of submissions, assignment of reviewers, and keeping track of the evaluation results. However, since EasyChair is made specifically for the process of handling conference papers, its workflow is

linear, and it is based on a set of roles such as Author, Reviewer, and Chair. Therefore, it is not equipped with an iterative supervision process when a manuscript goes through many rounds of revisions; thus, there are multiple roles and responsibilities performed by professors in relation to several student teams at the same time. The EasyChair system clearly shows how the process of academic reviews can be facilitated through a structured and role-based document routing process; nevertheless, its linear nature is not capable of handling the process of revisions that may be required in thesis supervision, an aspect addressed by the system through CAAC.

OpenProject

The OpenProject software is an open-source solution for project management that aims to help in planning, managing tasks, collaborating, and monitoring projects. Some of its capabilities include Agile boards, Gantt charts, work packages, milestones, document management, and role-based access control, which enables organizations to plan, organize, and collaborate on projects (OpenProject GmbH, n.d.). The main strength of this software is in offering comprehensive visibility into project timelines, task progress, and resource allocation within the integrated project management environment. As a generalized project management platform, however, OpenProject is not a product intended for management of academic theses and capstones. Though it allows attaching files to work packages, the software does not offer native solutions for processes of academic manuscript evaluation that include PDF annotations, adviser panel review processes, or specific thesis versioning. Therefore, some extra configuration and third-party tool integration would be required to use this software for those purposes. OpenProject serves as an example of the importance of

visualizing project timelines and milestones, but since there is no built-in functionality for manuscript assessment, the suggested system incorporates this type of functionality natively in its structure rather than by using external configuration options.

Microsoft SharePoint

Microsoft SharePoint is a collaborative and document management platform that is available in the Microsoft 365 product family. It provides such features as centralization of document storage, version control, access permissions, collaboration, and workflow routing, which made the program popular for managing digital documents securely in business organizations (Microsoft, n.d.). The first key advantage of SharePoint is the ability of the application to offer scalable document management solutions using versioning, permission management, and integration with other Microsoft products. Still, SharePoint was developed as a general solution for organizational document management and collaboration, not a specific academic workflow solution. Although organizations can implement their own approval processes and workflows in SharePoint, the development of workflows specific to the needs of undergraduate thesis and capstone management will require some development efforts. Moreover, permission management in SharePoint is based on roles defined in organizations, rendering it less suitable for the management of academic projects, in which professors may act as advisors, chairs, and panel members for different students. SharePoint explains why the above-mentioned features are important, but the permission model based on organizations cannot reflect the scenario where a faculty member plays different roles for various student groups, a problem that the proposed system solves using Context-Aware Access Control.

Google Classroom

Google Classroom is a Learning Management System (LMS) that is cloud-based, offered by Google, which allows teachers to form classes, assign coursework, communicate, organize educational material, and track the progress of learners using Google Workspace for Education (Google for Education, n.d.). Among the many advantages of the application is the friendly user interface, which integrates cloud-based productivity applications like Google Docs, Google Drive, Google Meet, and Google Forms, where students and teachers can conveniently submit coursework, provide feedback, and manage the documents. Nonetheless, the main focus of Google Classroom is on classroom teaching and assignment management in a one-to-many relationship rather than multi-step, multi-role processes needed for undergraduate thesis and capstone guidance (Raccoon Gang, 2026b). As such, Google Classroom is unable to provide functionality for sequential adviser and panel reviews, role allocation, version control, and workflow automation among other processes necessary for thesis management. Google Classroom highlights the value of a simple, user-friendly submission and communication experience; however, its one-to-many classroom model does not support the sequential, multi-role adviser-and-panel review process that thesis supervision requires, which the proposed system is purpose-built to handle.

Canvas LMS (HAU Incumbent System)

Canvas LMS is the official LMS of Holy Angel University that can be utilized for course administration, delivery of course content, assessment, scoring, and communication purposes. Some of the features available in Canvas LMS include SpeedGrader for marking

documents, modules for organization of content, and role-based access for administrative purposes (Instructure, n.d.). Although Canvas allows managing semester-based courses with grading, marking up, and collaborative functionality in an integrated manner, it is not designed for thesis and capstone processes (Raccoon Gang, 2026a). As a result, Canvas does not include features such as automatic workflow routing, project-specific faculty roles, multiple office integration with automatic routing, and manuscript versioning. The use of Canvas demonstrates the utility of a centralized document management and marking tool in a familiar university environment; however, since it is meant for courses as opposed to projects, it does not have workflow processes and project-related faculty roles, features introduced in the proposed system for theses and capstones.

In spite of the availability of multiple existing tools, including enterprise-level EDMS solutions and institutional Learning Management Systems, there is no system that can be configured without changes in order to serve the particular and multi-tiered bureaucratic procedure of thesis supervision in universities. General-purpose systems, such as SharePoint or OpenProject, would need to be heavily customized to simulate an academic procedure; instructional systems, like Google Classroom or Canvas LMS, would face problems with scaling from one instructor’s assignments to a panel evaluation process. The proposed system unites the most useful features of the above-listed tools: the version control of SharePoint, milestone control of OpenProject, routing features of EasyChair, and markups of Canvas LMS, together with Context-Aware Access Control and AI-based summarization as the added value.

From the literature reviews and the systems examined, it is clear that institutions in higher education are still focused on digital transformation through the replacement of disjointed paperwork and manual procedures with a central digital system. Key topics discussed in the literature include improved efficiency, increased transparency, enhanced collaboration between the involved parties, and proper document management throughout its entire lifecycle in a secure manner (Tiwari; Tsupari et al.; Leite et al.; Aliazas et al.). These findings indicate that digital transformation extends beyond mere digitization to encompass support for specialized academic workflows.

In order to attain these goals, there is a recurring theme of using Electronic Document Management Systems (EDMS), workflow automation, secure cloud-based storage, version control, automated notifications, and access control depending on roles and context in improving the processes of working with documents (Gökmen; Kayes et al.; Adegoke & Olalekan). Additionally, more recent advances in the field have shown how there is an increased application of Artificial Intelligence and annotations in documents in order to ease the process of document reviews and lower the effort needed to analyze large academic papers (Pandey et al.; Sun et al.; Zhang et al.).

However, even with such developments, the literature identifies a number of issues. The current systems for document management in enterprises emphasize storing and organizing documents, but fail to facilitate specialized academic workflows, whereas learning management systems were not developed with a view to facilitating multi-stage thesis/capstone supervision. Prior studies have also demonstrated that the current problems of inefficient static access control models, fragmented communication channels, and lack of

workflow automation are especially evident when a faculty member carries out multiple roles within various research projects (Aliazas et al.; Doni et al.). In spite of the advancements made in the field of document management using AI technologies, most systems give priority to the process of information retrieval and manipulation rather than generating concise summaries that would help academic evaluators understand long manuscripts.

These identified gaps emphasize the need for a centralized system that will be composed of centralized document management, workflow automation, access control flexibility, version tracking, collaboration, and secure authentication. The Development of a Web-Based Thesis and Capstone Project Management and Workflow Automation System for HAU-SOC will address these shortcomings through the integration of Context-Aware Access Control (CAAC), which is able to differentiate Global Roles and Project-Based Roles, workflow notifications, versioning, document annotation, localized AI-generated summaries, and domain-restricted user registration with email ownership verification through an approval button in the automated email. Through these components combined in a centralized web-based system, the proposed system intends to provide a more secure and efficient way of managing the whole process of thesis and capstone projects at the Holy Angel University – School of Computing.

Conceptual Framework

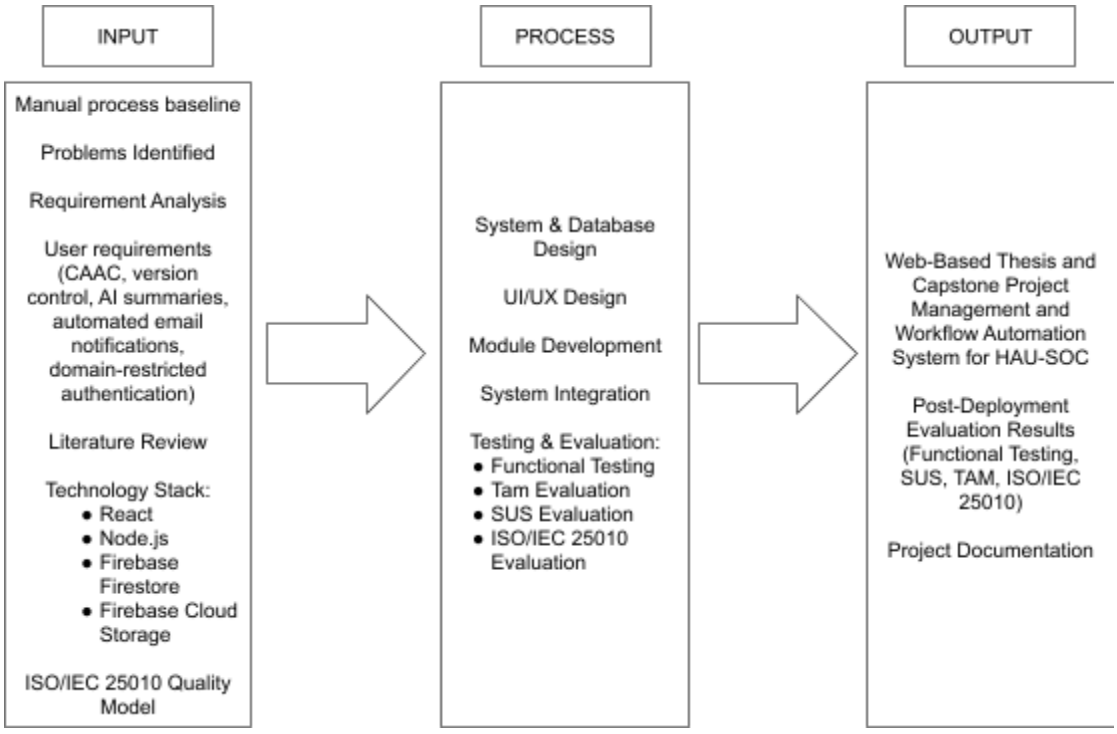
This capstone project uses the Input-Process-Output (IPO) Model as a conceptual framework, illustrating the transformation of user requirements, discovered workflow problems, literature results, and available technical resources into the developed Web-Based Thesis and Capstone Project Management and Workflow Automation System. Figure 1 maps out the conceptual framework underlying this study

The Input-Process-Output (IPO) model will be used as the conceptual framework for this study to present the development and evaluation cycle of the proposed system. The input stage sets the foundation by identifying the existing operational challenges—such as administrative delays, loss of revision history, and opacity of project progress—and establishing the functional baseline through requirement analysis grounded in an assessment of the School of Computing (SOC) Research Manual (May 2025 Revision) alongside interviews with involved faculty. This phase involves integration of the user requirements discovered (Context-Aware Access Control, versioning of documents, AI-based summary generator, and email notification feature), literature study outcomes, the technical requirements of the software stack (React, Node.js, Firebase Firestore, and Firebase Cloud Storage), the security constraints (restricted domain-based authentication), and the ISO/IEC 25010 Quality Model, which establishes the evaluation criteria to be applied later in the study. The Process phase includes designing the system/database, User Interface/User Experience Design, module development, and integration of the system, then the testing and evaluation through Functional Testing with a checklist, adapted Technology Acceptance Model (TAM) questionnaire, System Usability Scale (SUS) questionnaire, and ISO/IEC

25010 evaluation. This will yield the completed Web-Based Thesis and Capstone Project Management and Workflow Automation System for HAU-SOC that will have its finished UI design, together with a post-implementation report on the functional completeness, usability (SUS), user acceptance (TAM), software quality (ISO/IEC 25010), and full documentation of the project.

Figure 1

Conceptual Framework of the Development of a Web-Based Thesis and Capstone Project Management and Workflow Automation System for HAU-SOC (IPO Model)



Objectives of the Study

The general objective of this capstone project is to develop and evaluate a functional, usable, reliable, efficient in terms of performance, and secure web-based application for thesis and capstone project management and workflow automation system for HAU-SOC.

Specifically, this capstone project aims to:

1. Analyze the existing manual workflow by the policies set out in the SOC Research Manual (revised version May 2025), as well as interview relevant faculty to identify problems with the current workflow and determine the system requirements.
2. Design a responsive, single-page web application architecture that ensures usability and performance efficiency by employing React, Cloudflare edge caching, and NoSQL relational schema models.
3. Develop the main functional modules of the system (Context-Aware Access Control, AI summarization, and workflow automation) while ensuring its reliability and security using Firebase atomic transactions and domain-restricted authentication.
4. Evaluate the software technical quality of the proposed platform according to ISO/IEC 25010 standards (Functional Suitability, Usability, Reliability, Performance Efficiency, and Security) on the basis of IT expert evaluations.
5. Determine the perceived usability and acceptance of the platform by students, faculty, and administrators quantitatively via the System Usability Scale (SUS) and the Technology Acceptance Model (TAM).

Scope and Delimitations

The project plans to design and evaluate a central online management system for the thesis/capstone process at Holy Angel University – School of Computing (HAU-SOC). The scope covers the whole undergraduate research life cycle, which includes the Ideation Phase, Proposal Development, System Implementation (Capstone 2), Final Oral Defense, and Post-Defense Clearance. The system has been specifically designed for the student academic evaluation lifecycle, meaning that although the system would help the faculty in their supervisory and evaluative roles, it would not involve administrative processes of the faculty such as their publications or institutional KPIs. The target audience is HAU-SOC four-year undergraduate students conducting research projects and the faculty members who are going to supervise and examine the projects. Context-Aware Access Control (CAAC) is what will be utilized for permission management based on two separate role definitions: Global Roles (Student, Dean, Associate Dean, Program Chair/Coordinator, University Research Office, System Administrator) and Project-based Roles (Instructor 1, Instructor 2, Advisor, Panelist, Panel Chair).

In order to ensure that the user interface is secure and focused, the system uses state-based progressive visibility where project workspaces and documents are restricted only to the participants involved in the particular phase. The higher-level administrators such as the Dean, Associate Dean, Program Chair/Coordinator, and URO will be unable to see the project on their dashboard unless the project is formally approved and the workflow phase involves their role. In addition, in order to streamline the process of communication and avoid the use of chat modules within the system, the system uses an automated email notification module powered by Firebase that would alert users on the current status of the workflow or any actions that trigger it such as approvals or revisions.

The features which are significant include project registration, upload of PDFs in a secure way, version control, automated workflow which includes URO approval of manuscripts of the student, defense scheduling, AI summary, and a non-destructive annotation tool. Versioning of the document will be made immutable; in case of any submission from the student, a new version of the document will be created and not just an overwrite of the old one for auditing purposes during the review process. Only those documents which are officially submitted by the student via institutional review process will be tracked for their versioning. In terms of security, the domain restriction is made very strictly in such a way that only those who have @hau.edu.ph or @student.hau.edu.ph institutional emails will be able to register themselves for an account. The identity, credentials, and activation of accounts are managed by Firebase Authentication before full access of the system is granted. In terms of architecture, React is used for the frontend single-page application, while Node.js in Firebase Cloud Functions serves as the serverless backend API, Firebase Firestore handles real-time NoSQL database synchronization, and Firebase Cloud Storage ensures secure binary PDF storage.

This system is restricted to workflows in Holy Angel University - School of Computing (HAU-SoC). The scope of this system is limited to the workflow requirements of the School of Computing and may require further adaptation for use by other colleges or departments. It is only applicable to undergraduate theses and capstones that are done within the four-year curriculum of the School of Computing. All of the system's business rules and approval processes are guided by the School of Computing Research Manual (May 2025 Revision) and the interviews conducted among the faculty members before the title defense. It does not take into consideration any undocumented practices or variances of other

programs that may exist within the School of Computing. It does not cover the research workflow done by faculty members for their own personal research, nor does it cover research work beyond undergraduates, including graduate research work.

The project does not involve creating a native app but will have a responsive web interface. The system will not support direct, real-time editing of manuscript text within the application itself, since all editing should be done in external word processing software before re-uploading the edited document. However, the system will not monitor internal processes of manuscript editing before submission within the same research team, for example, who worked on what part of the manuscript and when the editing took place. Therefore, the system will not operate as a micro-task management tool or as a Kanban board for the students' internal group tasks. The system will consider only those documents that have been formally submitted. In addition, the system will not integrate third-party software plagiarism detector APIs; students must submit clearance certificates from external sources. Video presentations will not be stored in the system's storage infrastructure; users should submit links to the video files that are hosted externally.

The AI-generated summaries can be generated only from the PDF document uploaded by the user. Even though AI-generated summaries may help reviewers to navigate through long manuscripts, concerns have been raised regarding over-reliance and lack of proper engagement with the source material (Corbin & Walton). In addition to that, the AI model (Gemini Flash) could also result in occasional inaccuracies and hallucinations, which need to be taken into account when using the AI system. In order to address these issues, AI summaries will be clearly labeled as such and will only act as a supplemental navigation tool.

Though the system incorporates the University Research Office (URO) First and Second Level review routing, it strictly does not include the Institutional Review Board (IRB) endorsement process since, according to institutional guidelines, IRB clearance is only applicable to graduate studies and research by faculty and therefore cannot be incorporated into the four-year undergraduate capstone workflow. Additionally, even though the system is responsible for the full process of thesis and capstone evaluation from start to finish for all participants, it does not compute final marks for the panel and course instructors. It also does not include the recently created University Research Office (URO) Co-Authorship validation procedure, including the FM-AAC-URO-2001 Action Form and FM-AAC-URO-2014-2017 forms. The co-authorship of the faculty, which is done for faculty publication and for the university KPIs and not for the evaluation process itself, is not included in the system and will be performed through a separate administrative procedure within the URO office. Nevertheless, the system assists this process by providing an option for users to export their necessary FM-AAC-SOC-2003 monitoring logs as supporting documents.

METHODS

The Development of a Web-based System for Management and Workflow Automation of Thesis and Capstone Projects for HAU-SOC will employ the Iterative and Incremental Development (IID) model. The Iterative and Incremental Development is a software development method whereby a system is developed through iterative development processes where each process yields an increment that is evaluated, improved upon and further developed in the future iterations (Larman & Basili, 2003). Sequential development is different from Iterative and Incremental development in the sense that in sequential development all the components of a system are developed before evaluating it, meanwhile, IID develops the system progressively by acknowledging feedback and addressing problems that are encountered throughout the process.

The reason for using the Iterative and Incremental Development (IID) methodology is associated with the fact that the system consists of several modules which should be improved and validated continuously. Functional requirements like user authentication, role-based authorization, documents management, workflow automation, defense schedule management, notifications, and reports can be incrementally developed and then incorporated into the overall system. Every iteration involves testing and reviewing of developed modules so that necessary changes can be made and then other functional issues will be worked on. This methodology decreases the risks associated with the discovery of any defects only at the completion of the system, which is consistent with the emerging needs of a workflow system in an institution. The findings of the work by Pinheiro et al. (2010) also prove the appropriateness of the application of IID in the case of complex software projects. As per the

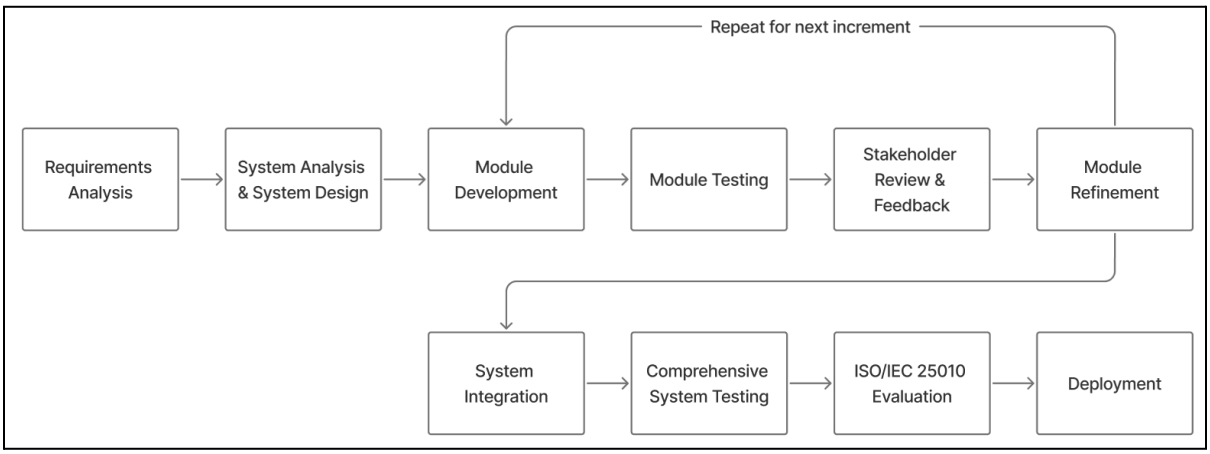
study, the usage of IID improved the early detection of issues, problem-solving capabilities, and the capability of developing software in iterations. Thus, the results of the study indicate the suitability of using the IID methodology in developing the described workflow management system due to its several workflow-oriented modules.

As shown in Figure 2, the adapted IID method starts with Requirement Analysis, whereby the functional and non-functional requirements of the system under consideration are identified. These are followed by System Analysis and System Design phases, in which the architecture, database, and specifications for each increment are defined. After this is the Module Development phase, where the proponents develop the system functionality, and this is followed by the Module Testing phase, where the developed increment is tested to determine whether it meets the required specifications. The developed increments undergo the Stakeholder Review and Feedback phase, where any problem or suggestion is collected and solved during the Module Refinement phase.

For the adapted IID approach, the increments created will be integrated progressively into the developing system. After all the necessary features have been developed, the process of testing, evaluation of software quality according to ISO/IEC 25010, and deployment of the system will come after that. Using the adapted IID approach, the proponents will be able to develop and evaluate the proposed system incrementally so that each increment contributes to achieving the overall goals of the project..

Figure 2

Adapted Iterative and Incremental Development (IID) Methodology for the Proposed Development of a Web-Based Thesis and Capstone Project Management and Workflow Automation System for HAU-SOC



Requirement Analysis

Requirement analysis involves the process of identifying, analyzing, validating, and documenting the functional and non-functional requirements of the desired system and is the foundation for building a solution to address the business needs of the intended end-users. For this study, requirement analysis establishes a thorough understanding of the thesis and capstone processes currently implemented at HAU-SOC by examining who participates in the process, what activities it comprises, where and when these activities occur within the academic calendar, how current procedures are carried out, and why the institution continues to rely on its present methods. The analysis covers the procedures for topic approval, manuscript submission, adviser assignment, document revision, proposal and oral defense, institutional approval, and archiving, and derives from these findings the functions and data that the proposed system must provide.

The target user organization for this system is Holy Angel University – School of Computing (HAU-SOC). HAU-SOC is a learning institution offering undergraduate courses concerning computing, including Bachelor of Science in Information Technology, Bachelor of Science in Computer Science, Bachelor of Science in Cybersecurity, and Bachelor of Science in Entertainment and Multimedia Computing. In connection with the requirements of these courses, students are required to conduct themselves either through theses or capstone projects prior to graduating from school. Thesis and Capstone Projects are conducted according to the guidelines stipulated in the School of Computing Research Manual (May 2025 Revision) (Appendix C-4).

The thesis and capstone process requires the involvement of academic staff in varied roles in the course of project completion. There is collaboration between students and their respective instructors in the process of writing a proposal; working together with advisers in the system development process; presenting work to members of a selected panel; seeking approval from Program Chairs/Coordinators, the Dean, and the Associate Dean; and ultimately submitting the finished document for institutional archiving.

There are eleven different stakeholders involved in the thesis/capstone process, whose roles change depending on their institutional position as well as the project they are assigned to: Students, Instructor 1, Instructor 2, Advisers, Panel Members, Panel Chairs, Program Chairs/Coordinators, the Dean, the Associate Dean, the University Research Office, and the System Administrator. Since one faculty member can have different roles at the same time depending on different projects, access to the proposed system is controlled using Context-Aware Access Control, whereby the institutional Global Roles are separated from the Project-Based Roles.

The existing thesis and capstone management process followed by Holy Angel University – School of Computing is composed of several interconnected academic and administrative activities beginning with project group formation and ending with institutional archival. To provide a comprehensive view of this workflow and of the permissions gathered for this study, supplementary documents are provided at the end of this proposal. These include the formal cover letters requesting permission from the host institution (Appendix A), a step-by-step visual flowchart of the current manual process (Appendix B), and the existing institutional files and official forms currently in use (Appendix C).

Based on the School of Computing Research Manual (2025 May Edition) and current documentation of the workflow process, as well as consultation with professors who teach theses and capstones, the Instructor for Thesis/Capstone 1 starts the process by forming project groups and assigning students to the groups. The list of project groups is then forwarded to the Program Chair/Coordinator, who assigns an adviser to each group. The Dean and Associate Dean subsequently review and approve the adviser assignments before the advisers are formally assigned. Once the adviser assignment process has been completed, student groups begin the ideation phase by preparing and submitting five proposed research topics for evaluation by both the Instructor for Thesis/Capstone 1 and their assigned adviser through the Canvas Learning Management System. In the course of face-to-face or online consulting activities as recorded in the Thesis/Capstone Monitoring Sheet (FM-AAC-SOC-2003), the students develop the topic proposals to arrive at one approved research topic. Once approved, the students will write and submit the concept paper via the Canvas portal for evaluation by the Instructor for Thesis/Capstone 1 and the assigned adviser.

In light of the feedback provided, the students will revise the concept paper as necessary prior to proceeding to chapter development and proposal manuscript writing.

After the completion of the proposal stage, the process goes on to Capstone/Thesis 2. The assigned Adviser continues to supervise the project during the implementation of the system, while Instructor 2 monitors the status of projects and validates the accomplishment of required course milestones. Adviser and student communication may take place through different channels depending on the platform the adviser prefers, including Microsoft Teams, Messenger, Canvas, university email, or personal meetings. Progress is reported weekly using the Attendance Monitoring and Activity Sheet (FM-AAC-SOC-2003). Once the assigned Adviser verifies that the project is ready for final evaluation, the required recommendation documents are completed, after which Instructor 2 prepares and publishes the official final oral defense schedule in coordination with the Program Chair/Coordinator.

During the process of oral defense, the document is physically handed over in a bound version at the Dean's Office, and the presentation of defense is made through Canvas or Microsoft Teams. The copies of the document are distributed to the members of the panel in physical form a few days before the defense. Following the panel's deliberation, the Panel Chair communicates the official verdict to the students, which may require minor revisions to be completed within one week, major revisions to be completed within two weeks, or a re-defense. Groups required to undergo a re-defense must complete the prescribed revisions before the rescheduled defense. The comments, suggestions, and recommendations of the panel are recorded in the Capstone Revision Form (FM-AAC-SOC-2004). In the process of iteration, the revised manuscript is submitted to the Adviser to check whether the criteria of the panel are met. Then the manuscript is presented to the Panel Members for their final

assessment. Only upon receiving full approval from the Adviser and Panel Members do the students present their finalized manuscript for English editing and proofreading. An Editor's Certificate from an accredited language editor should be issued to ensure the correctness of grammar and style of the document.

After the manuscript's quality is verified, the Plagiarism Clearance Certificate is released if the document passes the required plagiarism and artificial intelligence checks and meets the institution's originality threshold. Upon receipt of the Editor's Certificate and the Plagiarism Clearance Certificate, the manuscript undergoes institutional approval within HAU-SOC. The Approval Sheet is reviewed successively by the assigned Adviser and the Panel Members prior to the endorsement of the Program Chair/Coordinator. The Plagiarism Clearance Certificate and the Editor's Certificate are then submitted to the University Research Office (URO) for verification, after which the cleared manuscript is archived at the institutional level.

The process runs through two academic semesters. Group formation, adviser allocation, idea generation, concept paper assessment, and proposal defense are done during Thesis/Capstone 1, while the actual system development, weekly progress reporting, the final oral defense, post-defense modifications, and clearing are all done during Thesis/Capstone 2. Weekly achievements are noted every week during the development period via the Attendance Monitoring and Activity Sheet (FM-AAC-SOC-2003). Post-defense modifications are tied to definite time frames of one week for minor changes and two weeks for major ones based on the panel’s decision, while the final clearing is done prior to the end of the semester where the defense is conducted.

Although the present arrangement gives rise to the operational concerns discussed in the following subsection, several of its characteristics serve legitimate institutional purposes and were considered in designing the proposed system. The Canvas Learning Management System is the system of choice for submitting coursework by the university, and its continued use ensures that thesis and capstone requirements are in line with institutional academic policies. Current use of printed forms that have signatures in hand-writing gives an actual and auditable proof of academic decisions, which is especially critical for the Capstone Revision Form and Approval Sheet. Face-to-face and online consultations, likewise, are pedagogically deliberate, as they allow advisers to provide the sustained mentoring that written feedback alone cannot replace. Therefore, the system that is proposed is one which seeks to maintain these functionalities without disrupting them; the digital signature is meant to keep traceability similar to that of handwriting, while the consultation is excluded from the purview of the system.

The existing thesis and capstone management process of the Holy Angel University – School of Computing has supported undergraduate research activities for several years. However, because the ratio of faculty advisers to students remains uneven, faculty members holding global roles, such as Program Chairs/Coordinators and even the Dean, are likewise assigned as Advisers to address the shortage. Under these conditions, the absence of a centralized system gives rise to several operational problems, which are discussed in the following paragraphs.

One of the key concerns that have been mentioned above is the fragmentation of communication and submission of documents. Throughout the thesis and capstone course, students and professors use multiple communication tools such as Canvas LMS, Microsoft

Teams, Messenger, institutional email through Microsoft Outlook, printed copies and face-to-face interactions. As these tools work separately, important information, reminders about revisions and approvals, and submitted documents get scattered among different systems. Students and professors have to use multiple apps in order to find necessary documents or check on the updates of their conversations, which causes inefficiencies in searching for the information and its potential loss. There is no unified tool that sends workflow alerts; thus, notifications about defense schedules, revision deadlines, and approval results also depend on messaging tools or emails.

Another problem is about the lack of centralized storing of documents. During work with the proposals and the implementation of the system, students submit various versions of the manuscripts and supporting documents by the requests of their professors, advisers, and evaluators. There is no version control enforced by the system, which creates an opportunity to evaluate an outdated version of the manuscript or to receive a duplicated submission during evaluation.

A further concern is connected with manual control over the progress of the projects. Advisers, Instructors, and Program Chairs/Coordinators track the milestones of groups through consultations, printed monitoring forms, and manual verification of the requirements. Such a procedure is laborious but does not give an opportunity to track the status of individual projects, because delays in submission of the paper, waiting for the approval, and missing revision should be checked intentionally. This problem is also faced by the students themselves, who should ask their advisers or instructors if their paper was evaluated or returned for revision. Administrators, in turn, do not have the opportunity to

track the status of ongoing projects without examining several documents and messages manually.

The other element related to the operational problem refers to the manual routing of the approval process. Workflow steps such as approval of the proposal, recommendation of the adviser, scheduling of the defense, verification of the revisions, endorsement of the institution, and document approval require the use of hard copies and manual interaction between different departments. There being no workflow engine working automatically at this stage, every individual involved has to manually ensure that all required documents are available before going ahead to the next step.

The lack of context-based role management further adds to the administrative burden. A faculty member could be an Instructor, an Adviser, a Panel Member, a Panel Chair, a Program Chair/Coordinator, or even an administrator based on his or her involvement in the project. But the current system considers only one role for a user and does not take into account the dynamic changes in permissions according to his or her roles in the project.

Moreover, the existing process does not facilitate collaborative manuscript review. Advisers and Panel Members typically submit feedback through handwritten notes, separate documents, electronic mail, or face-to-face interaction. Because feedback originates from multiple sources, students find it difficult to consolidate the comments received and to verify that all required revisions have been addressed. Advisers and Panel Members face the corresponding difficulty of tracking which of their recommendations have been fulfilled.

All of the above-mentioned issues viewed together clearly underscore the need for a web-based centralized system, which will have features like workflow automation, document

management, access control, versioning, e-mail alerts, collaborative document review, and real-time project tracking within a single integrated environment. The solution to these problems is the primary rationale for the development of the proposed web-based thesis/capstone project management and workflow automation system for HAU-SOC.

The proposed system must capture several classes of information in order to support authentication, workflow execution, document management, monitoring, reporting, and institutional record-keeping. This includes user and account data based on institutional record information, such as student ID, institutional email address using either the @hau.edu.ph or @student.hau.edu.ph domain, program, and block section; role and permission data for Global Roles, Project-Based Roles, and the CAAC tags that regulate contextual access; data for programs, sections, and group assignments; project data including title, category, academic year, and workflow status; data for documents to facilitate version control; workflow data including submission, approval, and stage changes; annotation and review data including reviewer comments; and data for auditing system activities. The entities, attributes, data types, and relationships that have been utilized to meet the above mentioned requirements have been illustrated in the Entity Relationship Diagram (Appendix H) accompanied by a data dictionary (Appendix I).

Requirements Documentation

The proposed Development of a Web-Based Thesis and Capstone Project Management and Workflow Automation System for HAU-SOC is designed to assist throughout the whole research process. The central idea is to provide a tool that will integrate all fragmented procedures of document submission, milestone monitoring, manuscript

evaluation, and institutional approval into a singular centralized platform. Through a Context-Aware Access Control model that distinguishes institutional Global Roles from Project-Based Roles, the system ensures that its functionalities are exercised only by the participants authorized to do so at each stage. The above system is thus an additional institutional tool with features such as unchangeable document versions, automatic email alerts based on workflow processes, and an organized process from the idea generation stage through the archiving stage.

Global Roles

Student

Students are the primary users of the proposed system. They submit research topics, concept papers, manuscript revisions, weekly accomplishment reports, deployment information, editor's certificates, plagiarism certificates, and other required documents throughout the thesis and capstone process. The system allows students to track the status of their work, review any feedback, and get automatically notified via their emails according to workflow events. Students can upload topic proposals, upload their work in PDF files in a form of an immutable document, and update their entries for weekly progress on Attendance Monitoring and Activity Sheet (FM-AAC-SOC-2003). They are allowed to see summaries generated automatically using artificial intelligence and non-destructive annotations left by reviewers, but they are not allowed to change the original content of the submitted file. In the final clearing process, students upload Editor's Certificate and Plagiarism Clearance Certificate. Students do not have access to the approval of workflow gates. The system will

notify them electronically about approved documents, requests for revisions, appointment of advisers, and defense schedules.

Dean and Associate Dean

The Dean and Associate Dean are institutional approvers for different steps of the workflow in the School of Computing. The Dean and Associate Dean review and approve the adviser recommendations made by the Program Chair/Coordinator at the ideation stage. Once a project has completed all required internal academic processes and has been verified by the University Research Office, the Dean and the Associate Dean review the project, apply their digital signatures to the Approval Sheet, and grant final approval for the completed thesis or capstone. The system notifies them by electronic mail when the Program Chair/Coordinator submits the adviser assignment for approval and when a fully cleared project is endorsed for final institutional signature.

Program Chair/Coordinator

The Program Chair/Coordinator is responsible for managing institutional flow by allocating advisors in the ideation phase as well as the Panel Chair and Panel Members in the defense phases. This role verifies student and project information, reviews project endorsements, confirms compliance with institutional requirements, monitors program-level project progress, generates program-level reports, and endorses fully cleared projects to the Dean and the Associate Dean for subsequent approval. The Program Chair/Coordinator may assign project-based roles and may issue the formal endorsement for defense scheduling

upon receiving the Adviser's recommendation, but cannot act as the final executive signatory. The system will notify this individual through electronic mail following the endorsement of the Instructor 1, following the submission of the FM-AAC-SOC-2005 Capstone Recommendation Form by the Adviser, and after the signatures of the whole panel in the final Approval Sheet.

University Research Office (URO)

The University Research Office (URO) is a dedicated authority for clearance towards the end of the workflow of the project. Their main role is to receive, authenticate, and assess essential documents submitted by the students after defense. These include the editor's certificate, plagiarism check certificate, and manuscript, respectively. The clearance from URO ensures that the project satisfies all institutional research requirements to be approved finally. Under the proposed model, their permission will enable them to access and assess these particular uploaded documents and manuscripts. Having succeeded in the process of assessment, they shall give their approval by signing before returning the forms to the group carrying out the project. The URO clearance is necessary in the process of clearance towards getting final approval from the Dean through signing on the Approval Sheet.

System Administrator

The System Administrator manages the technical maintenance of the proposed system. The responsibilities of the System Administrator are managing users' accounts, configuration of the institution, system security, backups, and maintenance of all the software infrastructure. The most important function of the System Administrator is the configuration and maintenance of the Context-Aware Access Control (CAAC) tagging framework.

Administrators don't assess the manuscripts, but they have elevated privileges to audit tag assignments, deal with permission overriding, global system settings, and institutional thesis and capstone archive management.

Project-Based Roles

Instructor 1

Instructor 1 supervises students during the Thesis/Capstone 1 phase. This role creates project groups, assigns students to those groups, and forwards the finalized list to the Program Chair/Coordinator for adviser assignment during the ideation phase. Instructor 1 guides students through topic development by reviewing concept papers and title proposals, adding non-destructive highlights and comments to manuscripts, requesting revisions, and approving or rejecting proposed topics. The system helps Instructor 1 support the iterative drafting process, returns documents to the group for revisions, and approves groups that are qualified to present their proposal defense and be routed to Capstone 2. Instructor 1 is notified via email when the students post their concept papers, chapter drafts, and title proposals.

Instructor 2

Thesis/Capstone 2 is coordinated by Instructor 2. This function tracks the progress made on project implementation and confirms the accomplishment of all course requirements such as the delivery of the revised manuscript and system components. Instructor 2 verifies,

together with the assigned Adviser, that students are ready for the final defense, coordinates post-defense requirements, and confirms the completion of course-related requirements. The system allows Instructor 2 to create and publish the final oral defense schedule following the formal endorsement of the Program Chair/Coordinator. Instructor 2 confirms milestone achievement but does not make any annotations on the manuscripts, as the process of reviewing the substance of the manuscript is solely the responsibility of the Adviser at this stage of development. Notification of this approval will be sent to Instructor 2 through e-mail.

Adviser

The Adviser oversees the assigned thesis/capstone teams in the entire development life cycle of the system. The Adviser reviews, annotates the manuscript, monitors weekly accomplishments, and verifies the readiness of the project for oral defense. The system enables Advisers to make non-destructive comments and to send documents back to groups for editing. They will not be able to edit anything from the student's uploaded documents. Advisers can electronically sign the weekly logs of students through the Attendance Monitoring and Activity Sheet (FM-AAC-SOC-2003) and start the clearance process for the final defense through the digital Capstone Recommendation Form (FM-AAC-SOC-2005). They will also sign the Approval Sheet when the revisions have been done successfully. The system will alert Advisers via email whenever there are submissions of the weekly logs, manuscript drafts, and the final post-defense revisions.

Panel Members

The Panel Members assess the proposal and the final defense documents submitted by the group that is allocated to them. The Panel Member views the current version of the document being evaluated, the summary generated through artificial intelligence, and adds non-destructive annotations and comments for his/her reference in evaluating the document. The notes and comments are only seen by the individual who made them and not by other panelists or the students before the defense process. After the completion of the defense process, the Panel Members sign the Capstone Revision Form (FM-AAC-SOC-2004) after verifying the changes done by the students and the Approval Sheet. They cannot individually give out official verdicts for the defense. The system notifies Panel Members by electronic mail when they are assigned to a project panel and when manuscripts or presentation videos are uploaded for defense review.

Panel Chair

The Panel Chair performs all the responsibilities and holds all the system permissions assigned to a Panel Member. In addition to reviewing manuscripts and verifying post-defense revisions, the Panel Chair facilitates panel deliberations, records official defense verdicts, and documents revision requirements. The system allows the Panel Chair to record the official verdict, namely minor revisions, major revisions, or re-defense, by submitting the digital Capstone Revision Form (FM-AAC-SOC-2004). Recording the verdict determines the project's next workflow stage and initiates the system's automated revision countdown.

The functional capabilities of the proposed system are represented in Use Case Diagrams that define the system boundary and illustrate the major system functions alongside

their interacting human and system actors (Appendix D). These actors/systems, which include the Dean, Associate Dean, Program Chair/Coordinator, System Administrator, University Research Office, email system, and the external artificial intelligence system, are represented in relation to their functions, which relate to adviser and panel assignment, clearance validation, report creation, system configuration, and summary creation (Appendix D-1). The academic processes undertaken by project-based actors, namely Students, Instructor 1, Instructor 2, Advisers, Panel Members, and Panel Chairs, cover document submission, weekly logging, roster maintenance, topic registration, draft annotation, defense scheduling, verdict recording, and digital signing (Appendix D-2). A complete enumeration of system features is further provided in the Product Requirements Document (Appendix E).

The functional requirements define the capabilities that the proposed system must provide to support the complete thesis and capstone project workflow. The system shall allow the management of user access using role-based accounts and dashboards so that the System Administrator can manage accounts and validate official university emails, as well as allow users access to functionalities based on their roles. It shall support project group and topic registration, immutable document submission, version control, workflow tracking, and automated routing of project documents to authorized reviewers. Authorized users shall be able to add non-destructive annotations, while the system maintains an immutable audit trail of important activities and workflow transitions. The system shall also allow for adviser and panel assignment, manuscript summaries via the use of AI by reviewers, automated email notifications, weekly progress logging and validation, and scheduling and logging of defense (Table 1).

The system shall also facilitate the latter part of the process in project workflows with features such as official verdict recording, revision countdowns, digital signatures, final clearance, and archiving. The Panel Chair shall record the official defense verdict and any required revisions, at which point the system shall track any overdue revisions. The system shall allow for the digital signing of institutional forms as well as clearance from the Dean and the Associate Dean of any remaining requirements for the archiving of the projects. The system shall also generate reports, exports, and searchable/filterable records of thesis and capstone projects.

Table 1

Functional Requirements

Functional Requirement	User / Role	Description	Priority
The system allows users to manage user access	System Administrator	Enables administrators to assign roles of users, manage accounts, and verify official university emails.	High
The system allows immutable document submission and tracks progress of project workflow from ideation to archival	Student	Enable students to upload documents in PDF format, ensuring submitted versions are never overwritten or modified, and track the progress of the project workflow from ideation to archival.	High
The system allows project group and topic registration	Instructor 1	Enables Instructor 1 to register the groups, officially register the approved thesis or capstone project topic after the ideation phase,	High

		and view the final adviser after sending the group roster per block to the Program Chair/Coordinator.	
The system provides role-based dashboards.	All Users	Enables the system to display customized dashboards for each role to track active projects, pending tasks, and upcoming deadlines.	High
The system provides version control and history	All Users	Enable authorized users to compare previous and current iterations of documents as new versions are automatically created.	High
The system automates workflow routing	System	Enables the system to automatically route project documents to the next authorized reviewer based on the configured academic workflow.	High
The system allows non-destructive annotation	Instructor 1, Adviser, Panel Member, Panel Chair	Enable authorized reviewers to add comments and annotations without modifying the student's original uploaded file.	High
The system maintains an immutable audit trail	System	Enables the system to automatically record critical activities and workflow transitions into a protected audit log that cannot be modified by ordinary users.	High

Functional Requirement	User / Role	Description	Priority
The system allows adviser and panel assignment	Program Chair/Coordinator	Enable authorized coordinators to assign an Adviser to a project group and designate the Panel Chair and Panel Members.	Mid
The system generates AI-assisted manuscript summaries	Panel Chair, Panel Member	Enables the system to provide a structured AI summary strictly for complete manuscript submissions to aid reviewer evaluation.	Mid
The system triggers automated email notifications.	All Users	Enables the system to send automated email alerts for significant events like new assignments, revision requests, and schedule updates.	Mid
The system allows weekly progress logging and validation	Student, Adviser	Enable students to submit weekly accomplishment reports and let Advisers review, approve, return, and digitally sign these records.	Mid
The system provides defense scheduling	Instructor 2, Program Chair/Coordinator	Enables authorized personnel to create, record, and notify participants regarding proposal and final-defense schedules.	Mid
The system allows official verdict recording	Panel Chair	Enables the Panel Chair to exclusively record the official defense verdict and classify required revisions as minor or major.	Mid

The system automates revision countdowns	System	Enables the system to initiate a configurable revision countdown after the official verdict is recorded and flag overdue revisions.	Mid

Functional Requirement	User / Role	Description	Priority
The system allows digital signatures on forms	Dean, Associate Dean, URO, Adviser, Panel Member, Panel Chair, Program Coordinator	Enable authorized users to apply traceable digital signatures to digital versions of approved institutional forms.	Low
The system provides final clearance and archiving	Dean, Associate Dean	Enables the system to verify the completeness of final requirements for executive approval and transfer cleared projects to a records table archive.	Low
The system allows report generation and exporting	Dean, Associate Dean, Program Chair/Coordinator, System Administrator	Enable authorized users to generate, view, and export consolidated reports on project statuses, completion rates, and faculty workloads.	Low
The system provides records tables with search and filtering.	Dean, Associate Dean	Enable users to search and filter authorized records.	Low

The non-functional requirements define the quality standards and constraints that the proposed system must meet to ensure an effective, secure, reliable, and accessible user experience. These requirements are organized by quality attribute and priority level (Table 2). The system will offer an intuitive and responsive user interface that will require very little training, will work in both desktop and mobile browsers, will adhere to the standards of web accessibility, and will give clear notifications, such as the progress of uploading documents and confirming submission. The system will ensure high performance, with standard pages being loaded in less than three seconds under normal circumstances, will guarantee data integrity, with no incomplete transactions or duplicates, and will remain available throughout the academic periods that are required. The system will also guarantee security through authentication, HTTPS encryption, session timeout, input validation, privacy settings, and protection from any malware. The AI-generated content will be clearly marked with a mandatory disclosure confirming that the content is AI-generated information. In addition, manuscript confidentiality will be strictly maintained by executing summarization requests via the Gemini Flash API under an active paid billing tier, which contractually guarantees that student manuscript data will not be accessed by human reviewers or utilized by external AI services for model training and improvement. Finally, the system will allow regular backups, modularity for possible upgrades, and a permanent audit log of academic decisions and approvals made by a user.

Table 2

Non-functional Requirements

Quality Attribute	Non-functional Requirement	Description	Priority

Usability & Accessibility	The system should be easy to understand and use and compatible across devices.	Ensure the interface requires minimal training, utilizes responsive design for desktop and mobile browsers, follows standard web accessibility practices, and displays file upload progress.	High
Performance	The system shall provide fast response times and clear feedback.	Standard pages must load within three seconds under normal operating conditions, and document uploads shall display upload progress and confirm successful submission to maintain an efficient user experience.	High
Reliability & Availability	The system shall maintain consistent availability and data integrity.	Prevent incomplete workflow transactions and duplicate approval actions, preserve the complete accuracy of project records, and remain available during required academic periods.	High
AI Transparency & Protection	The system shall securely manage and label AI-generated content.	Clearly disclose when information is generated by artificial intelligence, and ensure student manuscripts submitted for summarization are protected by processing API calls exclusively under an active paid service tier	High
Security & Privacy	The system shall enforce strict data protection and access controls.	Use secure authentication, HTTPS encryption, session timeouts, input validation, restrict harmful file uploads, and strictly	High

		comply with institutional privacy policies.	

Quality Attribute	Non-functional Requirement	Description	Priority
Maintainability & Scalability	The system shall be resilient, restorable, and adaptable to growth.	Maintain regular backups for authorized restoration, use modular components for future enhancements, and reliably support an increasing volume of users and projects.	Mid
Auditability	The system shall trace all critical workflow and administrative actions.	Ensure that all academic decisions, approvals, and workflow changes are permanently and accurately traceable to the responsible user.	Mid

Design of Software, System, Product and Process

The proposed system is represented as a single, centralized process that facilitates and manages the end-to-end thesis and capstone workflow of the Holy Angel University – School of Computing (Appendix F). The system interacts with its external entities, namely the Student, Instructor 1, Instructor 2, Adviser, Panel Member, Panel Chair, Program Chair/Coordinator, Dean, Associate Dean, University Research Office, and System Administrator. These entities provide the major data inputs to the system, including authentication information, project registration information, project documents, review feedback, approval decisions, defense scheduling information, weekly progress records, user account management information, and URO verification decisions. The system in turn generates the major data outputs, namely authentication results, project status updates,

assigned project information, review feedback, artificial intelligence-generated manuscript summaries, defense and clearance information, reports, notifications, final project records, and system management information. By representing the proposed system as a single process, the context diagram establishes the overall system boundary and illustrates the primary data exchanges between the system and its external users before these interactions are decomposed in the succeeding Data Flow Diagrams.

The overall movement of data between the external entities and the proposed system is represented in the Level 0 Data Flow Diagram (Appendix G). The academic lifecycle begins when all user roles (System Admin, Student, Instructor 1, Instructor 2, Adviser, Panel Chair, Panel Member, Program Chair / Coordinator, Associate Dean, Dean, and URO) submit their login credentials to process 1.0, User Authentication & Role Access. These credentials are validated against D1, Users & Roles Database, and the corresponding hash data is fetched. Once validation is done, the Student sends the topic proposal along with group details to process 2.0, Project Registration & Topic Approval. Proposal registration details along with annotation are sent by Instructor 1 & Adviser, respectively. Process 2.0 then stores the project information and manuscript into data store D2, Projects Repository. As the study advances, the Student submits manuscript files to process 3.0, Document Submission & Version Control, which archives each versioned draft into data store D3, Manuscript & Versions. To facilitate automated assessment, process 4.0, AI Summarization & Document Review, queries data store D2 (Projects Repository) via a manuscript context API request and forwards this to the External AI Service (Gemini Flash). The external AI system delivers the processed context of the manuscript, allowing process 4.0 to deliver AI summarizations and

insights to the Adviser, Panel Chair, and Panel Member, and to receive progress milestones from Instructor 2.

Monitoring and Milestones administration take place through the following process: 5.0 Implementation & Weekly Progress Report. The Student submits weekly progress reports, whereas Instructor 2 is responsible for milestone verifications and sign-offs. The progress reports for clearance are entered directly into data store D5, Digital Forms & Clearances. Formal title defense workflows are managed by process 6.0, Defense Scheduling & Verdict Evaluation. The Program Chair / Coordinator enters the title defense schedule, panel assignments, adviser assignments, workflow approvals, and reports. During defenses, rubric evaluations and proposal verdicts are supplied by the Panel Chair and Panel Members, alongside panel comments from the Adviser. Data store D4 (Defense Records), Process 6.0 saves recommendations and decisions of the panel as well as revisions. The External Email Service will notify automatically. After defense, compliance will be taken care of through Process 7.0, Final Clearance. Recommendation and revision forms will be submitted by Panel Members, but the approval validations will be submitted by Instructor 2, Associate Dean, Dean, and URO. Process 7.0 will read/write the clearance status in D5 (Digital Forms & Clearances) and write project status in D2 (Projects Repository).

Lastly, after defense, the documentation and the decisions will go to process 8.0, Reporting & Audit Trail, along with the input from the Student. Process 8.0 will record/retrieve activity logs from data store D6 (Audit Logs), generate system activity logs for the System Admin, and get final approvals from the Dean and Associate Dean.

The structure of the logical database for the suggested system is shown on an Entity Relationship Diagram (see Appendix H). This diagram includes three parts that make up one single schema, with the core represented by USER and PROJECT entities. The first part describes identity, roles, and projects. The USER entity contains data on registered users, such as their name, email, study program, and status, and is distinguished by the user_id attribute. The ROLE entity includes both global and project-related roles, whereas USER_ROLE is an associative entity with USER and ROLE as foreign keys, which enables one user to have several roles at once. The PROJECT entity contains the project's title, category, term, and stage and is distinguished by the project_id attribute. It is linked to the project's members and assigned faculty via associative entities PROJECT_MEMBER and PROJECT_ASSIGNMENT. They have PROJECT and USER as foreign keys and make up one-to-many relations, as one project has many members and many assigned faculty.

The second section shows the process of a manuscript through DOCUMENT, AI_SUMMARY, REVIEW, WEEKLY_ACCOMPLISHMENT, DEFENSE, and EVALUATION_FORM. The DOCUMENT, which is distinguished by the unique document_id attribute, refers to the PROJECT and USER and has data on each version of the file uploaded into the system. AI_SUMMARY references a specific document version and stores the machine-generated summary produced only at the proposal or final defense milestone, while REVIEW references both DOCUMENT and USER to consolidate the category, decision, and student response associated with a given round of feedback. DEFENSE references PROJECT and records the date, verdict, and revision deadline of a proposal or final defense. EVALUATION_FORM references both DEFENSE and USER and stores the individual evaluation of each panel member. These relationships are likewise

one-to-many: a project accumulates many documents and weekly accomplishment records, a document receives many reviews, and a defense collects many evaluation forms.

The third section represents forms, workflow, and system records. DIGITAL_FORM references PROJECT and represents an institutional form such as a recommendation or clearance form, while DIGITAL_SIGNATURE references both DIGITAL_FORM and USER and stores the signatures required to authenticate it. WORKFLOW_HISTORY references PROJECT and USER and provides an append-only log of every stage transition a project undergoes, while NOTIFICATION and AUDIT_LOG reference USER directly rather than PROJECT, since they record system alerts and critical activity generated across the system as a whole. By employing the use of primary keys for unique identification, foreign keys for referential integrity, and associative tables for the many-to-many relationship within all three parts of the database, the schema is designed in such a way that it captures the multi-level process within the system while still maintaining integrity and traceability.

The data dictionary of the proposed system is presented in Appendix I. Whereas the Entity Relationship Diagram establishes the entities and their relationships, the data dictionary formalizes the schema by specifying, for each of the seventeen entities, the exact fields, data types, key constraints, and purpose of every attribute. Unique IDs in the schema will be implemented using UUIDs so that they remain unique, while the categorical attributes including account status, document status, review outcome, and defense decision will utilize enumeration rather than plain text. Nullable attributes describe attributes whose value depends on some condition, for example, a defense verdict which is initially not known, but becomes clear after being written by the Panel Chair, or a digital signature which does not

exist yet and needs to be added. The only associative table which uses a composite key is PROJECT_MEMBER. DOCUMENT achieves immutability by creating a new row each time its content is updated. Reference and configuration tables that would otherwise fall under the System Administrator role, such as academic term and workflow stage configuration tables, were deliberately excluded from the schema because they constitute administrative rather than academic workflow records.

The proposed system follows a three-tier, serverless architecture composed of a presentation tier, an application tier, and a data tier, together with two external services. The presentation tier consists of the React single-page application served as compiled static assets, which renders the role-based dashboards and mediates all user interaction. The application tier includes Firebase Cloud Functions running on Node.js and serves the endpoints where all the business logic is executed, such as request validation, state transitions of workflows, checking permissions with respect to Context-Aware Access Control tags, and sending notifications. The data tier consists of Cloud Firestore for structured records and Firebase Cloud Storage for uploaded manuscripts and certificates, both protected by security rules that deny direct client access. Two external services lie outside the system boundary: Gemini Flash, which is an artificial intelligence-based summary service that receives the manuscript content at certain milestone events, and Resend, which is an email service used for sending workflow notifications. All communications between the tiers are transmitted over the HTTPS protocol, and there is no communication between the application tier and the data tier via the client application.

Several design decisions required the proponents to weigh competing constraints, and the resulting trade-offs are summarized here in synopsis, with their detailed application to be discussed in the Results and Discussion section.

The first concerns the choice of a Backend-as-a-Service platform over a conventional server framework. A self-managed stack composed of Express and a relational database management system would have offered finer control over query optimization, schema constraints, and hosting configuration. Firebase was nevertheless selected because it removes the burden of server provisioning, scaling, and patching from a student development team working within the duration of two academic terms, and because its integrated authentication, storage, and serverless compute services shorten the development cycle required by the iterative approach adopted in this study. The trade-off accepted is a degree of vendor dependence and reduced control over the underlying infrastructure.

The second concerns the use of a document-oriented data store to realize a relational logical model. The Entity Relationship Diagram and data dictionary presented above specify primary keys, foreign keys, and referential integrity constraints, none of which Cloud Firestore enforces natively. The proponents still used Firestore because of its real-time synchronization capabilities allow changes in the workflow statuses to be reflected on connected dashboards without the need for polling. To preserve the integrity that a relational engine would otherwise guarantee, referential consistency is enforced at the application tier: all create, read, update, and delete operations are performed exclusively through server-side Cloud Functions, and operations that affect more than one collection are executed as atomic Firestore transactions. The trade-off accepted is that integrity depends on the correctness of

application logic rather than on database constraints, which increases the importance of the testing procedures described in the following subsection.

The third concerns the separation of frontend and backend hosting across two providers. Consolidating both on a single platform would have simplified deployment and monitoring. The proponents separated them because Hostinger provides managed hosting already familiar to the institution for static content delivery, while Firebase provides the serverless runtime and managed data services on which the application tier depends. The trade-off accepted is a more complex deployment pipeline, which is mitigated by the automation described below.

Development and Testing

Development will observe established conventions and standards appropriate to each layer of the system. The source code in JavaScript and React will adhere to the Airbnb JavaScript Style Guide (Airbnb, n.d.), which is a widely adopted coding convention guide used by the community that outlines coding conventions for naming variables and functions, organizing components, using ES6+ syntax, and organizing files. Adopting this style guide is necessary because the system's frontend is distributed across multiple independently authored React components, StudentDashboard.jsx, InstructorPanel.jsx, AdvisoryGuidance.jsx, and VerdictRecording.jsx, among others, that are developed, reviewed, and merged by different proponents through the pull-request workflow described later in this section; without a shared, automatically enforced convention, stylistic drift among contributors would slow down code review and increase the likelihood of overlooked defects during that review. This

is proven empirically by research in software engineering that has shown how adherence to coding conventions can lower defect density and improve understandability and modifiability of the code by programmers unfamiliar with it (Boogerd & Moonen, 2008). This is because one of the important non-functional requirements mentioned is that of maintainability and scalability, whereby the component must be legible even to those developers who have not authored it. This is enforced through ESLint and Prettier configurations committed to the repository so that formatting remains consistent across contributors regardless of individual editor settings. Security requirements will be considered taking into account the following elements of OWASP Top Ten (OWASP Foundation, 2025): Broken Access Control, Injection, Security Misconfiguration, all of which correspond with the above access control and input validation requirements. The accessibility of the interface will adhere to WCAG 2.1 (World Wide Web Consortium, 2018) at level AA as per the accessibility requirement listed in the non-functional requirements section. Software quality will be measured according to ISO/IEC 25010, as described in the Instrument section.

The system is built on JavaScript as the core programming language for both client-side and server-side programming in order to ensure consistent data types across the application. On the client side, JavaScript controls all user interaction within the React components, which include StudentDashboard.jsx, InstructorPanel.jsx, AdvisoryGuidance.jsx, and VerdictRecording.jsx. It performs client-side form validation when students submit concept papers, verifying that file types are permitted and that mandatory fields are complete before data is transmitted. JavaScript is responsible for handling the flow of data, where it handles the user input, serializes thesis metadata into a

JSON object, and sends it to the backend using HTTP POST requests, while real-time listeners handle the JSON response coming from the database.

HTML5 and CSS3 are directly integrated into React components via JSX syntax to create the visual layer of the client-side interface. HTML5 creates the semantics of the structure, where HTML forms define the document upload form, tables are used to record project milestones, and navigation menu structures are created. CSS3 styles the structure and creates responsiveness through media queries and the use of Flexbox and Grid layout, thus converting the complex interface structure, like the multi-column review criteria structure of the AdvisoryGuidance.jsx component, to a stacked layout on tablets and mobiles.

React is used as the frontend framework, helping the proponents to create a dynamic single-page application that updates dynamically without reloading the entire page (Hasan et al., 2022). React has been chosen because it allows creating efficient code due to its component-based structure and virtual DOM, which makes DOM manipulations unnecessary, thus making the process of rendering and state-based programming faster (Bai, 2022). Every interface component has its own layout and functionality, receives data via state management functions, and renders based on the JSON data provided by the server. The URLs correspond to certain components with the help of React Router.

On the backend, JavaScript executes within a Node.js environment to power serverless Firebase Cloud Functions, which act as the bridge between the frontend and the database by handling core business logic through an event-driven compute model (Li et al., 2021). Rather than a traditional server framework such as Express, Cloud Functions serve as the application programming interface endpoints that receive HTTP POST requests and

JSON payloads directly from the React frontend. When a form is submitted, the corresponding Cloud Function acts as a controller that parses the received data, performs server-side validation, issues secure database commands, and returns a structured JSON response to the client.

Backend responsibilities are separated by module. The file submissionHandlers.js parses JSON requests, validates document URLs, and creates the corresponding database records. The file authTriggers.js assigns role-based permissions automatically upon user registration. The file notificationService.js composes and dispatches workflow email notifications. Application configuration and credentials are isolated from application logic: the Firebase client configuration required by the React application is supplied through environment variables defined in a .env file, which is excluded from version control through .gitignore and injected at build time by the continuous deployment pipeline, while server-side secrets, namely the artificial intelligence service API key and the transactional email service API key, are stored in Firebase Functions configuration through the Firebase CLI and are read at runtime from the functions configuration object. Neither API key nor any other authentication credential is committed to the repository or revealed to the client package, while the Firebase account key that will be used for administration exists only in the deployment environment. Firebase Authentication authenticates institutional identity and generates JWTs with role claims, which are attached to subsequent requests from the React frontend application.

In relation to the manuscript summarization through automation, the system employs an integration with Google's Gemini Flash via the API in the Node.js Firebase Cloud

Functions backend. Gemini Flash is chosen because of the large context window provided by this model, which can handle lengthy academic manuscripts in a single pass without needing multi-step document splitting. Moreover, the inference latency is low, and the computational overhead is low enough to provide quick JSON-structured summaries. Architecturally and from a data governance perspective, all the back-end API calls are provisioned on an active paid billing tier. This satisfies institutional security requirements by ensuring that Google does not retain, inspect via human reviewers, or use submitted manuscript text and prompts to train foundational models. The generated JSON payload is subsequently stored in the AI_SUMMARY collection in Cloud Firestore and consumed by the React frontend during the proposal and final defense milestones. (*Gemini API Additional Terms of Service | Google AI for Developers*, n.d.).

Cloud Firestore, a NoSQL document database, manages and synchronizes data between the frontend and backend in real time. The database design comprises collections that function as tables and that are related through identifier fields in the manner described in the data model above. USERS collection contains information about credentials and permissions. It is stored as string fields user_id and email, as well as the CAAC_permissions object that holds the information about permissions. PROJECTS collection plays the role of the primary database for workflow statuses. The information about projects is stored as string fields project_id and group_name, an array field team_members, and milestone date fields. The SUBMISSIONS collection tracks document uploads through a project_id field referencing the parent document in PROJECTS, and implements immutable version control through the string fields submission_id and file_url, which points to secure Firebase Cloud Storage, together with an integer version_number. Further collections include

ANNOTATIONS for non-destructive reviews, WORKFLOWS for milestones, and AUDIT_LOGS, which enforce write-once behavior for all critical system operations.

All create, read, update, and delete operations are executed exclusively through server-side Cloud Functions in order to enforce security constraints and to prevent direct client manipulation of the database. Creating a new project group, for example, triggers a backend function that executes an atomic Firestore transaction, writing records to both the PROJECTS and WORKFLOWS collections simultaneously so that data integrity is preserved in the event of a network failure. For data retrieval, the React frontend subscribes to Firestore real-time listeners, so that when a backend function modifies a project's status, the database pushes the updated JSON payload to connected clients and the interface reflects the change without a browser refresh.

Various tools facilitate the development process and collaboration among the team members. Visual Studio Code is used as the primary integrated development environment to develop the React components and the Node.js backend, along with several extensions such as ES7+ React snippets and Firebase Explorer. Source control is maintained on GitHub using a branching strategy in which every change is reviewed through a pull request before being merged into the main branch. Figma is used to design prototypes and wireframes in advance of coding in order to establish the application's design system. The use of Docker containers is intended to standardize the development environment because containerization provides superior resource management features, and environment compatibility problems can be solved through complete isolation of portable application runtime (Hernandez & Rios, 2024). The Firebase Console provides web-based management of the database and authentication,

while the Firebase Emulator Suite allows the proponents to test endpoints, JSON requests, and database operations locally and safely.

Tests will be conducted in three stages, in line with the iterative and incremental methodology. The local test will be done for every increment using the Firebase Emulator Suite, which will test the Cloud Functions, Firestore Security Rules, and Storage Rules without any costs or changes to production data in the cloud environment. Module testing then verifies each developed increment against its documented functional requirement using the researcher-designed checklists described in the Instrument subsection. The next step is integration testing after increments have been integrated into the developing system with emphasis on workflow handoffs, permission assessments for Global and Project-Based roles, and atomicity of write operations involving multiple collections because these are where the application-level enforcement of referential integrity is most vulnerable. Once all the modules are completed, testing of the whole system will be done according to the Sample and Setting and Instrument subsections.

Deployment and hosting duties are shared between two different managed platforms. React UI is compiled to optimized static files via 'npm run build', where JSX components are compiled to HTML, CSS, and minimized JavaScript, which are then hosted on the Hostinger managed web hosting platform that will store compiled files, manage server maintenance and uptime, and provide HTTPS serving of files. Firebase hosts the backend consisting of Cloud Functions, Cloud Firestore, and Cloud Storage in Google Cloud infrastructure with built-in redundancy across Google's data centers, with no database server maintained by proponents.

Deployment follows an automated and decoupled continuous deployment pipeline. Code changes affecting either the user interface or the backend triggers are first tested locally using the Firebase Emulator Suite in order to prevent regressions and avoid unnecessary cloud invocation costs. After confirmation via peer-reviewed pull requests and merging into the master branch, the GitHub Actions workflow installs the necessary dependencies, adds the build-time environment variables, builds the single-page application, and moves the production files to Hostinger. The use of SFTP and Hostinger File Manager is kept as fallback options. The backend components, namely the Cloud Functions, Firestore composite indexes, and storage security rules, are deployed separately by the proponents through the Firebase CLI using firebase deploy and execute within Google Cloud's auto-scaling runtime.

The system will run using the domain hausoc-research.org, which is registered via Hostinger's DNS management service and routed via Cloudflare. Cloudflare provides the services of a reverse proxy and content delivery network that caches static files at edge servers to reduce load times, a web application firewall that filters out bad requests, DDoS protection, as well as automatic HTTPS encryption using free SSL certificates. Therefore, when requests are made to the domain, the requests are processed by Hostinger via DNS and then routed via Cloudflare to Hostinger and Firebase services.

The domain name itself was chosen according to the domain-naming principles covered in the Web and Search Engine Optimization course, namely relevance, memorability, simplicity, appropriate extension, ease of typing and sharing, and the avoidance of unnecessary complexity. It is relevant because hausoc identifies the Holy Angel University School of Computing and research reflects the system's role in supporting the thesis and

capstone process, so that both purpose and ownership are legible from the address alone. This name is clear and memorable, in that it is concise, straightforward, entirely in lowercase characters, and comprises two easily recognizable words without numbers, abbreviations, or hyphens that could be typed incorrectly or remembered. It is appropriate for the .org domain because it is not intended to be used in business operations, but for organizational and educational purposes. This association is supported by research on website credibility: Walther et al. (2004) found that among the top-level domains tested, .org websites were judged most credible and trustworthy provided that they carried no advertising, a condition the proposed system satisfies.

System introduction and long-term maintenance will take place via procedures defined within institutions aimed at securing user adoption and sustainability of the platform in the School of Computing. This system introduction process will be conducted in phases, starting with a pilot test done by the researchers, then practical training of the advisors and panelists in conducting online reviews and clearances.

Ongoing maintenance follows a shared-responsibility model. Infrastructure concerns, including physical server uptime, operating system updates, network routing, and daily database backups, are abstracted and managed by Hostinger's managed hosting platform and Firebase's multi-region cloud infrastructure. Departmental system administrators retain responsibility at the application level, which includes monitoring Cloudflare firewall analytics and Cloud Function runtime logs, performing periodic npm package audits to remediate frontend dependency vulnerabilities, auditing Firebase Security Rules against the

principle of least privilege, applying retention policies to the Cloud Storage files of graduated cohorts, and applying patched releases from the GitHub repository.

Sample and Setting

The study will use stratified random sampling for the sampling of students because of the necessity to maintain a proportionate representation of year levels and program participants in a large sample population, and simple random sampling will be utilized to sample the faculty members who had a smaller sample population that did not require sub-grouping. (Etikan & Bala, 2017). IT experts and University Research Office participants were selected through purposive sampling, given the small size and highly specialized composition of these groups (Memon et al., 2021). The selected students for participation in this research will come from third and fourth year students taking part in the various programs in the School of Computing, since they are the existing and potential users of the system. Faculty and administrative participants, comprising Instructors, Advisers, Panel Members, academic administrators, University Research Office personnel, and the System Administrator, will be selected on the basis of their involvement in thesis, capstone, research supervision, coordination, or related academic processes, and will evaluate the system according to their respective responsibilities.

The total population from which the participants will be selected consists of third- and fourth-year students currently enrolled in the School of Computing at Holy Angel University, together with the faculty members of the School of Computing and one personnel from the University Research Office. As of the first semester of Academic Year 2026–2027, the total population is 875 individuals, comprising 836 students, 38 School of Computing faculty members, and 1 URO personnel. The student population consists of 416 fourth-year students and 420 third-year students enrolled in the following programs: BS Information

Technology major in Web Development, BS Information Technology major in Network Administration, BS Entertainment and Multimedia Computing, BS Computer Science, and BS Cybersecurity. Third-year students are included as they constitute prospective users of the developed system upon reaching the thesis and capstone stages of their academic program. The faculty population consists of 38 members, including the Dean, the Associate Dean, program coordinators, and full-time and part-time faculty members of the School of Computing.

The calculated number of respondents among the students and faculty was based on the use of the finite population correction Cochran formula which was calculated using the Raosoft online calculator (Raosoft, Inc., 2004), at a 95% confidence level and 5% margin of error. The initial calculation was based on the formula for the infinite population:

$$n_0 = Z^2(p)(q) / e^2$$

$$n_0 = (1.96)^2(0.5)(0.5) / (0.05)^2$$

$$n_0 \approx 384$$

This was then adjusted for each population size with the use of the finite population correction:

$$n = n_0 / (1 + (n_0 - 1)/N)$$

For the student population (N = 836):

n = 384 / (1 + 383/836)

n = 384 / 1.458

n ≈ 264

For the faculty population (N = 38):

n = 384 / (1 + 383/38)

n = 384 / 11.08

n ≈ 35

This computation provided the sample size needed consisting of 264 students who will be selected using the stratified random sampling with strata by year level (and program, when appropriate) and 35 faculty members who will be selected using the simple random sampling.

As the only representative of the University Research Office members, the participant will be selected as the only person with this position in line with the inclusion criteria for the purposive sampling method being employed in this research. Should the participant not respond or withdraw from the study, additional participants will be selected based on the inclusion criteria to ensure that the sample size requirement will be attained.

Testing and assessment will be done inside the premises of the Holy Angel University School of Computing for the first semester of the academic year 2026–2027. The system will be accessed by participants via the use of institutional or personal devices that are connected either to the internet of the university or of their own, with test accounts prepared by the proponents in order for the system to undergo testing in its usual context of operation. These

sessions may be done inside computer labs or the offices of the participants, or online, depending on their availability.

Instruments

For end-user testing, the proponents will utilize a combination of researcher-developed and literature-based instruments.

For functional testing, the proponents will develop researcher-designed checklists derived from the approved functional requirements of the proposed system (Appendix J). This approach is appropriate because the system implements institution-specific workflows that require verification against the project's documented specifications. The developed test cases will evaluate whether each implemented function performs according to its intended behavior using a binary Pass or Fail metric, consistent with requirements-based testing practices (Mustafa et al., 2021).

The process of checking content validity will involve having the researcher-developed checklist of functional testing examined by the thesis adviser and analyzed for relevance, clarity, and consistency with the already accepted functional requirements. The necessary adjustments will be made based on the advice given by the adviser before the final use of the checklist. The approval by the adviser will be recorded for documentation purposes. A pilot test will also be performed using some selected respondents in order to check the clarity and functionality of the checklist.

The System Usability Scale created by Brooke (1996) will be used in testing for usability, and this is comprised of ten questions which were formulated in such a way as to capture users' perception of usability in interactive systems (Appendix K). This questionnaire

measures various parameters relating to usability such as ease of learning, efficiency of use, confidence during interaction, and overall system usability. In order to maintain its psychometric reliability, the tool will be used without any modifications from its original state. The adopted questionnaire is provided in Appendix L.

For user acceptance testing, the proponents will adapt the Technology Acceptance Model developed by Davis (1989), particularly its twelve items measuring Perceived Usefulness and Perceived Ease of Use (Appendix M). These constructs assess whether users perceive the developed system as beneficial to their work and sufficiently easy to use, both of which are recognized determinants of user acceptance of information systems (Davis, 1989). The item wording will be adapted to reflect the academic roles of the students and faculty evaluating the system, while the original response format is retained to ensure statistical validity. The adapted questionnaires are presented in Appendix N.

For expert evaluation, the evaluation will be done through the use of ISO/IEC 25010:2011 Software Product Quality Model (Appendix O). The framework gives an internationally accepted guide for the assessment of software product quality in terms of various characteristics of software quality. Although ISO/IEC 25010:2023 introduced a revised nine-characteristic model, this study adopts the ISO/IEC 25010:2011 model, as its five characteristics — Functional Suitability, Usability, Reliability, Performance Efficiency, and Security — directly align with the technical and security evaluation objectives of this study and remain consistent with prior software quality evaluation research in similar academic information systems. These characteristics are the most relevant to the technical objectives and security requirements of the proposed system (Table 3).

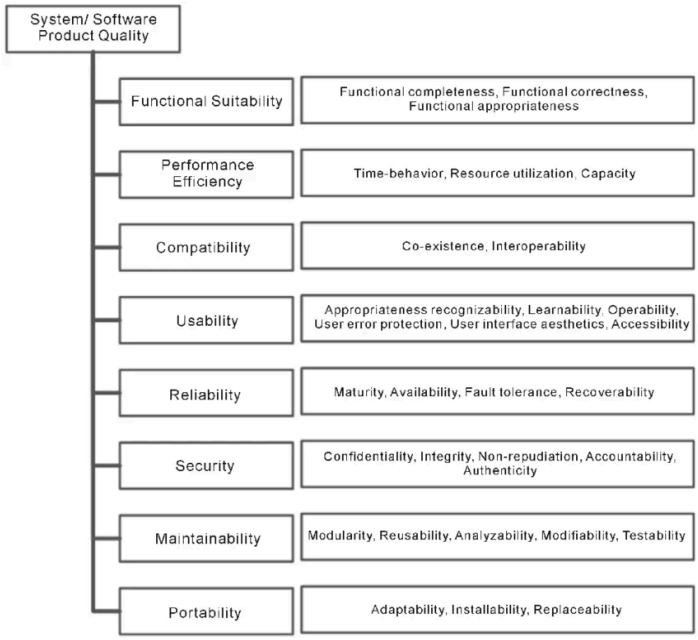
Table 3

ISO/IEC 25010 Quality Characteristics and Their Definitions

ISO/IEC 25010 Quality Criterion	Meaning
Functional Suitability	Assesses whether the system provides functions that meet the specified requirements and supports users in completing their intended tasks accurately and appropriately.
Usability	Assesses the ease with which users can learn, understand, operate, and interact with the system effectively and efficiently.
Reliability	Assesses the system's ability to perform consistently and maintain stable operation under specified conditions.
Performance Efficiency	Assesses the system's responsiveness, processing speed, and efficient utilization of system resources during operation.
Security	Assesses the system's ability to protect information and data from unauthorized access while ensuring confidentiality, integrity, and appropriate user authentication and authorization.

Figure 3

ISO/IEC 25010:2011 Software Product Quality Model



Data Collection Procedures

The process of data collection begins with an official invitation to the chosen respondents with an explanation of the goals of the evaluation. Prior to taking part in the evaluation activities, the participants will be offered an informed consent form and will be given all necessary information about the research. They will sign the form of voluntary participation and will know that their answers will remain confidential.

Before conducting the test, the proponents will conduct a video presentation showing the working of the web application and its major features like role-based dashboards, document submission process, approvals, and notifications. Participants will be given

sufficient time to use the system independently and perform the predefined workflows corresponding to their assigned roles. Student end-users will access their project workspace, submit project documents, review document versions, and receive or respond to workflow notifications. The Instructor, Advisor, Panel Member, academic administrator, URO representative, and the System Administrator shall perform the responsibilities that are relevant to his/her position. These include document review, annotation, approval, routing, project management, and system configuration.

There will be adequate time given to the IT experts to engage with and assess the functioning of the system using criteria of functional suitability, usability, reliability, performance efficiency, and security based on the chosen ISO/IEC 25010 quality criteria. Having completed the activities of the system, the participants will complete the evaluation forms according to the functions they perform. End users such as students and faculty members, along with the System Administrator, will complete the functional testing checklist prepared by the researcher along with the System Usability Scale (SUS) and TAM questionnaire using the online Google Form, while IT experts will complete the ISO/IEC 25010 evaluation form using the online Google Form.

Data Analysis Plan

The data collected through the various testing and evaluation instruments will be analyzed using statistical tools appropriate to their respective scales of measurement. The instruments adopted in this study employ different response scales, each retained in its original form to preserve established validity. Functional testing uses a binary Pass or Fail criterion. The System Usability Scale uses its original five-point Level of Agreement scale,

having a range from Strongly Disagree to Strongly Agree, yielding a composite score ranging from 0 to 100. The adapted Technology Acceptance Model uses its original seven-point Level of Likelihood scale, ranging from Extremely Unlikely to Extremely Likely. The expert evaluation based on ISO/IEC 25010 uses a five-point Level of Excellence scale, a scale length supported by recent survey research indicating that it achieves an appropriate balance among cognitive effort, response bias, and information quality (Taufique et al., 2026), thereby allowing the IT experts to rate the technical criteria without the effects of survey fatigue.

Responses to the functional testing checklist will be analyzed using descriptive statistics, specifically frequency counts and percentages, in order to determine the functional completion rate of the system workflows. The percentage of successful task completions is calculated using the following formula:

$$P = \frac{f}{N} \times 100$$

Where:

P - represents the percentage of successful task completions

f - represents the frequency of "Pass" responses, which is the total count of successfully executed tasks.

N - represents the total number of items evaluated, which is the absolute total of all tasks tested on the checklist.

For instance, when there are 8 specific functionalities in the evaluation checklist and 10 individuals participate in the test process, the number of items evaluated becomes 80 (N = 80). When seven out of the eight items are completed successfully by the test subjects (f = 72), then dividing 72 by 80 gives us 0.90, and multiplying by 100 gives us a 90% successful task completion rate (P = 90%). This process is systematically applied across the evaluation checklists for each respective user role to measure functional reliability. The 100% rate of functionality completion will be deemed to be the acceptance criterion for the system since each and every entry in the checklist of functional testing is a direct representation of a functional requirement that has been pre-approved (Appendix J). Any failed item therefore indicates an unmet requirement rather than an optional or exploratory feature.

The System Usability Scale will be scored using Brooke's standard evaluation formula to generate a composite usability score ranging from 0 to 100. For odd-numbered items, namely items 1, 3, 5, 7, and 9, the score contribution is the scale position minus one. For even-numbered items, namely items 2, 4, 6, 8, and 10, the contribution is five minus the scale position. The sum of these calculated scores is then multiplied by 2.5 to obtain the final overall score.

$$SUS = \left(\sum (R_{odd} - 1) + \sum (5 - R_{even}) \right) \times 2.5$$

Where:

SUS - refers to the final usability score of the system, which ranges from 0 to 100.

R_{odd} - refers to the adjusted scores from odd-numbered questions, where 1 is subtracted from each response.

R_{even} - refers to the adjusted scores from even-numbered questions, where responses are reversed by subtracting each answer from 5.

For example, if the sum of all the scores of the items after adjusting is 22, the final SUS score is then obtained by multiplying 22 by 2.5:

$22 \times 2.5 = 55$

Therefore, the total SUS score of the system from the respondent is 55. The same procedure is followed for other respondents to determine the overall usability perception of the system. To interpret the resulting SUS scores, this study adopts the acceptability benchmarks established by Bangor et al. (2008), based on their analysis of scores from over 2,300 SUS evaluations, with an average SUS score across studies of approximately 68 (Table 4):

Table 4

System Usability Scale (SUS) Score Interpretation

SUS Score Range	Interpretation
70 and above	Acceptable
50 – 69.99	Marginal
Below 50	Not Acceptable

Responses to the adapted Technology Acceptance Model questionnaire will be analyzed by computing the construct mean, that is, the arithmetic average, for each group of related items, namely Perceived Usefulness and Perceived Ease of Use.

$$\bar{x} = \frac{\sum x_i}{n}$$

Where:

$\bar{X}$ - represents the construct mean or the average score of a TAM factor.

$\sum x_i$ - represents the total sum of responses from all items under a specific construct.

n - represents the total number of questionnaire items included in that construct.

For example, if a construct contains 5 items and the total score of a respondent's answers is 30, the construct mean is calculated as:

30 ÷ 5 = 6

This indicates that the response from the respondent had an average score of 6 out of 7 in terms of that construct. This procedure is followed for all respondents tto determine the overall level of acceptance of the system based on its Perceived Usefulness and Perceived Ease of Use.

The interpretation of the computed construct means involves calculating the interval width by finding the difference between the maximum and minimum score divided by the number of scale ratings:

(7 − 1) ÷ 7 = 0.86

Using the calculated interval, the construct means will be translated into verbal interpretations following the original seven-point Likelihood Rating scale (Table 5):

Table 5

Technology Acceptance Model (TAM) Construct Mean Interpretation Scale

Range	Interpretation
6.15 – 7.00	**Extremely Likely**
5.30 – 6.14	**Quite Likely**
4.44 – 5.29	**Slightly Likely**
3.58 – 4.43	**Neither Likely nor Unlikely**
2.72 – 3.57	**Slightly Unlikely**
1.87 – 2.71	**Quite Unlikely**
1.00 – 1.86	**Extremely Unlikely**

Responses to the IT expert evaluation will be aggregated and analyzed using the weighted mean in order to determine the experts' assessment of each software quality sub-characteristic under the five selected characteristics. The weighted mean will first be computed for each sub-characteristic; the weighted means of all sub-characteristics under a given characteristic will then be averaged to obtain that characteristic's overall assessment score. For example, the weighted means of Confidentiality, Integrity, Non-repudiation, Accountability, and Authenticity will be averaged to determine the overall score for Security.

$$\bar{x}_w = \frac{\sum f_i x_i}{N}$$

Where

$\bar{x}_w$ - represents the weighted mean or the overall average evaluation score.

f_i - represents the frequency or number of responses for a specific rating category.

x_i - represents the assigned weight of each rating category.

N - represents the total number of expert respondents.

For example, if most experts give a higher rating, the weighted mean will reflect a higher overall evaluation score.

To interpret the computed weighted mean, the interval width is calculated by finding the difference between the highest and lowest possible scores and dividing it by the number of rating categories.

(5 - 1) ÷ 5 = 0.80

The resulting interval values are used to create consistent ranges for interpreting the evaluation results and assigning their corresponding verbal descriptions (Table 6).

Table 6

ISO/IEC 25010 Weighted Mean Interpretation Scale

Range	Scale	Interpretation
4.21 - 5.00	**5**	**Very Good**
3.41 - 4.20	**4**	**Good**

2.61 - 3.40	**3**	**Fair**
1.81 - 2.60	**2**	**Poor**
1.00 - 1.80	**1**	**Very Poor**

With only three IT experts involved, each weighted mean is drawn from a small set of ratings and should be read with caution rather than as a stable statistic. The individual ratings for each sub-characteristic will be reported alongside the mean so the underlying scores remain visible.

Ethical Consideration

The participants will be given adequate information about the aim of the study, the procedures for testing and assessment, what they are supposed to do, and the kinds of information that will be gathered from them before their involvement in the study. Before the conduct of the study, the researchers will ensure that all the necessary institutional approvals have been obtained. Participation will always be voluntary, and the participants will be at liberty to decide whether or not they want to participate in the research.

For system testing, users will access the system via their emails at Holy Angel University, and dummy data will be used where applicable. Information related only to the objectives of the study will be collected and will include only the following: task completion outcomes, errors, usability issues, evaluation ratings, comments, and recommendations. Where necessary, screenshots, activity logs, or screen recordings may be included in the documentation. Users will be informed beforehand, and all sensitive information will be removed or blurred out.

The privacy of all data collected will be ensured through the use of participant codes of the participants and will be kept in a secure and password-protected setting. Access to the data collected will be restricted to the researchers and authorized persons who are part of the research study. The collection, processing, storage, and usage of the data collected will conform to the rules outlined in the Data Privacy Act of 2012 (Republic Act No. 10173). The information collected will be utilized only for the purpose of research, specifically to assess the performance of the system, and will not be disclosed in such a way as to identify the individual participant. Data will be retained only for the duration needed to conduct the research study.

The results obtained through system testing and evaluation will be analyzed in relation to the functionality, usability, and quality of the developed system. It will not be considered a measure of technical knowledge, scholastic achievements, or individual performance of the participants. Such procedures aim to protect the autonomy, confidentiality, and rights of all participants involved in the study.

References

Adegoke, T. B., & Olalekan, A. A. (2026). Design and development of a Workflow-Driven Online student Clearance System for seamless Multi-Departmental Integration. Open MIND. https://doi.org/10.5281/zenodo.18506360

Airbnb. (n.d.). *Airbnb JavaScript style guide*. GitHub. Retrieved September 21, 2026, from https://github.com/airbnb/javascript

Aliazas, J. V., dela Cruz, R., & Ilagan, N. (2024). Enhancing university operations: A study of the electronic document management systems (EDMS) of one higher education institution. TWIST, 19(3), 229–237. https://doi.org/10.2139/ssrn.5714542

Bangor, A., Kortum, P. T., & Miller, J. T. (2008). An empirical evaluation of the system usability scale. International Journal of Human-Computer Interaction, 24(6), 574–594. https://doi.org/10.1080/10447310802205776

Bai, A. (2022). Million.js: A Fast Compiler-Augmented Virtual DOM for the Web. *arXiv*. https://doi.org/10.1145/3555776.3577683

Benavides, L., Arias, J. T., Serna, M. A., Bedoya, J. B., & Burgos, D. (2020). Digital Transformation in Higher Education Institutions: A Systematic Literature review. Sensors, 20(11), 3291. https://doi.org/10.3390/s20113291

Boogerd, C., & Moonen, L. (2008). Assessing the value of coding standards: An empirical study. In H. Mei & K. Wong (Eds.), Proceedings of the 24th IEEE International

Conference on Software Maintenance (pp. 277–286). IEEE. https://doi.org/10.1109/ICSM.2008.4658076

Brooke, J. (1996). SUS: A “quick and dirty” usability scale. In P. W. Jordan, B. Thomas, B. A. Weerdmeester, & I. L. McClelland (Eds.), Usability evaluation in industry (pp. 189–194). Taylor & Francis. https://doi.org/10.1201/9781498710411-35

Bumalod, M. C., & Velasco, R. M. A. (2024). Synergistic Information Security Design Implementation based on Role-Based Access Control, Information Classification, and AES Cryptographic Encryption. International Journal in Information Technology in Governance Education and Business, 6(1), 68–85. https://doi.org/10.32664/ijitgeb.v6i1.136

Corbin, T. A., & Walton, J. (2025). The missing story of GenAI summarisers: a critical research agenda. Higher Education Research & Development, 44(7), 1655–1668. https://doi.org/10.1080/07294360.2025.2486185

Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. MIS Quarterly, 13(3), 319–340. https://doi.org/10.2307/249008

Doni, A. W., Darwel, Ningsih, M. P., Ponda, A., Sugriarta, E., & Susanti, D. (2026). Design and implementation of an online thesis management system using the waterfall model at a health polytechnic in Indonesia. Indonesian Journal of Learning Education and Counseling, 8(2), 153–169. https://doi.org/10.31960/ijolec.v8i2.3343

EasyChair. (n.d.). EasyChair conference management system. https://easychair.org/

Etikan, I., & Bala, K. (2017). Sampling and Sampling Methods. Biometrics & Biostatistics International Journal, 5(6), 215–217. https://doi.org/10.15406/bbij.2017.05.00149

Gökmen, A. M. (2025). Strategic Trends in Digital Document Management. Hazine-i Evrak Arşiv Ve Tarih Araştırmaları Dergisi, 7(8), 393–420. https://doi.org/10.59054/hed.1737798

Gemini API Additional Terms of Service | Google AI for Developers. (n.d.). Google AI for Developers. https://ai.google.dev/gemini-api/terms#data-used-paid

Google for Education. (n.d.). Google Classroom Help Center. https://support.google.com/edu/classroom/

Grassi, P. A., Fenton, J. L., Newton, E. M., Perlner, R. A., Regenscheid, A. R., Burr, W. E., Richer, J. P., Lefkovitz, N. B., Danker, J. M., Choong, Y.-Y., Greene, K. K., & Theofanos, M. F. (2017). Digital identity guidelines: Authentication and lifecycle management (NIST Special Publication 800-63B). National Institute of Standards and Technology. https://doi.org/10.6028/nist.sp.800-63b

Hasan, M., Rahman, M., Chowdhury, M., Rahman, M., Abdulle, K., Sadia, F., & Hasan, M. (2022). Testing React Single Page Web Application using Automated Testing Tools. *Proceedings of the 17th International Conference on Evaluation of Novel Approaches to Software Engineering*, 469–476. https://doi.org/10.5220/0011077900003176

Hernandez, L., & Uc Rios, C. E. (2024). Docker Optimization of an Automotive Sector Virtual Server Infrastructure. *Knowledge Engineering and Data Science*, *7*(1), 71–85. https://doi.org/10.17977/um018v7i12024p71-85

Ifeoma, O. S., & Hillary, A. N. (2025). Paperless office management and the performance of state owned tertiary institutions in Enugu State. International Journal of Humanities, Literature and Art Research. https://doi.org/10.70382/mejhlar.v7i6.030

Instructure. (n.d.). Canvas LMS. https://www.instructure.com/canvas

International Organization for Standardization, & International Electrotechnical Commission. (2011). ISO/IEC 25010:2011: Systems and software engineering—Systems and software Quality Requirements and Evaluation (SQuaRE)—System and software quality models. https://www.iso.org/standard/35733.html

International Organization for Standardization, & International Electrotechnical Commission. (2023). ISO/IEC 25010:2023: Systems and software engineering—Systems and software Quality Requirements and Evaluation (SQuaRE)—Product quality model (2nd ed.). https://www.iso.org/standard/78176.html

Kayes, A. S. M., Han, J., Rahayu, W., Islam, M. S., & Colman, A. (2017). A policy model and framework for context-aware access control to information resources. arXiv preprint. https://doi.org/10.48550/arxiv.1703.02162

Larman, C., & Basili, V. (2003). Iterative and incremental developments. a brief history. Computer, 36(6), 47–56. https://doi.org/10.1109/mc.2003.1204375

Leite, L. O., Lagstedt, A., Kolog, E. A., & Tsupari, K. (2023). Contextualising thesis process digitalisation at a university in Ghana. Australasian Journal of Educational Technology, 39(4), 33–47. https://doi.org/10.14742/ajet.8801

Li, Z., Guo, L., Cheng, J., Chen, Q., Yao, B., & Ji, M. (2021). The serverless computing survey: A technical primer for design architecture. *ACM Computing Surveys*, *54*(10s), 1–34. https://doi.org/10.1145/3508360

Memon, M. A., Thurasamy, R., Ting, H., Cheah, J.-H., Chuah, F., & Cham, T. H. (2021). The inconvenient truth about convenience and purposive samples. *BMC Medical Research Methodology*, *21*(117). https://doi.org/10.1186/s12874-021-01331-9

Microsoft. (n.d.). SharePoint documentation. Microsoft Learn. https://learn.microsoft.com/sharepoint/

Mohammed, A. H. Y., Dziyauddin, R. A., & Latiff, L. A. (2023). Current multi-factor of authentication: approaches, requirements, attacks and challenges. International Journal of Advanced Computer Science and Applications, 14(1). https://doi.org/10.14569/ijacsa.2023.0140119

Mustafa, A., Wan-Kadir, W. M. N., Ibrahim, N., Shah, M. A., Younas, M., Khan, A., Zareei, M., & Alanazi, F. (2021). Automated Test Case Generation from Requirements: A Systematic Literature Review. *Computers, Materials & Continua/Computers, Materials & Continua (Print)*, *67*(2), 1819–1833. https://doi.org/10.32604/cmc.2021.014391

OpenProject GmbH. (n.d.). Open source project management software. https://www.openproject.org/

OWASP Foundation. (2025). OWASP Top 10:2025. https://owasp.org/Top10/2025/

Pandey, M., Arora, M., Arora, S., Goyal, C., Gera, V. K., & Yadav, H. (2023). AI-based Integrated Approach for the Development of Intelligent Document Management System (IDMS). Procedia Computer Science, 230, 725–736. https://doi.org/10.1016/j.procs.2023.12.127

Pinheiro, C., Maurer, F., & Sillito, J. (2010). Improving responsiveness, bug detection, and delays in a bureaucratic setting: a longitudinal empirical IID adoption case study. Lecture Notes in Business Information Processing, 214–219. https://doi.org/10.1007/978-3-642-13054-0_21

Raccoon Gang. (2026a). Canvas LMS pros and cons: Features, limitations, and use cases. Raccoon Gang Blog. https://raccoongang.com/blog/canvas-lms-pros-and-cons/

Raccoon Gang. (2026b). Moodle vs Google Classroom: Features and LMS comparison. Raccoon Gang Blog. https://raccoongang.com/blog/moodle-vs-google-classroom/

Raosoft, Inc. (2004). Sample size calculator [Online calculator]. Retrieved from http://www.raosoft.com/samplesize.html

Republic of the Philippines. (2012). *Republic Act No. 10173: Data Privacy Act of 2012*. National Privacy Commission. https://privacy.gov.ph/data-privacy-act/

Sandhu, R., Ferraiolo, D., & Kuhn, R. (2000). The NIST model for role-based access control. Proceedings of the Fifth ACM Workshop on Role-Based Access Control - RBAC '00, 47–63. https://doi.org/10.1145/344287.344301

Schott, K., Silva, K., Frommholz, I., Mayr, P., Kern, D., & Hienert, D. (2026). AI overviews in Academic Search: Evaluating AI-generated summaries of search results in a domain-specific search engine. arXiv (Cornell University). https://doi.org/10.48550/arxiv.2607.03421

Sun, C., Hwang, G.-J., Yin, Z., Wang, Z., & Wang, Z. (2023). Trends and issues of social annotation in education: A systematic review from 2000 to 2020. Journal of Computer Assisted Learning, 39(2), 329–350. https://doi.org/10.1111/jcal.12764

Taufique, K. M. R., Sabbir, M. M., & Rabbanee, F. K. (2026). A guide to key decision Criteria for Likert‐Scale use in survey research. *Global Business and Organizational Excellence*, *45*(5), 493–508. https://doi.org/10.1002/joe.70032

Tiwari, S. P. (2024). Digital Transformation Framework for Higher Education: Principles, Guidelines, and Actionable Recommendations. International Journal of Social Science Research and Review, 7, 132–139. https://doi.org/10.47814/ijssrr.v7i9.2299

Tsupari, K., Lagstedt, A., & Kauppinen, R. (2024). Revealing the digital transformation, uncovering hidden process dimensions of education. Business Process Management Journal, 30(8), 260–283. https://doi.org/10.1108/bpmj-09-2023-0748

Walther, J. B., Wang, Z., & Loh, T. (2004). The effect of top-level domains and advertisements on health web site credibility. Journal of Medical Internet Research, 6(3), e24. https://doi.org/10.2196/jmir.6.3.e24

World Wide Web Consortium. (2018). Web Content Accessibility Guidelines (WCAG) 2.1. https://www.w3.org/TR/WCAG21/

Zaman, F., Kamiran, F., Shardlow, M., Hassan, S., Karim, A., & Aljohani, N. R. (2024). SATS: simplification aware text summarization of scientific documents. Frontiers in Artificial Intelligence, 7, 1375419. https://doi.org/10.3389/frai.2024.1375419

Zhang, H., Yu, P. S., & Zhang, J. (2024). A systematic survey of text summarization: From statistical methods to large language models. arXiv preprint. https://doi.org/10.48550/arxiv.2406.11289

Zheng, R., Jiang, J., Hao, X., Ren, W., Xiong, F., & Zhu, T. (2019). CaACBIM: A Context-aware Access Control Model for BIM. Information, 10, 47. https://doi.org/10.3390/info10020047

Appendix A

Cover Letter to Host/Locale

HOLY ANGEL UNIVERSITY SCHOOL OF COMPUTING

27 July 2026

DR. MARLON I. TAYAG
Dean
Holy Angel
University – School
of Computing,
Angeles City

Dear **Dr. Tayag**:

A contingent part of the requirements for the Bachelor of Science in Information Technology with area of specialization in Web Development is Capstone Project, a thorough study with the development of a web application and/or mobile application. Students enrolled are required to identify respondents for their study, including data gathering through interviews and surveys. The objective of the course is to provide the students the opportunity to design and develop a web application and/or mobile application.

The following students have expressed their intention to consider your prestigious institution for their project proposal as respondents:

Curley, Adrian D.	Dungca, Cedric Luis R.
Dungca, Karl Andrei T.	Sarmiento, Sofia Anne T.

Through this letter, I would like to seek your approval for the above undertaking. Rest assured that the information that will be given to them will be for academic purposes only and will be treated with utmost confidentiality.

Sincerely,

ASST. PROF. KEVIN ALDRIN G. ESPINOSA, MIT
Instructor

#1 Holy Angel Avenue, Sto. Rosario, Angeles City, Philippines 2009
Tel. Nos.: (045) 888-8691; 888-2902; 887-5748; 887-2455; 624-5277; 625-9619 | Fax: (045) 888-1754; 888-2514
Email: hau@hau.edu.ph | www.hau.edu.ph

HOLY ANGEL UNIVERSITY SCHOOL OF COMPUTING

27 July 2026

MS. CARISMA A. CARO
Chair, Computer Science,
Holy Angel University – School of Computing,
Angeles City

Dear **Ms. Caro**:

A contingent part of the requirements for the Bachelor of Science in Information Technology with area of specialization in Web Development is Capstone Project, a thorough study with the development of a web application and/or mobile application. Students enrolled are required to identify respondents for their study, including data gathering through interviews and surveys. The objective of the course is to provide the students the opportunity to design and develop a web application and/or mobile application.

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EMAIL: HAU@HAU.EDU.PH | WWW.HAU.EDU.PH

HOLY ANGEL UNIVERSITY SCHOOL OF COMPUTING

27 July 2026

DR. EVERLY M. CHUA
Chair,
Cybersecurity,
Holy Angel
University – School
of Computing,
Angeles City

Dear **Dr. Chua**:

A contingent part of the requirements for the Bachelor of Science in Information Technology with area of specialization in Web Development is Capstone Project, a thorough study with the development of a web application and/or mobile application. Students enrolled are required to identify respondents for their study, including data gathering through interviews and surveys. The objective of the course is to provide the students the opportunity to design and develop a web application and/or mobile application.

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EMAIL: HAU@HAU.EDU.PH | WWW.HAU.EDU.PH

HOLY ANGEL UNIVERSITY SCHOOL OF COMPUTING

27 July 2026

MR. CHRIS ALMOCERA
Program Coordinator, Information Technology,
Holy Angel University – School of Computing,
Angeles City

Dear **Mr. Almocera**:

A contingent part of the requirements for the Bachelor of Science in Information Technology with area of specialization in Web Development is Capstone Project, a thorough study with the development of a web application and/or mobile application. Students enrolled are required to identify respondents for their study, including data gathering through interviews and surveys. The objective of the course is to provide the students the opportunity to design and develop a web application and/or mobile application.

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Email: hau@hau.edu.ph | www.hau.edu.ph

HOLY ANGEL UNIVERSITY SCHOOL OF COMPUTING

27 July 2026

MR. TIMOTHY JAMES S. CAMILING
Chair,
Entertainment and
Multimedia
Computing,
Holy Angel
University – School
of Computing,
Angeles City

Dear **Mr. Camiling**:

A contingent part of the requirements for the Bachelor of Science in Information Technology with area of specialization in Web Development is Capstone Project, a thorough study with the development of a web application and/or mobile application. Students enrolled are required to identify respondents for their study, including data gathering through interviews and surveys. The objective of the course is to provide the students the opportunity to design and develop a web application and/or mobile application.

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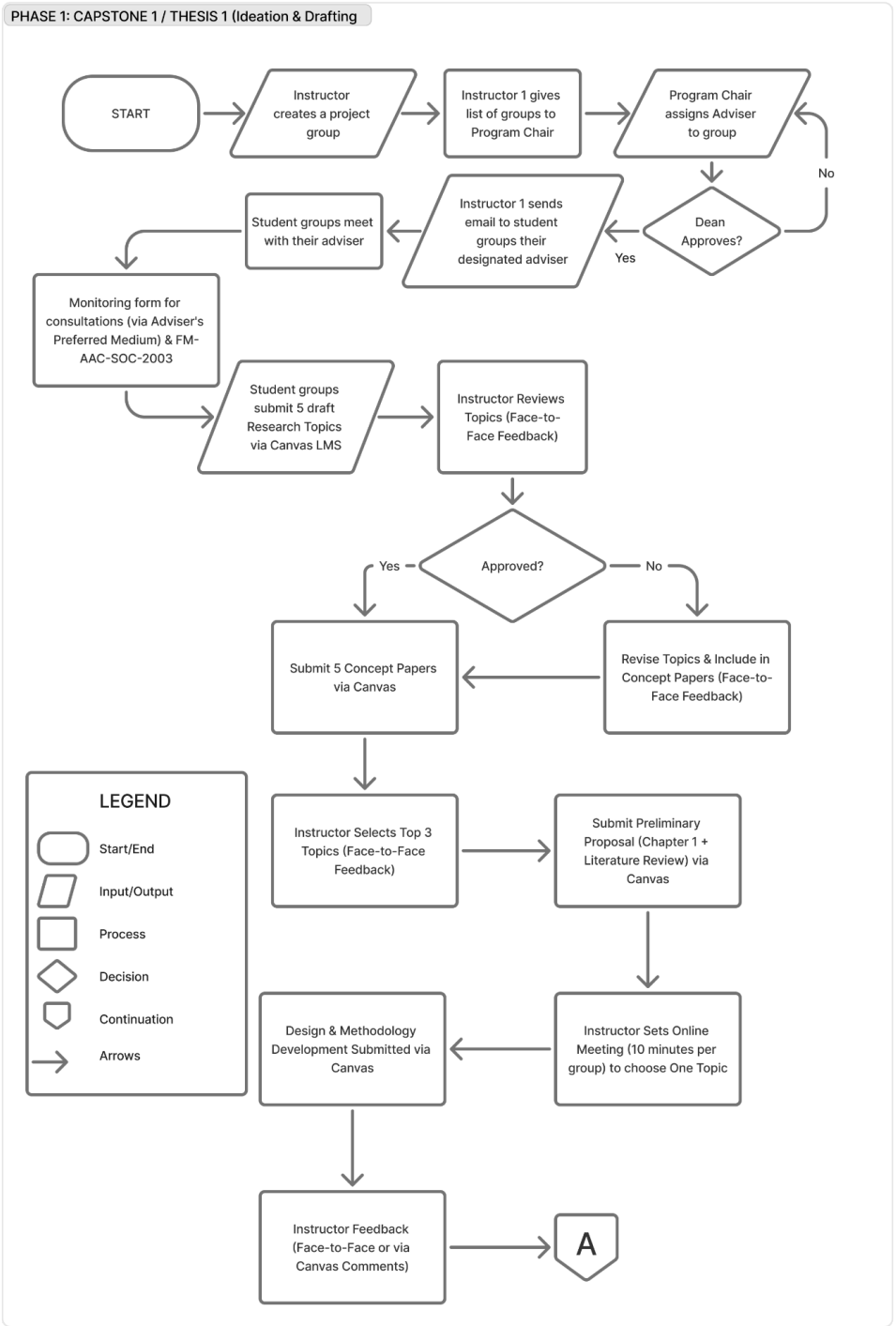
Sincerely,

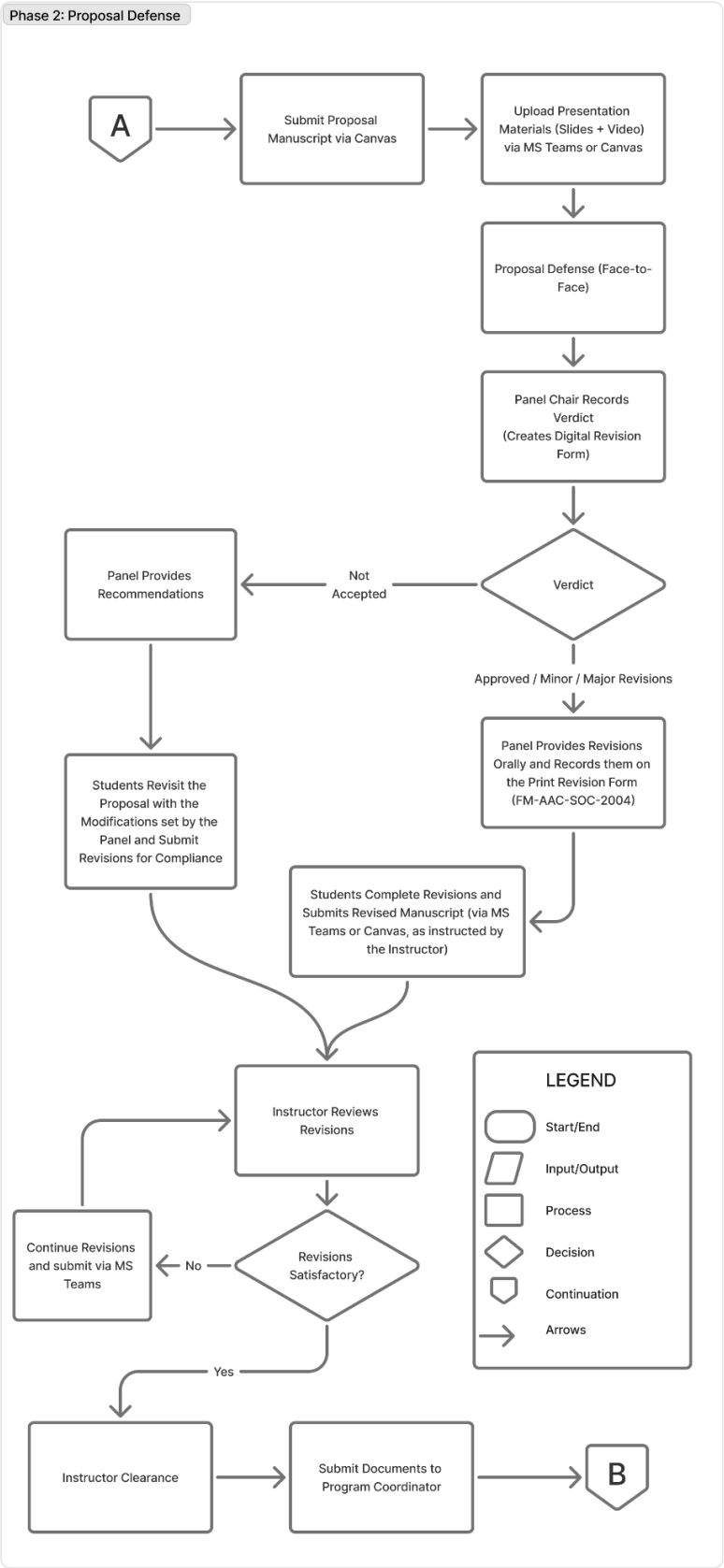
ASST. PROF. KEVIN ALDRIN G. ESPINOSA, MIT
Instructor

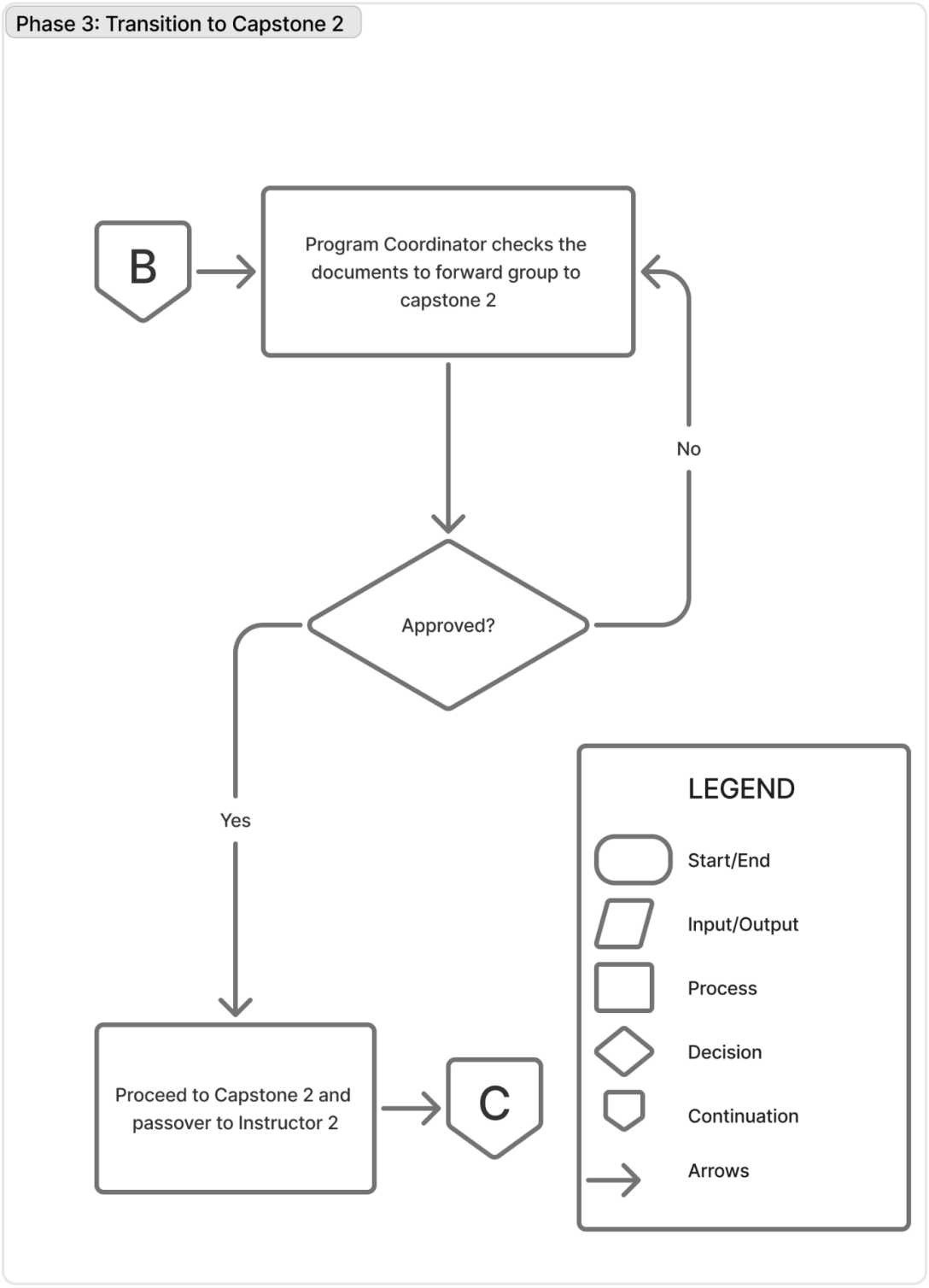
#1 HOLY ANGEL AVENUE, STO. ROSARIO, ANGELES CITY, PHILIPPINES 2009
TEL. NOS.: (045) 888-8691; 888-2902; 887-5748; 887-2455; 624-5277; 625-9619 | FAX: (045) 888-1754; 888-2514
EMAIL: HAU@HAU.EDU.PH | WWW.HAU.EDU.PH

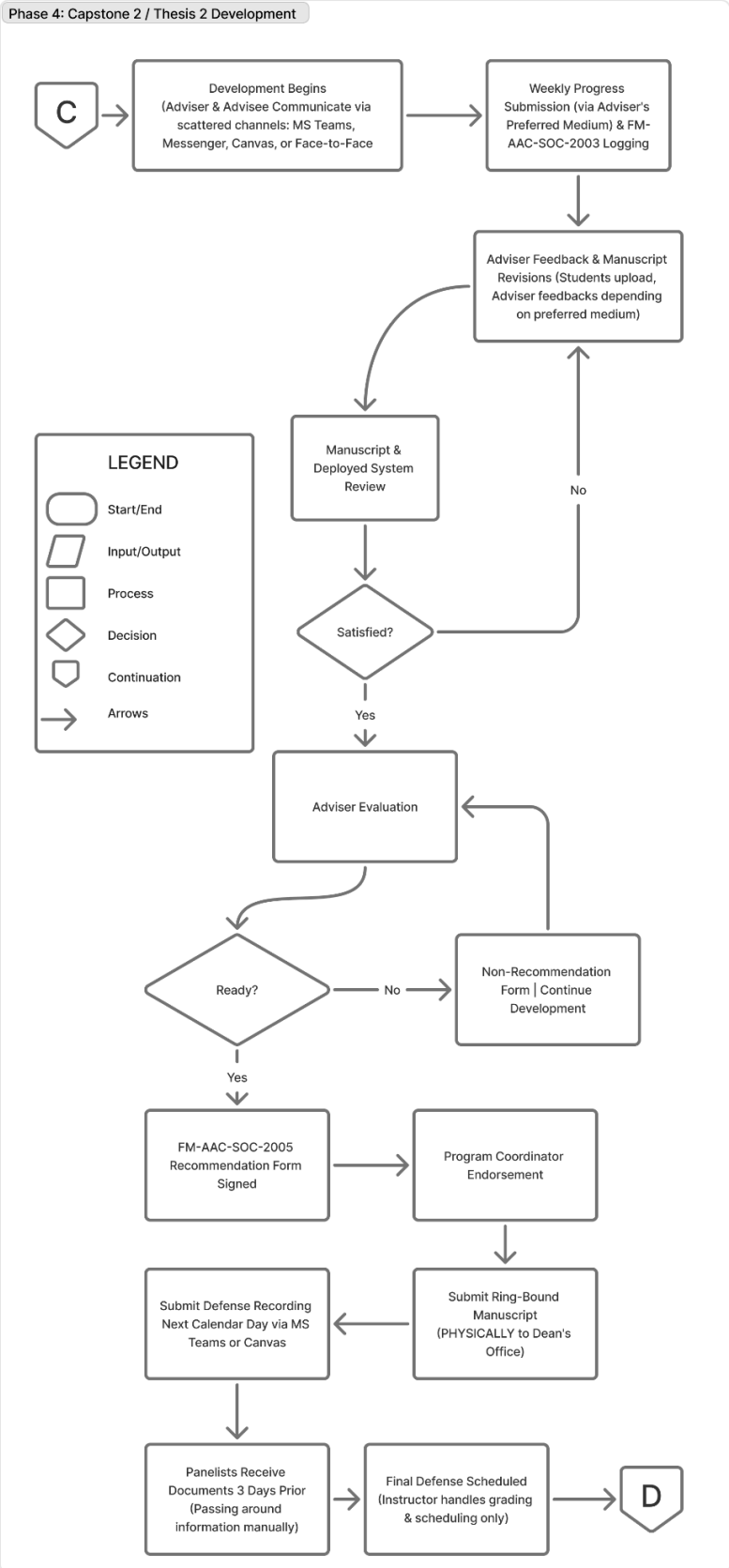
Appendix B

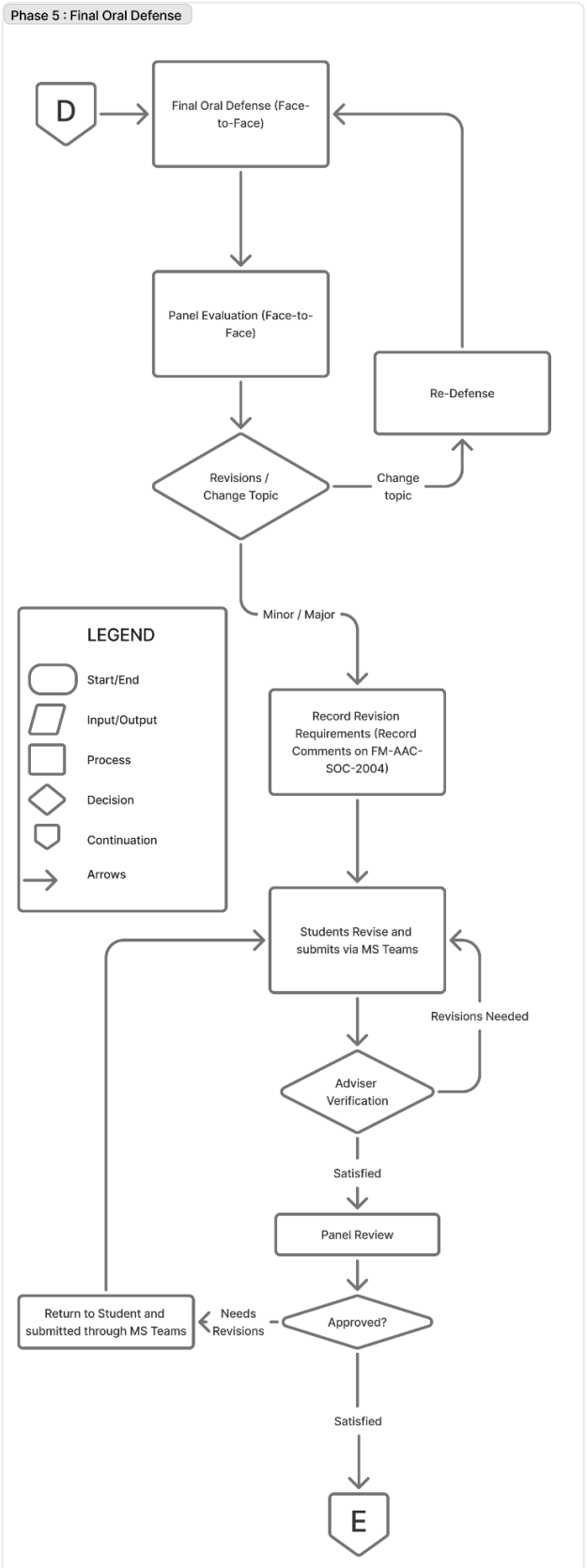
Current Process Flowchart

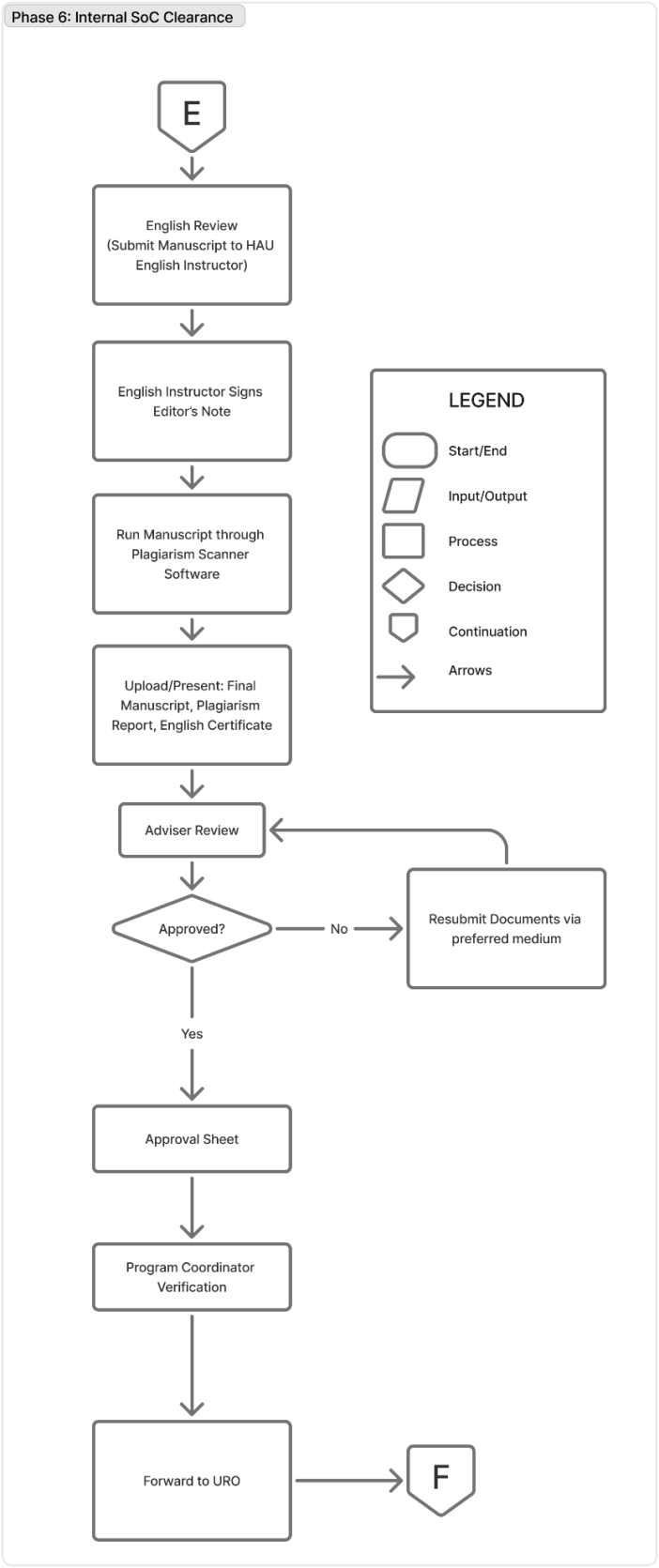


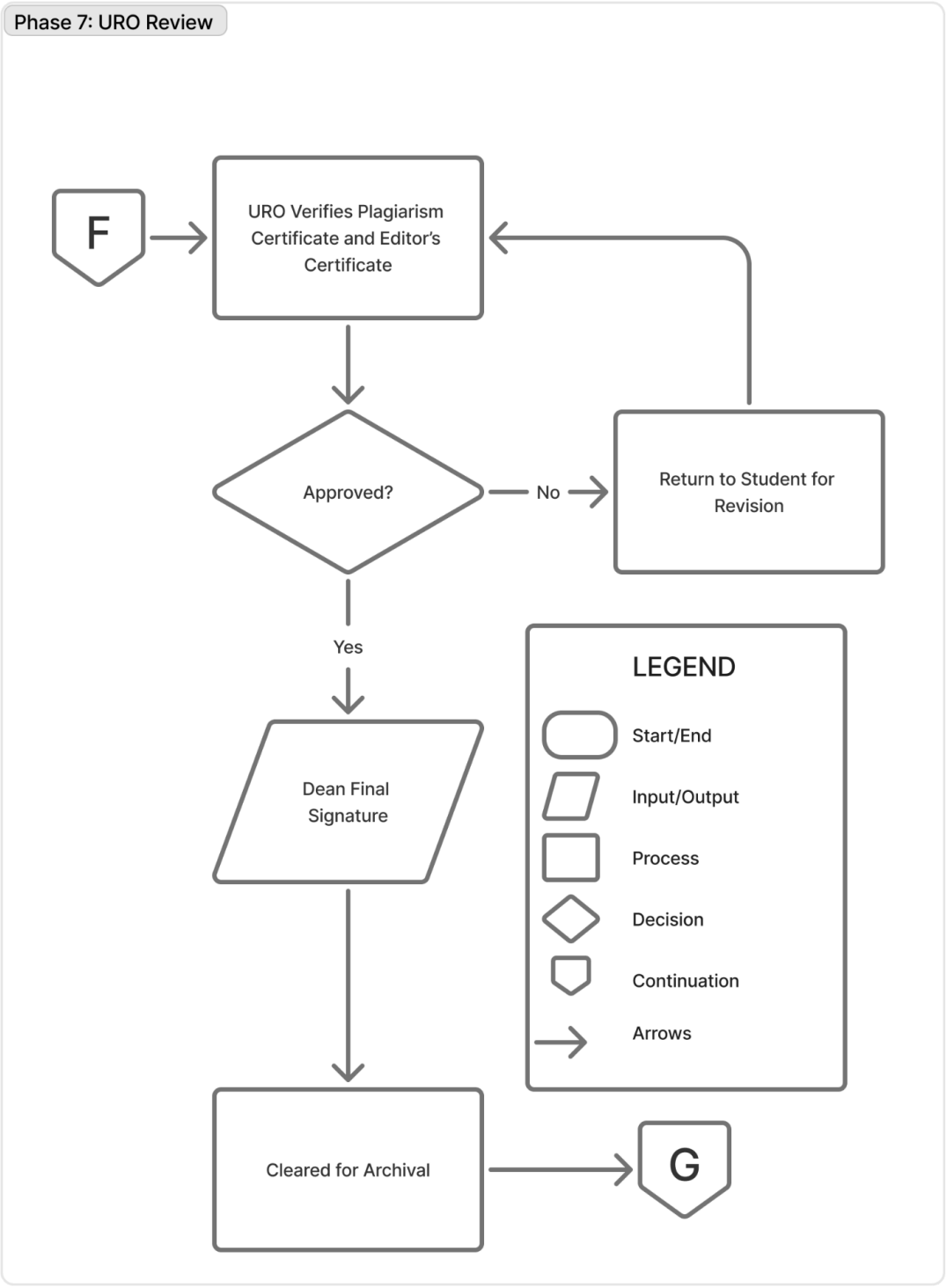

Phase 2: Proposal Defense
A
Submit Proposal Manuscript via Canvas
Upload Presentation Materials (Slides + Video) via MS Teams or Canvas
Proposal Defense (Face-to-Face)
Panel Chair Records Verdict (Creates Digital Revision Form)
Verdict
Not Accepted
Panel Provides Recommendations
Approved / Minor / Major Revisions
Panel Provides Revisions Orally and Records them on the Print Revision Form (FM-AAC-SOC-2004)
Students Revisit the Proposal with the Modifications set by the Panel and Submit Revisions for Compliance
Students Complete Revisions and Submits Revised Manuscript (via MS Teams or Canvas, as instructed by the Instructor)
Instructor Reviews Revisions
Revisions Satisfactory?
No
Continue Revisions and submit via MS Teams
Yes
Instructor Clearance
Submit Documents to Program Coordinator
B
LEGEND
Start/End
Input/Output
Process
Decision
Continuation
Arrows

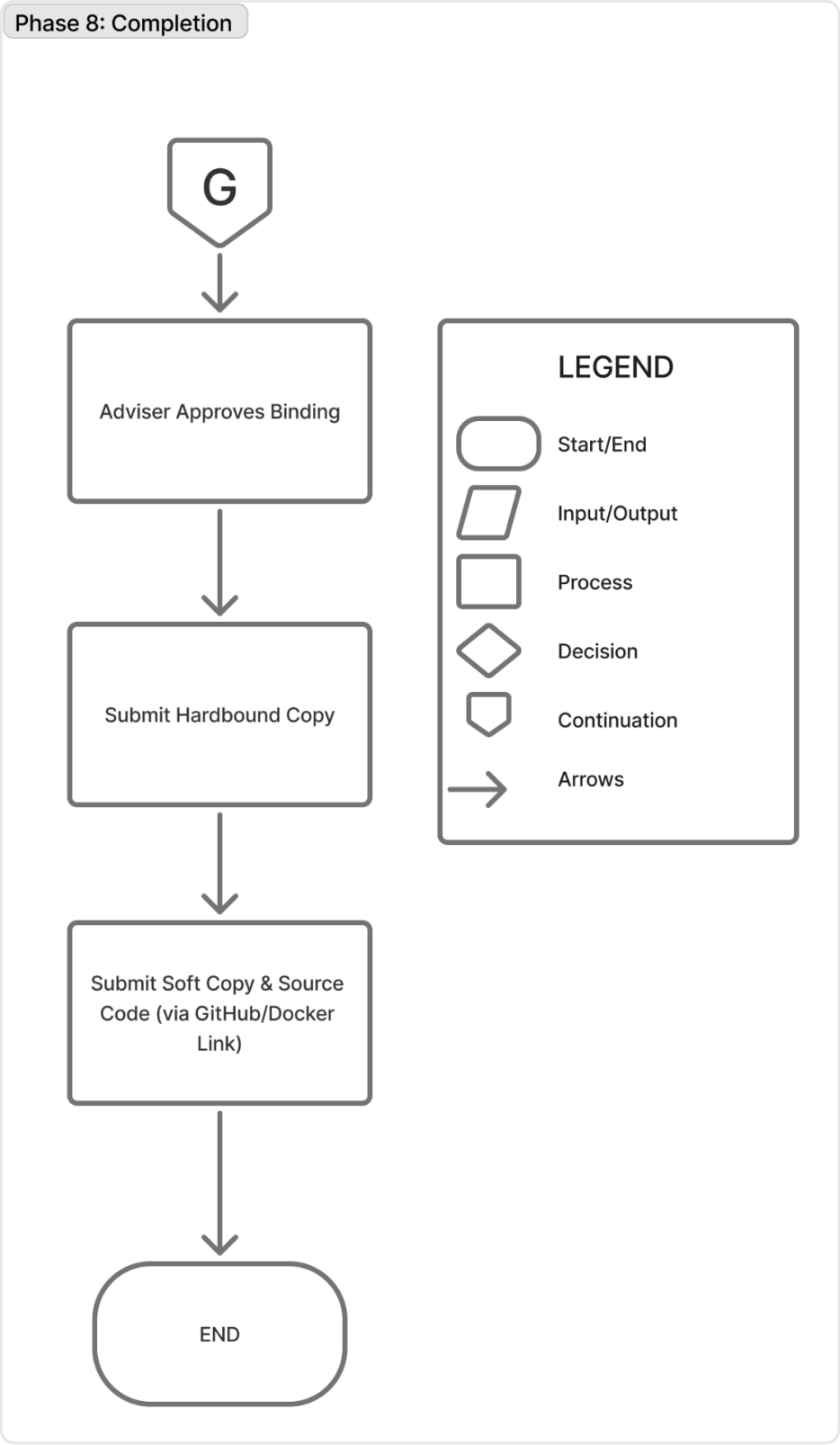

Phase 3: Transition to Capstone 2
B
Program Coordinator checks the documents to forward group to capstone 2
No
Approved?
Yes
Proceed to Capstone 2 and passover to Instructor 2
C
LEGEND
Start/End
Input/Output
Process
Decision
Continuation
Arrows


Phase 4: Capstone 2 / Thesis 2 Development
C
Development Begins (Adviser & Advisee Communicate via scattered channels: MS Teams, Messenger, Canvas, or Face-to-Face
Weekly Progress Submission (via Adviser's Preferred Medium) & FM-AAC-SOC-2003 Logging
Adviser Feedback & Manuscript Revisions (Students upload, Adviser feedbacks depending on preferred medium)
Manuscript & Deployed System Review
No
LEGEND
Start/End
Input/Output
Process
Decision
Continuation
Arrows
Satisfied?
Yes
Adviser Evaluation
Ready?
No
Non-Recommendation Form | Continue Development
Yes
FM-AAC-SOC-2005 Recommendation Form Signed
Program Coordinator Endorsement
Submit Ring-Bound Manuscript (PHYSICALLY to Dean's Office)
Submit Defense Recording Next Calendar Day via MS Teams or Canvas
Panelists Receive Documents 3 Days Prior (Passing around information manually)
Final Defense Scheduled (Instructor handles grading & scheduling only)
D


Phase 5 : Final Oral Defense
D
Final Oral Defense (Face-to-Face)
Panel Evaluation (Face-to-Face)
Re-Defense
Revisions / Change Topic
Change topic
Minor / Major
LEGEND
Start/End
Input/Output
Process
Decision
Continuation
Arrows
Record Revision Requirements (Record Comments on FM-AAC-SOC-2004)
Students Revise and submits via MS Teams
Revisions Needed
Adviser Verification
Satisfied
Panel Review
Return to Student and submitted through MS Teams
Needs Revisions
Approved?
Satisfied
E


Phase 6: Internal SoC Clearance
E
English Review (Submit Manuscript to HAU English Instructor)
English Instructor Signs Editor's Note
Run Manuscript through Plagiarism Scanner Software
Upload/Present: Final Manuscript, Plagiarism Report, English Certificate
Adviser Review
Approved?
No
Resubmit Documents via preferred medium
Yes
Approval Sheet
Program Coordinator Verification
Forward to URO
F
LEGEND
Start/End
Input/Output
Process
Decision
Continuation
Arrows

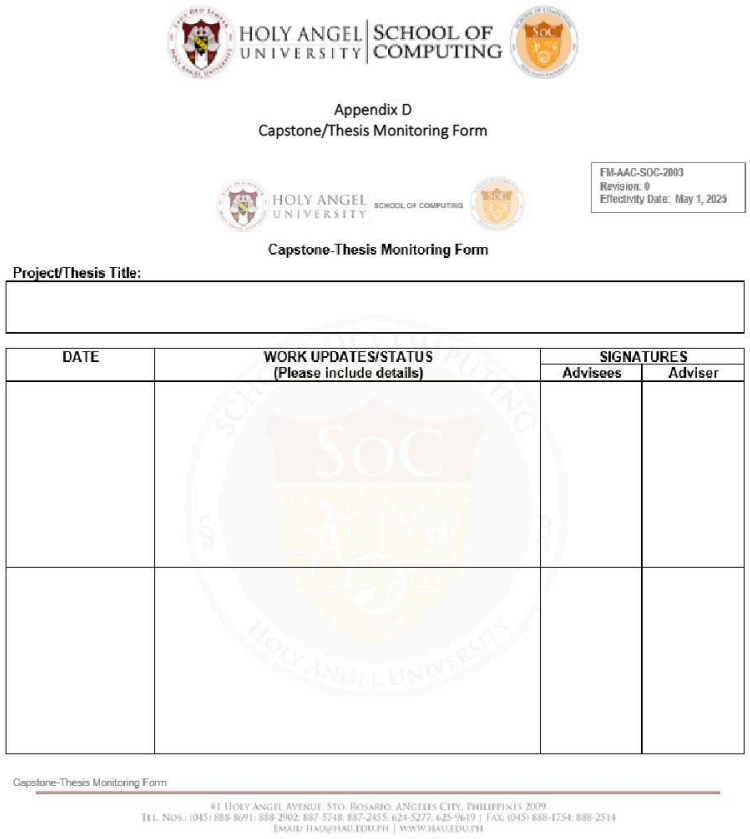

Phase 7: URO Review
F
URO Verifies Plagiarism Certificate and Editor's Certificate
Approved?
No
Return to Student for Revision
Yes
Dean Final Signature
Cleared for Archival
G
LEGEND
Start/End
Input/Output
Process
Decision
Continuation
Arrows


Phase 8: Completion
G
Adviser Approves Binding
Submit Hardbound Copy
Submit Soft Copy & Source Code (via GitHub/Docker Link)
END
LEGEND
Start/End
Input/Output
Process
Decision
Continuation
Arrows

Appendix C

Existing Files, Forms, or Manuals

C-1: FM-AAC-SOC-2003



HOLY ANGEL UNIVERSITY | SCHOOL OF COMPUTING

Appendix D
Capstone/Thesis Monitoring Form

FM-AAC-SOC-2003
Revision: 0
Effectivity Date: May 1, 2025

HOLY ANGEL UNIVERSITY SCHOOL OF COMPUTING

Capstone-Thesis Monitoring Form

Project/Thesis Title:

DATE	WORK UPDATES/STATUS (Please include details)	SIGNATURES	
		Advisees	Adviser

Capstone-Thesis Monitoring Form

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Email: hau@hau.edu.ph | www.hau.edu.ph

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C-2: FM-AAC-SOC-2004

HOLY ANGEL UNIVERSITY SCHOOL OF COMPUTING

FM-AAC-SOC-2004
Revision: 0
Effectivity Date: May 1, 2025

Capstone Revision Form

Project/Thesis Title: _____

_____

Proponents:

Student #	Name	Signature

Date: _____ **Time:** _____ **Room:** _____

Topic Recommendations:

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Remarks:

☐ Approved with Minor Revisions
☐ Approved with Major Revisions
☐ Change Topic Schedule of Re-defense on: _____

Panel Chair: _____ Signature: _____
Members of the Panel:

_____ _____
_____ _____
_____ _____

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C-3: FM-AAC-SOC-2005

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Appendix A
Recommendation Form

HOLY ANGEL UNIVERSITY | SCHOOL OF COMPUTING

FM-AAC-SOC-2005
Revision: 0
Effectivity Date: May 1, 2025

Capstone Recommendation Form

Project/Thesis Title:

_____

Proponents:

Student #	Name	Student #	Name

[ ] Recommended
[ ] Not Recommended
Reason(s) for non-recommendation:

_____
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_____
Adviser

Noted by:

_____
Program Chairperson/Coordinator, _____

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Tel. Nos.: (045) 888-8691; 888-2902; 887-5748; 887-2455; 624-5277; 625-9619 | Fax: (045) 888-1754; 888-2514
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C-4: School of Computing Research Manual (May 2025 Revision)



School of Computing
Research Manual

HOLY ANGEL UNIVERSITY | SCHOOL OF COMPUTING

I. Introduction

The commission on Higher Education has approved programs for Computer Science, Cybersecurity, Information Technology and Entertainment and Multimedia Computing. This shall also include any programs that may be endorsed by the Technical Panel for Information Technology Education (TPITE), and subsequently approved by CHED.

Cited in CHED CMO 25 S. 2015 are the following descriptions of the thesis/capstone requirements:

BS Computer Science students are required to complete a thesis that is focused on the theories and concepts of computing in the form of a scientific work.

BS Information Technology students must complete a capstone project in the form of an IT application, a Multimedia Systems development, or an IT Management project.

BS EMC for Digital Animation-Technology specialization, the capstone project should be in the form of animation content or an interactive project, done independently by the student. The animation project shall involve full production of original content, not limited to full-length animated video.

In the absence of the CMO for Cybersecurity, the School provides the following description:

BS Cybersecurity students must complete a capstone project focusing on operations on Cybersecurity that will examine trends and operations of cyber-attacks, threats, and vulnerabilities.

The Thesis/Capstone Project is required for candidates for graduation across all Computing Programs. Both the thesis and capstone projects are terminal project requirements that would not only demonstrate a student's comprehensive knowledge of the area of study and research methods used but also allow them to apply the concepts and methods to a specific problem in their area of specialization.

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II. Suggested Areas of Research / Capstone Project

Research/Capstone Project Categories

The research or capstone project must be applicable to certain processes, events, and or systems to produce innovative results, generate new knowledge or theories. Novelty is welcomed in the development of existing research projects. The suggested research project/output may fall in any of the current technologies, but not limited to:

BS Computer Science	BS Information Technology	BS EMC	BS Cybersecurity
Trends Computer Science Topics • Software and Development Theory • Mobile Computing Systems • Software Extensions or Plug-ins • Expert Systems and Decision Support Systems • Systems Software (software tools/utilities, interpreters, simulators, compilers, security aspect) • Intelligent Systems • Game Development • Computer Vision • Image/Signal Processing • Natural Language Processing • Pattern recognition and data mining. • Bioinformatics • Graphics Applications • Cloud Computing • Parallel Computing • Embedded Systems • Emerging Technologies • Artificial Intelligence	**Software Development** • Software Customization • Web Development (with at least Alpha Testing with Live Servers) • Web Applications Development (Full stack) • Development Operations (DevOps) • Mobile Computing Systems • E-learning Systems • Interactive Systems • Information Kiosks **Network Design and Implementation** • IT Management o System Administration o Cloud Infrastructure o Edge Computing • IT Strategic Plan • IT Security Analysis, Planning and Implementation o Cloud Security • Software Defined Network (Network Automation)	**Digital Animation** • Game Design Documentation (Game planning/setting, game prototype, game play) • Promotion/ Marketing, Patriotism (Philippine Culture in General), Computing Literacy) • Social Issues on gender and safe spaces • State of the Philippine Market/Industry for Multimedia • Perception of Creative Industries and Career • Open-Source vs Proprietary License for Multimedia Platform • Skills vs Talent • Hybrid Live Action (30%) and Animation (70%) or frame by frame. • Multimedia Approach (LFX, DPP,2D/3D Animation and others)	**Network Security** • Develop network intrusion detection system. • Comprehensive Network Security Audit **Web Application Security** • Security Assessment **Mobile Application Security** • Investigate Mobile Malware threats and develop countermeasures. **Digital Forensics** • Conduct Forensic Analysis of Digital evidence from a simulated cybercrime scenario. **Internet of Things Security** • Conduct security analysis of IOT devices and propose strategies for securing IOT. • Design and implement a secure communication protocol for IOT devices. • AI in Cybersecurity



III. Guidelines

A. Groupings

CAPSTONE1 / THESIS1

- Students who are qualified to enroll Capstone/Thesis must be in their 4th year standing/3rd Year standing (CS) and have taken 100% of their major professional courses and 80% specialization courses.
- The students report to their capstone/thesis instructor in their given schedule.
- Group members should be identified in Thesis/Capstone and may be formed with three (3) to five (5) members for EMC, three (3) to four (4) members for CS, IT and CBS. Group size exceeding the maximum number of members should be properly justified in writing with the approval of the Dean.
- Students who wish to pursue a solo Capstone/Thesis project are required to submit an undertaking signed by their parent or guardian, affirming their consent and the student's commitment to independently complete all project requirements.

CAPSTONE2 / THESIS2

- Only students who passed Thesis/Capstone1 can take Thesis/Capstone2.
- Strictly NO regrouping policy, group members are already identified from Thesis/Capstone 1.
- A list of recommended advisers is submitted to the dean by the chairperson for approval.
- The students report to their assigned capstone adviser and are required to fill out the advising monitoring sheet (*See Appendix D*).

Students whose co-member/s did not enroll

1. If the group has three (3) members remaining, they are still required to continue work on the approved proposal.
2. If the group has less than three (3) members remaining, the group may adopt 1 or 2 members either from returning, re-enrolled or students with the same case as theirs to form another capstone group.
3. The student has the option to continue with the approved title or look for a group with less than four (4) members and is willing to accept him/her as a new member.
   - Dissolved group
     - The returning student finds a group with less than four (4) members and is willing to accept him/her as a new member. Group size exceeding 5 should be properly justified in writing.
     - If no group is willing to accept the said student, he/she proceeds working on his/her own and submits a proposal and undergoes a title defense within two weeks after the start of classes.

B. Proposal Defense:

a. On the sixth (6th) week during the Prelim Exam Period, the group should present three (3) different topics to their instructor to seek initial approval of the topics *(based on the list provided in the previous section).*
b. The approved topic will be presented in the proposal defense.



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c. Proposal defense is scheduled on the 13th to 14th week of the semester, after the Midterm Exam Period.
d. The groups (**all members**) are advised to be present at least 30 minutes prior to their oral defense schedule. A 15% deduction is given to groups who fail to arrive on time.
e. A group who failed to attend their scheduled proposal defense without any valid reason will automatically be rescheduled once and shall get a grade of 3.0.
f. On a case-to-case basis a group may be allowed to be rescheduled once there is a valid reason approved by the Dean.
g. A group whose proposal was not accepted will be given recommendations to revisit their proposal with the modifications set by the panel and submit revisions for compliance.
h. The group will be given a grade of 3.0 once they have complied with proposal modification.
i. The groups should submit a 10–15-minute video presentation of the topic as recommended by the instructor; the presentation should include but not limited to the following:

BS Computer Science	BS Information Technology	BS EMC	BS Cybersecurity
Abstract Introduction (includes problem statement) **Review of Related Literature** **Conceptual or Theoretical Framework** **Objectives of the Study** *Scope and delimitations* **Methods** • Research Design o *Sources of Data (Data Sets and all relevant data sources)* • Research Instrument o *Participants/Sample and Settings* o *Data Analysis or Statistical Treatment of Data* • Research Procedures o *Data Treatment* o *Algorithm Selection* o *Algorithm Application* o *Testing* • Ethical Consideration **Results** **Discussion** *Conclusion* *Recommendation* **References** **Appendices**	**Abstract** Introduction (includes problem statement) **Review of Related Literature** **Conceptual Framework** **Objectives of the Study** *Scope and delimitations* **Methods** • *Requirement Analysis* • *Requirement Documentation* • *Design of Software, System, Product and/or Process* • *Development and Testing (includes Sample and Settings, Data Analysis)* • *Ethical Consideration* **Results** (Output) **Discussion** (Evaluation) *Conclusion* *Recommendation* **References** **Appendices**	***Abstract*** Introduction (includes problem statement) **Review of Related Literature** **Conceptual Framework** **Objectives of the Study** *Scope and delimitations* **Methods** • *Requirement Design* • *Research Procedure* • *Sample and Setting* • *Instrument / Simulation Tool* • *Equation Derived / Used* • *Data Analysis* • *Ethical Consideration* *Output deliverables:* • Script • Character Design/Model • Background or Environment Design • Model Sheets • Storyboard • Animatic **Results** (Output) **Discussion** (Evaluation) *Conclusion* *Recommendation* **References** **Appendices**	**Abstract** Introduction (includes problem statement) **Review of Related Literature** **Problem Statement** **Conceptual or Theoretical Framework** **Objectives of the Study** *Scope and delimitations* *Significance of the study* **Methods** • Research Design • Data o Instrument/Simulation Tool o Equations Derived/Used o Data Analysis or Statistical Treatment of Data • Ethical Consideration • Research Procedures o Data Treatment o Testing **Results** **Discussion** *Conclusion* *Recommendation* **References** **Appendices**



j. Another fifteen minutes (15 minutes) will be allotted for questioning.
k. Recommendations should be written in the "FM-AAC-SOC-2005 Rev 0_Capstone Recommendation Form" which will be the sole basis for checking the revised manuscript and output.
j. The above-mentioned requirements should be submitted to the instructor. The instructor should then submit the following to the Program Chairperson which will then be turned over to the instructor/adviser of the Thesis2/Capstone 2.
k. These will be the basis of the Instructor and Adviser to guide the students for the development and completion of their project.

C. Final Oral Defense

a. Prior to the oral defense, advisers are required to assess their assigned group's paper and output and submit the recommendation form to their respective chairpersons for formal endorsement of groups to be scheduled for Final Oral Defense.
b. Approved proposal topics will be presented and completed for final oral defense.
c. Changes to the approved proposal, such as title, scope and limitations will not be allowed. This is to prevent untoward revisions in the prior agreement made during the proposal presentation.
d. The group submits the bound(ring-bound) documentation to the Dean's office on the given deadline before the scheduled oral defense from 8:00AM to 5:00PM. Failure to submit on time will result in a 10-point deduction on the manuscript. Therefore, a group who submits the next day is only allowed until 12:00 noon. Consequently, for groups who were not able to make it to the given extension, the re-defense policy shall be applied.
e. The submission should include the Association of Computing Machinery (ACM) format, and the Program Heads of each program will provide the link.
f. Video presentation recording submission shall be on the next calendar day after the manuscript submission deadline. Contents of the 10–15-minute video presentation shall be as follows:
   c.1 Prayer
   c.2 Introduction of the group and each student's assigned task
   c.3 Statement of the Problem or Research Background and why the research is significant
   c.4 Objectives
   c.5 Scope and Limitations
   c.6 Method
   c.7 Results
   c.8 Discussion of Findings
   c.9 Conclusion
   c.10 Recommendations
g. Submitted documents shall be distributed by the adviser to the panel members 3 working days before the scheduled defense.
h. The groups (**all members**) are advised to be present at least 30 minutes prior to their oral defense schedule. Failure to come on time during their oral defense will be considered for re-defense.
i. If the group decided not to pursue the oral defense on the date/schedule assigned to them, no rescheduling will be granted, they will automatically be considered for re-defense.
j. In cases where a group wishes to remove a member, the concern must first be discussed with the adviser and all group members. Following this, clear and sufficient evidence must





be presented to support the recommendation for removal, based on the grounds of the member's lack of participation. However, should the member appeal, the group's request for removal will be considered void until the matter is fully reviewed.

k. The group is given the following presentation flow during the oral defense:

Program	Presentation Flow	Time Allowed
BS Computer Science (2.5 hrs)	Presentation Algo Process and Implementation Question and Answer Deliberation	60 mins 60 mins 30 mins
BS Cybersecurity (2.5 hrs)	Presentation of Topology Checking of Configuration Question and Answer Deliberation	45 mins 45 mins 45 mins 15 mins
BS IT Web Development (2 hrs)	Presentation System Question and Answer Deliberation	45 mins 60 mins 15 mins
BS IT Net Ad (2.5 hrs)	Presentation of Topology Checking of Configuration Question and Answer Deliberation	45 mins 45 mins 45 mins 15 mins
BS Entertainment and Multimedia Computing (1.5 hrs)	Presentation of Output Question and Answer Deliberation	15 mins 45 mins 30 mins

l. The group's presentation, manuscript, and output will be assessed based on the rubric.
m. The submitted video prepared will be distributed by the adviser three days before the scheduled Final defense. It is the responsibility of the assigned panel to review and watch the video. It will not be played during the actual oral defense.
n. Revisions and suggestions may be raised during presentation; these will be documented properly in the revision forms (FM-AAC-SOC-2004 Rev 0_Capstone Revision Form).
o. During the deliberation the adviser together with the students are going to be requested to be outside of the defense room.
p. In the event of clarifications during deliberation, the adviser may be called in by the panel.
q. Panel members shall be required to sign off on the document revision form.
r. The members of the group should sign the revision form to signify their agreement to the revisions stated in the form.
s. The grades decided by the panel during the deliberation are **final and irrevocable.**

D. Post Defense Requirements:

a. With Revisions

- The group is given a time frame recommended by the panel to comply with revisions. Grace time may fall under the following category:
    1. Minor revision – 1 week grace period for compliance to accomplish revisions in documentation and output.
    2. Major revision – 2-week grace period for compliance to accomplish major revisions in documentation and minor revisions in their output or vice versa.





- The revised documentation and output are presented to the adviser and the panelists for checking within the given time frame. Failure to submit on time earns a 5% deduction per week on the computed final grade before the final exam week.
- The group submits a hardbound copy of the documentation one week before Final Exam week or on the specified submission date.
- Groups who fail to comply with the ***complete/revised document requirements*** (**hardbound**) one week before Final Exam week or on the specified submission date will automatically get a grade of 5.0 for the semester. However, they are required to comply with the requirements within the next semester (grace period). The grade given to these groups can only be a passing rate of 3.0. In this case, the use of rubric as basis for grading does not apply. **The adviser applies for a change of grade from 5.0 to 3.0.**
- Groups who fail to comply within the grace period are required to re-enroll the course and submit a new proposal for approval. The grade of 5.0 given in the previous semester is retained.

b. Re-defense

- When the group fails to pass during the final oral defense, they will be subjected to a re-defense that is scheduled two weeks after but not later than two weeks prior to the submission of the final manuscript (hardbound).
- When a group fails to submit their manuscript within the specified deadline, they will be scheduled for oral defense not later than two weeks prior to the submission of the final manuscript (hardbound).
- When a group fails to present during their scheduled re-defense, the group may opt to present on the next term or semester provided they submit a letter of undertaking addressed to the Dean through the Program Chairperson.
- A grade of 3.0 is awarded to groups who pass the re-defense. The use of rubric as basis for grading does not apply.
- Failure to comply with revisions recommended within the grace period shall deem the groups non-compliant and shall fail the course. The students are required to re-enroll on the course and submit a new proposal for approval. The grade of 5.0 given in the previous semester is retained.
- The group shall notify their adviser in writing if they choose to revise or complete their thesis or capstone after the grace period of one term or semester or should they wish to re-enroll instead.

b.1. Verdict

The groups are given the following extension for re-defense based on the following verdict.

- 1 week – minor revision in documentation and output.
- 2 weeks – major revision in documentation and minor output or vice versa.

Students who re-enroll:

1. Student who failed their Capstone2/Thesis2.
2. Students who re-enroll should submit a request letter of undertaking stating that a new proposal will be submitted, addressed to the Dean through the Program Chairperson. The request shall be subject to the approval of the Dean.





E. Attendance:

a. For Thesis/Capstone 1:

- Each member of the group is required to attend the class as prescribed by the instructor.
- Attendance will be recorded and graded by the instructor using a prescribed rubric.

b. For Thesis/Capstone 2:

- Each member will be monitored by the Thesis/Capstone Adviser.
- A regular meeting will be scheduled.
- Failure to meet and comply with recommendations of the Thesis/Capstone Adviser shall deem the group unqualified for recommendation for the Final Defense.

I. Revisions:

- Proposal Defense:
  a. **Approved.** Minor revisions are necessary, but they do not have to be presented in front of the panel and checked by all panelists.
  b. **Approved with Revisions.** Major revisions shall be incorporated in the final copy of the revised Project Proposal summary. These must be checked by the panelists.
  c. **Change Topic.** The Proponents/Researchers failed to propose a feasible or researchable or scholarly Research / Capstone Project.

- Final Oral Defense

  Revisions recommended by the oral defense panel fall into the following category:

  a. Minor Revisions
     Grade: 100 – 85
  b. Major Revisions
     Grade: 84 - 75
  c. Re-defense
     Grade: below 75

  All revisions shall be checked by the adviser and panel members (*see Section D*), once checked, panel members are requested to sign off on the Document Revision form to signify that revisions were made and completed.

IV. Completion and Submission:

1. The groups are to submit the revised documentation for final editing to identified English Instructors within the University. The list of English instructors disseminated to students shall be requested from the respective Chairpersons or department heads handling English.
2. The English Instructor from this university certifies editing by signing the Editor's Note.
3. Have your documentation run through Plagiarism Scanner Software *(see flowchart pdf file).*
4. The approval sheet signifies approval of the adviser, panel, chairperson, and dean that the research project is ready for final submission. It should only be signed when:
   a. All the recommendations in the revision form have been met and approved by the panel members.
   b. The duly signed editor's note had been presented by the group.





c. The document has passed plagiarism check.

5. The Approval sheet should be duly signed by the adviser, members of the panel, Program Chairperson, and the Dean once all clauses in 4 have been met.
6. The group seeks approval from their adviser before final binding of the documentation for submission.
7. The group submits a hardbound and softcopy of the documentation with the provided link by the Program Chairs/Coor through Github or Docker.
8. The group who fails to submit the above requirements gets a grade of 5.0. When these groups have complied with the requirements, the Adviser may later apply for a change of grade following University change of grade policy.

V. Colloquium

All research groups will be required to participate in the School of Computing Colloquium. The activity is prescribed in CMO for all programs. Presentation of research papers are to be presented in a forum where there is an audience that consists of stakeholders in the field where the program is concentrated on. However, for the EMC program in lieu of presentation, students will be asked to display a poster on their study.

VI. Duties and Responsibilities:

A. Researchers

a. The Research Study is an original work, and the researchers are solely responsible for its content.
b. The group is anticipated to report and work with the duly assigned adviser and submit all necessary requirements.
c. The group is expected to do the necessary revisions recommended by the instructors during the class, the adviser, and the panel during and after Final Oral Defense.
d. Groups are to comply with all the requirements stated in the SoC Capstone/Thesis guidelines.
e. The researcher will format the paper using APA 7th edition, which will be submitted for the hardbound copy, while a journal type format will be prepared for publication purposes either school base journal or external journals. For BSCS, BSIT, and BSCYB programs ACM format will be used and APA format for BSEMC program.

B. Instructor for Thesis/Capstone1

a. Present Research Writing and Methods.
b. Discuss the guidelines in THESIS/CAPSTONE project.
c. Facilitate grouping or student group assignments.
d. Responsible for scheduling the title and pre-oral defense.
e. Compute and submit grades of students.
f. Ensure compliance of students from proposal to pre-oral defense.
g. Discuss with students the document content and format APA 7th format and the ACM format.
h. Inform students on the rubrics used in the title and pre-oral presentation.
i. Write all the recommendations from the panelists in the FM-AAC-SOC-2004 Rev 0_Capstone Revision Form.
j. Inputs the final grade of the students for the course.





C. Adviser – Thesis/Capstone 2

a. Sets regular meetings with the group (the team must seek proper appointment) to answer questions and help resolve bottlenecks and conflicts.
b. Duly sign the Thesis/Capstone Monitoring Sheet (FM-AAC-SOC-2003 Rev 0_Attendance-Monitoring and Activity Sheet) accomplished during the regular meetings.
c. Monitor group progress in completing the project and documentation.
d. Discuss to the group the rubrics used for Final Defense.
e. Check and correct the research project documentation before the oral defense.
f. Evaluate the project and documentation for recommendation to oral defense.
    i. If the group is ready for oral defense and is recommended, the adviser must complete the recommendation form (See Appendix A) FM-AAC-SOC-2005 Rev 0_Capstone Recommendation Form and submit it to the Chairperson for the final schedule of oral defense.
    ii. If the group is not ready and is not recommended for oral defense, the adviser must fill out the non-recommendation form and state the reason/s why the group is not recommended. All members of the group must sign the form as well.
g. During the oral defense, the adviser is required to be present.
h. The adviser may clarify, paraphrase questions, and in some cases interpret panel questions to the group as allowed by the panel. Defending the study for the students or answering for them is not allowed.
i. The adviser must see to it that all comments, suggestions, and recommendations by the panelist are taken down or noted in the Thesis/Capstone Revision Form (FM-AAC-SOC-2004 Rev 0_Capstone Revision Form). The adviser must also secure a copy of the said form at the Dean's Office.
j. The adviser should not reveal the names of panel members assigned to the group.
k. The adviser should not reveal the grade of the group after the oral defense, only a pass or fail verdict given by the panel shall be made known to the group.
l. During document revisions for final submission of the document, the adviser should be the first to sign the approval sheet, given that he/she attests that the research document is complete, in proper format, proof-read and counterchecked for plagiarism.
m. The adviser will check both APA 7th and ACM formats.
n. The adviser should submit the students' grades to the Chairperson immediately after the oral defense.
o. When thesis groups assigned to the adviser were not able to comply within the regular period of Thesis/Capstone, and have failed during that period, the adviser should be the one to accomplish the change of grade when the group has completed and fulfilled the Thesis/Capstone requirements.

D. Panelist Composition/Duties

a. The panel shall consist of three members that consist of the following:
    i. Faculty Member from SoC
    ii. Invited Expert in the field
b. There shall be one (1) Chairperson and two (2) members on the panel; the panelists will decide who among themselves will be the panel Chairperson.
c. Prior to the oral defense, panelists cannot interact with the group/s assigned to them.
d. The panel members are required to objectively evaluate the students' performance using the rubrics for:
    i. Documentation – research paper





ii. Presentation – video presentation and oral defense
iii. Output – system / application / website / mobile app / network design / security implementation / forensic analysis / security analysis / animated film / movie / infomercial / game, etc.

e. During the oral defense, panel members are asked to deliver questions objectively in a way that would encourage fruitful interchange of ideas. It is discouraged to deliver questions that are meant to embarrass or degrade the students.
f. The panelists should evaluate the performance of the group and grade them based on the rubric.
g. After the oral defense, the panelists must convene themselves in a closed-door deliberation and decide whether the study has minor or major revisions.
h. Once the panel has completed their deliberation, the adviser will be called in so the panel can discuss the results. Clarifications can be discussed before students are called in.
i. The panel chairperson shall deliver the results of the deliberation to the group.
j. Panelists should welcome questions from the group that pertain to results of the deliberation and the revisions and recommendations if there are any.
k. In case of conflict in decision among the panelists, the majority rule shall be followed.
l. Panelists should see to it that all the requirements in the revision form are met before signing the approval sheet.

E. Panel Chairperson

a. The panel Chairperson delivers to the group the recommendations of the panel.
b. He/she precedes the deliberation. Once decision of deliberation is issued, it is final and irrevocable.
c. The panel Chairperson should be responsible to remind the adviser that the grading sheet should be sent after verdict is delivered.

VII. Grading System:

In grading both courses, rubrics shall be placed in a shared drive where access shall be given to panel members during evaluation/grading.

CAPSTONE1 / THESIS1

Proposal Defense

Topic Proposal	30%	*(Prelim submission)*
Proposal Defense	40%	*(Final Presentation)*
Instructor	30%	*(Capstone/Thesis component submissions, Group participation/cooperation)*
Total	100%	
Passing Average		**60%**

CAPSTONE2 / THESIS2

Final Oral Defense

Research Output	40%
Documentation	30%
Presentation	20%
Adviser	10%
Passing Average	**60%**



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VIII. Rubrics:

For First Semester Proposal Defense:

- Appendix B.1: Rubrics for CS, CYB and EMC
- Appendix B.2: Rubrics for IT

For Second Semester Final Oral Defense:

- Appendix C1: Rubrics for EMC
- Appendix C2: Rubrics for BSIT – WebDevt
- Appendix C3: Rubrics for BSIT – NetAd
- Appendix C4: Rubrics for BSCS
- Appendix C5: Rubrics for BSCyb

IX. American Psychological Association (APA) 7th Edition Format

General Guidelines:

1. Use white 8.5 x 11-inch paper.
2. The margins should be 1-inch on the top, bottom, right and left is 1.5 inch.
3. Border line should be inserted as prescribed in Appendix C.
4. The first word in every paragraph should be indented one half inch.
5. APA recommends using Times New Roman font, size 12.
6. Double space the entire research paper.
7. To create the running head/page header:
    a. insert page numbers justified to the right-hand side of the paper (do not put p. or pg. in front of page numbers)
    b. Then type "TITLE OF YOUR PAPER" justified to the left using all capital letters.
    c. If your title is long, this running head title should be a shortened version of the title of your entire paper.

APA Paper Components:

Your paper should include these major sections:

1. Title Page
2. Executive Summary (for IT)
3. Abstract (EMC)
4. Main Body
5. References

X. Association for Computing Machinery (ACM) Format – General Guidelines

1. Paper and Margins

- The journal format should have 5 to 6 pages.
- Bio-note is 1 page only.
- Use standard white 8.5 x 11-inch paper.
- Set margins as follows: Top, bottom, and right: 1 inch and left: 1.5 inches

2. Paragraph Format

- Indent the first line of every paragraph by 0.5 inch.
- Use 1.5 line spacing throughout the document.

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3. Font Requirements

Component	Font Name	Size	Style/Notes
Title	Times New Roman	12 pt	Bold
Author's Name	Times New Roman	11 pt	Arrange in ascending order by surname
Affiliation	Times New Roman	9 pt	—
Spacing (Title to Affiliation)	—	—	1.5 line spacing
Abstract, CCS Concepts, Keywords	Times New Roman	8 pt	Keywords in bold
Headings	Times New Roman	9 pt	Use numbered heading style
Subheadings	Times New Roman	9 pt	Use sub-list numbering style
References Heading	Times New Roman	9 pt	—
List of References	Times New Roman	7 pt	—
About the Authors (Heading)	Times New Roman	9 pt	Include a 1x1 inch photo; name in ALL CAPS
Bio-note	Full name: Times New Roman (9 pt) Paragraph: Times New Roman (8 pt)		—
Page Numbers	Times New Roman	9 pt	Center-aligned at the bottom of the page

4. Required ACM Paper Sections

Your manuscript should include the following components in order:

1. Title and Author(s)
2. Introduction
3. Methodology
4. Results and Discussion
5. Conclusion and Recommendations
6. References
7. About the Authors

See file 1: SOC Official Template Preliminary
See file 2: SOC Official Capstone/Thesis Body
See file 3: SOC Template Appendices
See file4: Github and Docker (for the submission of all documents)

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Appendix A
Recommendation Form

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FM-AAC-SOC-2005
Revision: 0
Effectivity Date: May 1, 2025

Capstone Recommendation Form

Project/Thesis Title:

_____

Proponents:

Student #	Name	Student #	Name

[ ] Recommended
[ ] Not Recommended
Reason(s) for non-recommendation:

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_____
Adviser

Noted by:

_____
Program Chairperson/Coordinator, _____

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Appendix B.1
Rubrics for Proposal Defense (CS, CYB and EMC)

Name: _____

Title: _____

Degree: _____

Direction: Please encircle the number which best applies to the material being evaluated.
5 (Excellent), 4 (Very Good), 3 (Good), 2 (Satisfactory) and 1 (Poor)
Evaluation Categories:

A. The Problem and Its Background

1. Issues and trends related to the problem are carefully presented to establish its validity.	5	4	3	2	1
2. The study is timely and relevant to the author's field of specialization.	5	4	3	2	1
3. Theoretical/Conceptual Framework is valid and adequately presented.	5	4	3	2	1
4. The problem statement or research background is sufficiently congruent with the theoretical/conceptual framework.	5	4	3	2	1
5. The sub-problems or specific research topics are anchored to the major problem or topic.	5	4	3	2	1
6. The state assumptions, if any, are adequate and relevant to the problem.	5	4	3	2	1
7. Whenever applicable, the stated hypotheses are testable.	5	4	3	2	1
8. The scope of the study is properly identified and delineated.	5	4	3	2	1
9. The study adds new insights to current practices and/or theories.	5	4	3	2	1
10. The sources and documents used are properly acknowledged, correctly and accurately cited.	5	4	3	2	1

B. Related Materials

11. The review of related literature and studies effectively discusses significant findings and existing published articles and projects relevant to the present study.	5	4	3	2	1
12. The similarities and differences between and among materials reviewed and the present study are clearly and correctly identified.	5	4	3	2	1

C. Method

13. The research design is appropriate and adequate for seeking



the answers to the stated problem/s or topic.	5	4	3	2	1
14. The data, document, and instruments used are sufficient, reliable, and valid.	5	4	3	2	1
15. Appropriate or applicable data sample, technique and size is used.	5	4	3	2	1
16. The procedures for data collection are coherently and logically explained.	5	4	3	2	1
17. The techniques for data analyses are adequate and Appropriate.	5	4	3	2	1

D. Other Requirements of the Manuscript

18. The report is free from padding with irrelevant words, phrases, quotes, statistics, examples, etc. that are not essential for accuracy, clarity, and completeness.	5	4	3	2	1
19. The language is precise, and emphasis is duly observed.	5	4	3	2	1
20. The APA Journal format is correct and consistent.	5	4	3	2	1
21. Documentation is complete, and references include all materials/sources cited.	5	4	3	2	1
22. The appended materials are helpful and pertinent to the report.	5	4	3	2	1

TOTAL POINTS _____/110

Name and Signature of Evaluator

Date Accomplished



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Appendix B.2
Rubrics for Proposal Defense (IT)

Name: _____

Title: _____

Degree: _____

Direction: Please encircle the number which best applies to the material being evaluated.
5 (Excellent), 4 (Very Good), 3 (Good), 2 (Satisfactory) and 1 (Poor)
Evaluation Categories:

A. The Problem and Its Background

1. Issues and trends related to the problem are carefully presented to establish its validity.	5	4	3	2	1
2. The study is timely and relevant to the author's field of specialization.	5	4	3	2	1
3. Theoretical/Conceptual Framework is valid and adequately presented.	5	4	3	2	1
4. The problem statement or research background is sufficiently congruent with the theoretical/conceptual framework.	5	4	3	2	1
5. The sub-problems or specific research topics are anchored to the major problem or topic.	5	4	3	2	1
6. The state assumptions, if any, are adequate and relevant to the problem.	5	4	3	2	1
7. Whenever applicable, the stated hypotheses are testable.	5	4	3	2	1
8. The scope of the study is properly identified and delineated.	5	4	3	2	1
9. The study adds new insights to current practices and/or theories.	5	4	3	2	1
10. The sources and documents used are properly acknowledged, correctly and accurately cited.	5	4	3	2	1

B. Related Materials

11. The review of related literature and studies effectively discusses significant findings and existing published articles and projects relevant to the present study.	5	4	3	2	1
12. The similarities and differences between and among materials reviewed and the present study are clearly and correctly identified.	5	4	3	2	1

C. Method

13. Stakeholders' and users' requirements are properly



identified and aligned with the defined objectives.	5	4	3	2	1
14. The chosen design/model is appropriate for complying with the identified requirements.	5	4	3	2	1
15. The standards for development and testing should be Appropriate to meet defined requirements.	5	4	3	2	1
16. The procedures for data collection are coherently and logically explained. The data, document, and instruments used are sufficient, reliable, and valid.	5	4	3	2	1
17. The techniques for data analyses are adequate and appropriate.	5	4	3	2	1

D. Other Requirements of the Manuscript

18. The report is free from padding with irrelevant words, phrases, quotes, statistics, examples, etc. that are not essential for accuracy, clarity, and completeness.	5	4	3	2	1
19. The language is precise, and emphasis is duly observed.	5	4	3	2	1
20. The APA Journal format is correct and consistent.	5	4	3	2	1
21. Documentation is complete, and references include all materials/sources cited.	5	4	3	2	1
22. The appended materials are helpful and pertinent to the report.	5	4	3	2	1

TOTAL POINTS _____/110

Name and Signature of Evaluator

Date Accomplished



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Appendix C.1
Rubrics for EMC Oral Defense

First Semester

A. Proposal Defense (70%) – CAPSTONE 1

Criteria	Percentage Weight
I. Presentation (30%)	
Knowledge of the Research Contents	30
Delivery	20
Attitude	10
Knowledge of Research Methodology and Techniques	20
Presentation Tools *	20
II. Research Paper (35%)	
Introduction	15
Review of Related Literature and Studies	10
Conceptual Framework	10
Objective of the Study	15
Scope and Delimitation	10
Research Design	10
Respondents and Sources of Data	10
Research Instrument	10
Research Procedure	10
III. Technical Aspect (35%)	
Originality and Concept	15
Story/Message/Information	15
Background Setting	10
Character Design	10
Story Board	20
Script	15
Animatic	15

B. Instructor (30%)

Criteria	Percentage Weight
Follows assigned task	20
Participation	20
Cooperation	20
Completeness assigned task on time	20
Shows punctuality	10
Regular attendance in group meetings	10
Total	100

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C. Final Oral Defense (90%) – CAPSTONE 2

Criteria	Percentage Weight
I. Presentation (20%)	
Organization	20
Delivery	20
Content	20
Response to Questions	20
Format Presentation	20
II. Research Paper (30%)	APA
Introduction Background Rationale (Statement of Problem) Research Purpose and Plan for School Improvement Research Questions (Hypotheses, as appropriate)	15%
Literature Review Critical Review of Literature & Relevance Conceptual / Theoretical Framework	15%
Statement of the Problem or Objectives of the Study Statement of the Problem or Objectives of the Study Statement of the Problem or Objectives of the Study	15%
Methodology *Research Approach & Design* Data Collection *Data Analysis* *Research Design and Execution*	20%
Results Result Interpretation Correctness of Analysis Presentation Format	15%
Discussion & Conclusion Supported by Results Evaluation of Results Theoretical & Practical Implications Recommendations and future directions	10%
Presentation and Writing Writing Structure Writing Mechanics APA Style	10%
III. Technical Aspects (40%)	
Originality and Concept	10
Story/Message/Information (Demo Reel)	10
Setting and Character Design	15
Color Rendering, Texturing and Lighting	10
Animation	20

Character Gestures and Emotions	10
Voice Recording and Lip Syncing/Musical Score	5
Editing/Special Effects/ Transitions	5
Camera Works	5
Musical Score	5
Spelling and Grammar	5
I. Appendices	
Character Modelling, Object Modelling, Texturing Procedures, rigging Procedures, Lighting Procedures, Animating Procedures	40
Story Board	20
Script	20
Animatics	10
Completeness and Formatting of Appendices	10
Total	**100%**

D. Adviser (10%)

Criteria	Percentage Weight
Follows Assigned Role	20%
Participation	20%
Cooperation	20%
Completes Assigned Task on Time	20%
Displays Punctuality	10%
Regular Attendance in Group Meetings	10%

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Appendix C.2
Rubrics for BSIT – WebDev Capstone 2 Final Oral Defense

I. Rubric for Procedure (Group Cooperation)	**10%**
Follows assigned role	20
Participation	20
Cooperation	20
Completes assigned task on time	20
Displays punctuality	10
Regular attendance in group meetings	10
II. Rubric for Presentation (Individual)	**20%**
Organization	20
Delivery	20
Content	20
Response to Questions	20
Format of Presentation	20
III. Rubric for Manuscript	**30%**
Introduction	15
Literature Review	15
Statement of the Problem or Objective of the Study	15
Methodology	20
Results	10
Discussion & Conclusion	15
Presentation & Writing	10
IV. Rubrics on Web Application (Technology, Function, and Database)	**50%**
Web Application functions without errors	15
Reliability of Web Application functions	15
Web Technologies (Innovations) e.g. Augmented Reality/Virtual Reality/SEO/Dynamic or Responsive or Interactive Pages	15
Information Download	5
Security	10
Validation	5
Format and Platform Transferability	10
Database Structure	10
Table and Field Labeling	5
Queries and Reports	10
V. Rubric on Web Application (Content, Layout, and Design)	**50%**
Organization and Content	20
Graphics	10
Color Scheme	10
Fonts	10
Layout	10
Navigation and Ease of Use	10
Spelling and Grammar	10
Website Credibility	10
Browser and Resolution Compatibility	10

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Appendix C.3
Rubrics for BSIT – NetAd Capstone 2 Final Oral Defense

I. Rubric for Procedure (Group Cooperation)	**10%**
Follows assigned role	20
Participation	20
Cooperation	20
Completes assigned task on time	20
Displays punctuality	10
Regular attendance in group meetings	10
II. Rubric for Presentation (Individual)	**20%**
Organization	20
Delivery	20
Content	20
Response to Questions	20
Format of Presentation	20
III. Rubric for Manuscript	**30%**
Introduction	15
Literature Review	15
Statement of the Problem or Objective of the Study	15
Methodology	20
Results	10
Discussion & Conclusion	15
Presentation & Writing	10
IV. Rubrics on Technical Paper	**40%**
Executive Summary	15
Requirements	15
Proposed Physical Design	20
Proposed Logical Design	20
Security Policy	10
Implementation Plan	10
Cost Proposal	10
V. Rubrics on Simulation	**30%**
Architecture (physical and logical)	25
Operations (appropriate configuration and logical processes)	25
Functionality	25
Security	25
VI. Technology Implementation	**30%**
Platform	30
Applications	25
Model (Implementation of Architecture)	25
Relevant Tools	20

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Appendix C.4
Rubrics for BSCS

I. Presentation (15%)	Weight(%)
Knowledge of the Research Contents	30
Delivery	20
Attitude	10
Knowledge of Research Methodology and Techniques	20
Presentation Tools *	20
Total	100
II. Research Paper (35%) APA 7th Ed.	
Introduction	10
Review of Related Literature and Studies	5
Conceptual Framework	10
Objective of the study	10
Scope and Delimitation	5
Research Design	5
Respondents and Sources of Data	10
Research Instrument	5
Research procedure	5
Results	10
Discussion	10
Recommendations	10
Mechanics of writing and format	5
III. Software (40%)	
Algorithm Application (30)	
Algorithm is applied appropriately	15
Algorithm functions are integrated in the software's objective	15
Performance Effectiveness (20)	
Software functions without errors	10
Reliability of Software functions	10
Performance Efficiency (20)	
Results and processes are consistent	10
Computations and output made are accurate	10
Ease of Use (15)	
Interface and Menus are easy to follow	10
Aesthetics and Typography	5
Flexibility (15)	
Platform Independent	10
Device Independent	5

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Appendix C.5
Rubrics for BSCyb

I. Rubric for Procedure (Group Cooperation)	**10%**
Follows assigned role	20
Participation	20
Cooperation	20
Completes assigned task on time	20
Displays punctuality	10
Regular attendance in group meetings	10
II. Rubric for Presentation (Individual)	**20%**
Organization	20
Delivery	20
Content	20
Response to Questions	20
Format of Presentation	20
III. Rubric for Manuscript	**30%**
Introduction	15
Literature Review	15
Statement of the Problem or Objective of the Study	15
Methodology	20
Results	10
Discussion & Conclusion	15
Presentation & Writing	10
IV. Rubrics on Technical Paper	**40%**
Executive Summary	15
Security Requirements	15
Security Analysis	20
Proposed Design and Implementation	20
Proposed Devices (Cost) and Configuration	20
Implementation Plan	10
V. Rubrics on Simulation	**30%**
Device setup	25
Configuration	25
Functionality	25
Security	25
VI. Security Analysis and Implementation	**30%**
Application and its Platform	30
Analysis Tools	25
Testing	25
Network Monitoring Tool	20

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Appendix D
Capstone/Thesis Monitoring Form

FM-AAC-SOC-2003
Revision: 0
Effectivity Date: May 1, 2025

HOLY ANGEL UNIVERSITY SCHOOL OF COMPUTING

Capstone-Thesis Monitoring Form

Project/Thesis Title:

DATE	WORK UPDATES/STATUS (Please include details)	SIGNATURES	
		Advisees	Adviser

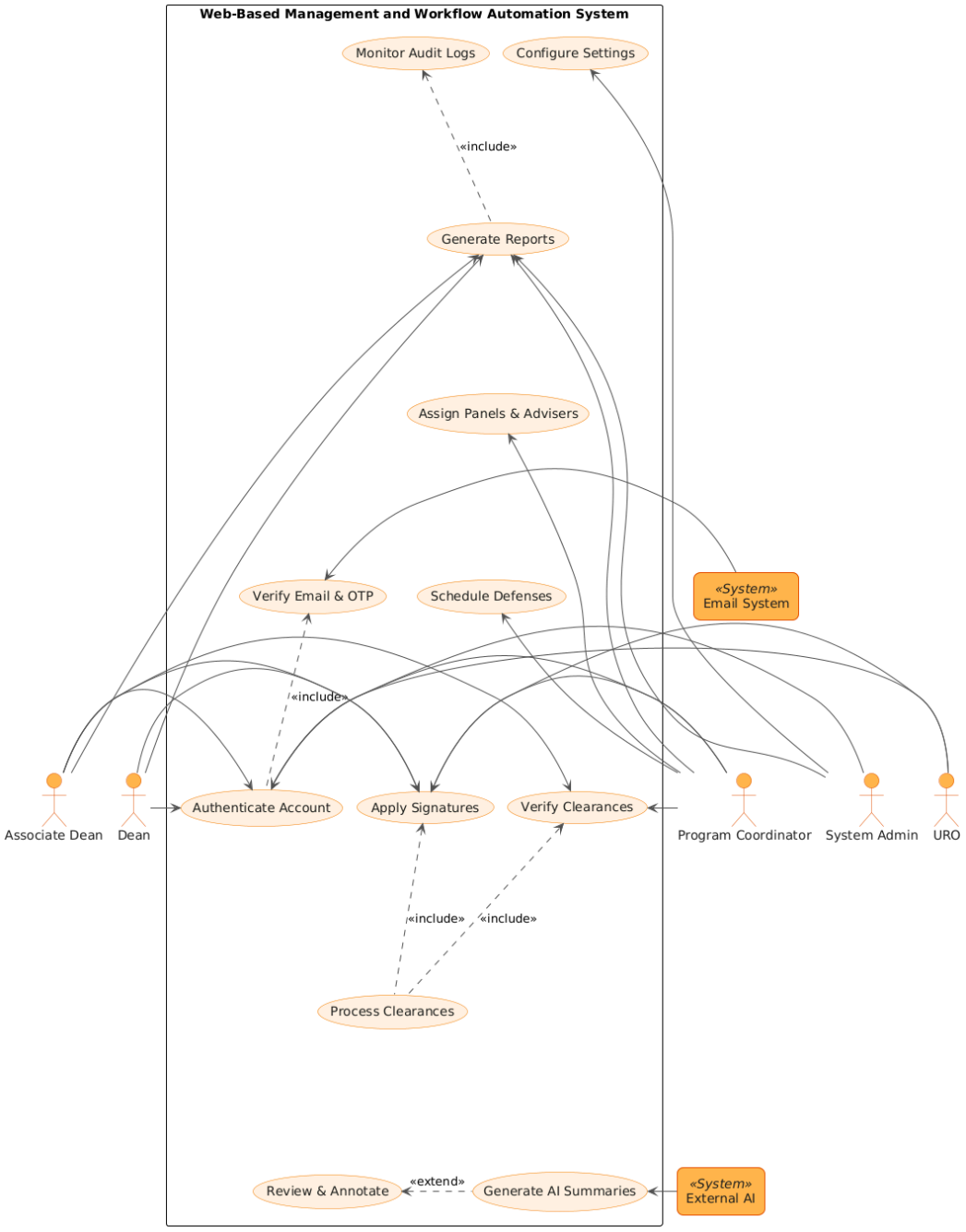
Capstone-Thesis Monitoring Form

#1 Holy Angel Avenue, Sto. Rosario, Angeles City, Philippines 2009
Tel. Nos.: (045) 888-8691; 888-2902; 887-5748; 887-2455; 624-5277; 625-9619 | Fax: (045) 888-1754; 888-2514
Email: hau@hau.edu.ph | www.hau.edu.ph

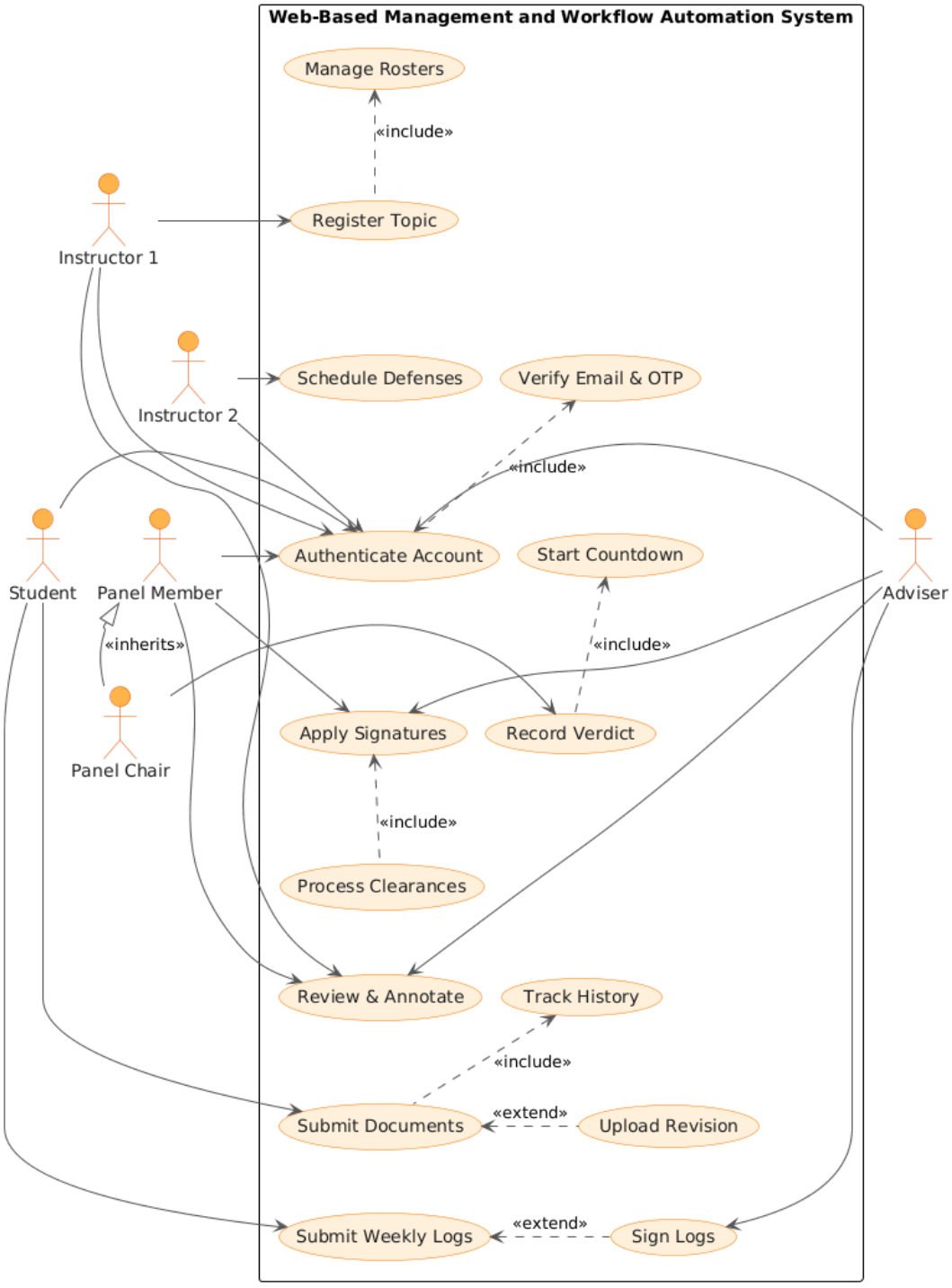
Appendix D

Use Case Diagram

D-1: Global Administrative Roles



D-2: Project-Based Roles



Appendix E

Product Requirements Document

PRODUCT REQUIREMENTS DOCUMENT

Web-Based Management and Workflow Automation System for Thesis and Capstone Projects

Document Version: 1.3

Document Status: Initial Product Requirements

Prepared for: School of Computing

Date: September 22, 2026

---

1. Product Overview

1.1 Product Name

Web-Based Management and Workflow Automation System for Thesis and Capstone Projects

1.2 Product Description

The proposed system is a centralized web-based platform designed to manage the complete lifecycle of thesis and capstone projects. It will support project conceptualization, document submission, academic review, adviser guidance, defense scheduling, evaluation, revision monitoring, approval routing, final clearance, and archival.

The system will replace fragmented processes involving email, printed forms, spreadsheets, separate document repositories, and manual follow-ups. It will provide a unified workspace where students, instructors, advisers, panel members, coordinators, academic administrators, and authorized research personnel can perform their assigned responsibilities.

The system will also incorporate artificial intelligence-assisted document summarization to help academic reviewers understand submitted manuscripts efficiently. AI summaries shall only be generated for complete manuscript submissions during the Proposal Defense and Final Defense stages. AI-generated content will serve only as decision support and will not replace human evaluation, academic judgment, or official approval.

---

2. Product Vision

To provide the School of Computing with a secure, transparent, efficient, and auditable digital platform that streamlines thesis and capstone project management from initial conceptualization to final institutional archiving.

---

3. Product Objectives

The system shall:

1. Centralize thesis and capstone project records, documents, evaluations, and approvals.
2. Automate document routing according to the established academic workflow.
3. Provide structured adviser guidance during project conceptualization and development.
4. Maintain immutable submission records and complete document version histories.
5. Improve communication through automated notifications and reminders.
6. Allow authorized reviewers to annotate and evaluate documents without modifying the original files.
7. Facilitate adviser, panel, coordinator, associate dean, and dean approvals.
8. Support defense scheduling, evaluation, verdict recording, and revision monitoring.
9. Generate AI-assisted summaries of submitted manuscripts for authorized reviewers.
10. Digitize institutional forms, recommendations, certifications, and signatures.
11. Maintain a complete audit trail of system activities and academic decisions.
12. Preserve approved final documents in a secure institutional archive.

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4. Problem Statement

The current management of thesis and capstone projects may involve multiple disconnected platforms and manual procedures. Documents may be submitted through email, messaging applications, printed copies, or separate cloud storage services. Evaluations, signatures, revision requests, and defense schedules may also be handled independently.

These practices may result in:

- Lost or duplicated documents;
- Difficulty identifying the latest manuscript version;
- Delayed reviews and approvals;
- Limited visibility of project status;
- Inconsistent application of academic workflows;
- Difficulty monitoring student progress;
- Incomplete documentation of revisions and decisions;
- Manual preparation and routing of institutional forms;
- Lack of a centralized archive; and
- Limited accountability and traceability.

The proposed system will address these concerns through centralized records, automated workflow routing, role-based access, document version control, digital evaluation, and structured monitoring.

5. Product Scope

5.1 In Scope

The system will include the following major functions:

- User registration and authentication;
- Role-based and project-based access control;
- Project group creation and approved project topic registration;
- Project conceptualization and adviser guidance;
- Concept paper and title proposal submission;
- Manuscript and supporting-document submission;
- Immutable document versioning;
- AI-assisted summarization of complete proposal and final-defense manuscripts;
- Document annotation and reviewer comments;
- Evaluation and recommendation forms;
- Automated workflow routing;
- Weekly accomplishment and attendance monitoring;
- Digital signatures and approvals;
- Adviser and panel assignment;
- Proposal and final defense scheduling;
- Defense evaluation and verdict recording;
- Revision classification and countdown monitoring;
- Final document verification and clearance;
- Plagiarism certificate and editor's note submission;
- University Research Office-related routing, when applicable;
- Institutional archiving;
- Reporting and audit trails; and
- Automated email notifications and reminders.

5.2 Out of Scope

The initial version of the system will not include:

- Real-time chat or instant messaging;
- Video conferencing;
- Grading or scoring of thesis and capstone projects;

- Replacement of institutional plagiarism-detection software;
- Public editing of academic manuscripts;
- Direct editing of uploaded PDF files;
- Learning management system functions unrelated to thesis or capstone management; and
- Mobile applications for Android or iOS.

The system may use responsive web design to support access through mobile browsers.

6. Stakeholders

6.1 Primary Stakeholders

- School of Computing Dean;
- Associate Dean;
- Program Coordinators;
- Thesis and Capstone Instructors;
- Project Advisers;
- Panel Chairs;
- Panel Members;
- Students;
- University Research Office personnel; and
- System Administrators.

6.2 Secondary Stakeholders

- Academic editors;
- External evaluators;
- Research ethics reviewers;
- Intellectual property personnel;
- Accreditation teams;
- School administrative staff; and
- University information technology personnel.

7. User Roles

The system shall support global roles and project-specific roles.

7.1 Global Roles

7.1.1 Student

The student shall be able to:

- Register and authenticate using an authorized university account;
- Submit project ideas, concept papers, title proposals, manuscripts, and supporting documents;
- View project status and current workflow stage;
- Receive comments, annotations, decisions, and revision instructions;
- Upload revised document versions;
- Submit weekly accomplishment records;
- View defense schedules;
- Upload final requirements;
- Monitor revision deadlines; and
- View approved and completed project records.

7.1.2 Program Coordinator

The Program Coordinator shall be able to:

- View projects within the assigned academic program;
- Verify student and project information;
- Assign or recommend advisers;
- Assign panel chairs and panel members;
- Review project endorsements;

- Approve projects for defense scheduling;
- Monitor program-level project progress;
- Verify completion of forms and requirements;
- Generate program-level reports; and
- Route projects to the Associate Dean or other authorized offices.

7.1.3 Associate Dean

The Associate Dean shall be able to:

- Review endorsed projects;
- Verify compliance with school-level requirements;
- Approve or return requests;
- Review defense schedules and panel assignments;
- Monitor projects across academic programs;
- Review project completion status; and
- Generate consolidated reports.

7.1.4 Dean

The Dean shall be able to:

- View all thesis and capstone projects under the school;
- Review projects requiring executive approval;
- Approve, return, or reject routed documents;
- Review final project completion records;
- Access school-level dashboards and reports;
- Review and compliance information; and
- Authorize institutional endorsements when required.

7.1.5 System Administrator

The System Administrator shall be able to:

- Manage user accounts;
- Manage role assignments;
- Configure academic terms and programs;
- Configure workflow stages;
- Manage templates and institutional forms;
- Manage document categories;
- Configure revision periods and deadlines;
- View audit logs;
- Perform authorized system maintenance;
- Manage reference data; and
- Generate administrative reports.

7.1.6 University Research Office (URO)

The University Research Office shall be able to:

- Review the submitted editor's certificate and plagiarism certificate;
- Verify the finalized manuscript for compliance before institutional sign-off;
- Approve or return the clearance package; and
- Endorse the project for the Dean's final signature.

7.2 Project-Based Roles

A user may hold a global role while also serving in one or more project-specific roles.

7.2.1 Instructor 1

Instructor 1 shall be responsible for the

conceptualization and initial development stage.

The role shall be able to:

- Create project groups;
- Assign students to project groups;
- Review project concepts and proposed titles;
- Guide students during project conceptualization;
- Add comments and annotations;
- Request revisions;

- Approve or reject proposed topics;
- Register the group's approved thesis or capstone project topic;
- Review initial manuscript drafts;
- Monitor required milestones;
- Endorse qualified projects for adviser or coordinator review; and
- Verify completion of course-related requirements.

7.2.2 Instructor 2

Instructor 2 shall be responsible for the

succeeding implementation or completion stage.

The role shall be able to:

- Review continuing project records;
- Monitor implementation progress;
- Review updated manuscripts and system deliverables;
- Validate completion of required milestones;
- Endorse projects for final defense;
- Review post-defense requirements; and
- Verify course completion requirements.

7.2.3 Adviser

The Adviser shall be able to:

- Access assigned projects;
- Guide students during conceptualization and development;
- Review submitted manuscripts;
- Add non-destructive annotations;
- Provide comments and recommendations;
- Approve or return document versions;
- Review weekly accomplishment reports;
- Apply a digital signature to validated reports;
- Recommend a project for defense;
- Submit the required recommendation form;
- Verify post-defense revisions; and
- Endorse the final manuscript for clearance.

7.2.4 Panel Chair

The Panel Chair shall be able to:

- Access assigned defense documents;
- Review AI-generated manuscript summaries;
- Review adviser and instructor comments;
- Add annotations and evaluation comments;
- Record the official defense result;
- Consolidate panel recommendations;
- Classify revisions as minor or major;
- Initiate the revision countdown;
- Verify panel compliance requirements; and
- Endorse the revised manuscript after panel confirmation.

7.2.5 Panel Member

The Panel Member shall be able to:

- Review assigned manuscripts;
- View AI-generated summaries;
- Add annotations and comments;
- Complete evaluation forms;
- Submit recommendations;
- Review revised documents; and
- Confirm whether required revisions were addressed.

8. Role and Access Management Requirements

8.1 Multiple Role Support

The system shall allow a single user to hold multiple roles. For example, a faculty member may simultaneously serve as:

- Instructor;
- Adviser;
- Panel Member; and
- Panel Chair.

The interface shall display functions according to the user's active role and project assignments.

8.2 Contextual Access Control

Access shall depend on:

- The user's global role;
- The user's project-specific role;
- The academic program;
- The current workflow stage;
- The document status;

- The user's assignment to the project; and
- The approval authority required.

8.3 Progressive Visibility

Users shall only see records and functions relevant to their responsibilities. For example:

- Students shall only access projects in which they are registered members.
- Advisers shall only access projects assigned to them.
- Panel members shall only access projects assigned for their review.
- Program Coordinators shall only access projects under their assigned programs.
- The Dean and authorized administrators may access school-wide records.

9. Product Workflow

9.1 Project Conceptualization

1. Instructor 1 creates the project group.
2. Students submit project ideas during the ideation phase.
3. Students upload a concept paper or title proposal.
4. Instructor 1 reviews the project concept.
5. Instructor 1 guides the students during conceptualization.
6. Instructor 1 adds comments or annotations.
7. Students upload a revised concept.
8. Instructor 1 approves or returns the proposal.
9. Instructor 1 registers the group's approved project topic.
10. The approved concept is routed to the next authorized reviewer.

9.2 Adviser Assignment

1. The Program Coordinator reviews approved project concepts.

2. The Coordinator identifies an appropriate adviser.
3. The adviser assignment is recorded in the system.
4. The assigned adviser receives a notification.
5. The adviser accepts or acknowledges the assignment.
6. Students gain access to the adviser-guided development workflow.

9.3 Proposal Development and Review

1. Students upload the proposal manuscript.
2. The system creates an immutable document version.
3. Once the complete proposal manuscript is submitted, the system generates an AI-assisted summary.
4. The adviser reviews the manuscript.
5. The adviser adds comments and annotations.
6. Students revise and upload a new version.
7. The adviser recommends the project for proposal defense.
8. The Coordinator verifies the requirements.
9. The project proceeds to panel assignment and defense scheduling.

9.4 Proposal Defense

1. The Program Coordinator assigns the Panel Chair and Panel Members.
2. The defense schedule is recorded.
3. Students, adviser, and panel members receive notifications.
4. Panel members review the manuscript and AI summary.
5. The defense is conducted.
6. Panel members submit evaluation forms.
7. The Panel Chair records the official verdict.
8. The Panel Chair classifies the required revisions.
9. The revision countdown begins.

9.5 Project Implementation and Monitoring

1. Students continue project development.
2. Students submit weekly accomplishments or activity records.
3. The adviser reviews and digitally signs the weekly records.
4. Instructor 2 monitors implementation milestones.
5. The system sends reminders for pending requirements.
6. Students upload updated manuscripts and project deliverables.
7. Students attach a revision note or changelog describing the modified pages or completed system components.

8. The adviser and instructor verify readiness for final defense.

9.6 Final Defense

1. The adviser submits the final-defense recommendation.
2. The Coordinator verifies eligibility.
3. A final-defense schedule is created.
4. Panel members review the latest manuscript together with the AI-generated summary.
5. Panel members complete the final evaluation.
6. The Panel Chair records the official result.
7. Required revisions are classified.
8. The system starts the final revision countdown.

9.7 Final Revision and Clearance

1. Students upload the revised final manuscript together with a revision note or changelog identifying the modified pages or system components.
2. Panel members verify whether revisions were addressed.
3. The adviser validates the final manuscript.
4. Students upload the editor's note, plagiarism certificate, and other required documents.
5. The Coordinator verifies completeness.
6. The University Research Office verifies the editor's certificate and plagiarism certificate.
7. The final manuscript is marked as cleared.
8. The approved records are archived.

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10. Functional Requirements

FR-01: User Registration

The system shall allow authorized users to register using their official university email addresses.

FR-02: Email Domain Validation

The system shall restrict registration to approved institutional email domains.

FR-03: One-Time Password Authentication

The system shall send a one-time password to the registered email address for account verification or login confirmation.

FR-04: Secure Login

The system shall provide secure login and logout functions.

FR-05: Password Management

The system shall provide password creation, password reset, and account recovery functions.

FR-06: User Profile

The system shall allow users to manage basic profile information, subject to administrator verification.

FR-07: Role Management

The system shall allow administrators to assign one or more global roles to a user.

FR-08: Project-Specific Role Assignment

The system shall allow authorized personnel to assign Adviser, Panel Chair, and Panel Member roles for individual projects.

FR-09: Project Group Creation

The system shall allow Instructor 1 to create project groups.

FR-10: Group Membership

The system shall allow Instructor 1 or authorized administrators to add and verify group members.

FR-11: Project Registration

The system shall allow Instructor 1 to register the approved thesis or capstone project after successful completion of the ideation phase.

- Proposed project title;
- Project category;
- Academic program;
- Academic term;
- Project members;
- Project description;
- Proposed technology;
- Research area; and
- Initial project objectives.

FR-12: Concept Paper Submission

The system shall allow students to upload a concept paper or title proposal in PDF format.

FR-13: Conceptualization Guidance

The system shall allow Instructor 1 and the assigned Adviser to provide structured guidance during conceptualization.

Guidance may include:

- Comments;
- Recommendations;
- Required actions;
- Approval decisions; and
- Revision requests.

FR-14: Document Uploading

The system shall allow students to upload authorized academic documents in supported file formats.

FR-15: Required Document Validation

The system shall verify whether required documents have been submitted before allowing the project to proceed to the next stage.

FR-16: Immutable Submission

A submitted document version shall not be overwritten or permanently modified.

FR-17: Document Versioning

The system shall create a new document version for every revised submission. Each version shall contain:

- Version number;
- File name;
- Submission date and time;
- Submitting user;
- Document category;
- Workflow stage; and
- Submission remarks.

FR-18: Version Comparison

The system shall allow authorized users to view previous and current versions of a document.

FR-19: Document Status

Each document shall have a status, such as:

- Draft;
- Submitted;
- Under Review;
- For Revision;
- Approved;
- Rejected;
- Superseded; or
- Archived.

Project archive result (recorded once, at final clearance): Pass or Fail. No intermediate status labels are shown at the archive stage.

FR-20: AI-Assisted Summary

The system shall generate a structured summary only for complete manuscript submissions during the Proposal Defense and Final Defense stages.

The summary may include:

- Project background;
- Problem statement;
- Objectives;
- Proposed methodology;
- Expected output;
- Technology or tools;
- Target users;
- Scope;
- Limitations; and
- Key contributions.

FR-21: AI Output Labeling

All AI-generated content shall be clearly labeled as machine-generated information based on the manuscript.

FR-22: Human Review Requirement

AI-generated summaries shall not automatically approve, reject, grade, or evaluate a project. Manuscripts submitted for AI summarization shall only be sent to an AI service that has an institutionally approved data-handling agreement, and each transmission and its purpose shall be logged in the audit trail.

FR-23: AI Summary Refresh

The system shall generate the AI summary once per defense milestone (one for Proposal Defense, one for Final Defense), triggered only by the complete manuscript submission for that milestone. If a project is required to re-defend within the same milestone, the summary shall be regenerated only upon explicit reviewer request, not automatically, in order to limit repeated exposure of manuscript content to the AI service and control processing cost.

FR-24: Non-Destructive Annotation

Only the Panel (Panel Chair and Panel Members), Adviser, and Instructor 1 shall be authorized to add annotations, without modifying the original uploaded document. Panel Member annotations shall remain private to the submitting Panel Member until the defense has been conducted.

FR-25: Annotation Categories

Annotations may be categorized as:

- General Comment;
- Technical Concern;
- Methodology Concern;
- Documentation Issue;
- Formatting Issue;
- Required Revision; or
- Recommendation.

FR-26: Reviewer Comments

The Panel, Adviser, and Instructor 1 shall be able to submit general comments for a document version. The Program Coordinator, URO, and Dean do not add content-level comments; they act only at the approval/routing stage.

FR-27: Comment Attribution

Every comment and annotation shall display the author, role, date, and time.

FR-28: Revision Response

Students shall be able to provide responses or resolution notes for reviewer comments.

FR-29: Review Decision

Only the Panel, Adviser, and Instructor 1 shall be able to select a review decision:

- Approve;
- Approve with Minor Revisions;
- Return for Major Revisions;
- Reject; or
- Endorse to Next Stage.

FR-30: Automated Routing

The system shall route project documents based on the configured academic workflow. A typical route may include:

Instructor 1 → Adviser → Program Coordinator → Panel → Associate Dean → University Research Office → Dean

The actual route may vary according to the project type and institutional requirements.

FR-31: Stage Completion Control

The system shall prevent a project from moving to the next stage until required approvals and documents are complete.

FR-32: Return for Revision

An authorized reviewer shall be able to return a document to the student with revision instructions.

FR-33: Workflow History

The system shall record every workflow movement, including:

- Previous stage;
- New stage;
- User responsible;
- Date and time;
- Decision; and
- Remarks.

FR-34: Weekly Accomplishment Submission

Students shall be able to submit weekly accomplishment or activity reports.

FR-35: Weekly Attendance Record

The system shall support weekly attendance or consultation records when required by the institution.

FR-36: Adviser Weekly Validation

The Adviser shall be able to review, approve, return, and digitally sign weekly records.

FR-37: Progress Monitoring

The system shall display completed, pending, overdue, and upcoming project milestones.

FR-38: Adviser Assignment

Authorized coordinators shall be able to assign an adviser to a project.

FR-39: Panel Assignment

Authorized coordinators shall be able to assign a Panel Chair and Panel Members.

FR-40: Assignment Conflict Checking

The system shall identify possible duplicate assignments or scheduling conflicts.

FR-41: Defense Scheduling

The system shall allow authorized personnel to create proposal and final-defense schedules.

FR-42: Defense Information

A defense schedule shall contain:

- Project;
- Defense type;
- Date;
- Time;
- Venue or online meeting information;

- Adviser;
- Panel Chair;
- Panel Members; and
- Additional instructions.

FR-43: Schedule Notification

The system shall notify all assigned participants when a defense is scheduled or modified.

FR-44: Defense Evaluation

Panel members shall be able to complete institutional evaluation forms.

FR-45: Official Verdict

Only the authorized Panel Chair shall be able to record the official defense verdict.

FR-46: Revision Classification

The Panel Chair shall classify required revisions as minor or major.

FR-47: Revision Countdown

The system shall start a configurable revision countdown after the official defense verdict.

FR-48: Revision Deadline Monitoring

The system shall display the remaining number of days before the revision deadline.

FR-49: Overdue Revision Flagging

The system shall automatically mark revisions as overdue when the deadline has passed.

FR-50: Revision Verification

Authorized advisers and panel members shall be able to verify that required revisions have been addressed.

FR-51: Digital Forms

The system shall provide digital versions of

approved institutional forms. These may include:

- Project registration form;
- Adviser assignment form;
- Weekly accomplishment form;
- Capstone recommendation form;
- Defense evaluation form;
- Revision compliance form;
- Final clearance form; and
- Endorsement form.

FR-52: Digital Signature

Authorized users shall be able to digitally sign approved institutional forms.

FR-53: Signature Traceability

Each digital signature shall include:

- Signatory name;
- Role;
- Date and time;
- Associated project;
- Associated form; and
- Signature status.

FR-54: Final Requirement Uploading

Students shall be able to upload final requirements, including:

- Final manuscript;
- Editor's note or certification;
- Plagiarism certificate;
- Revision compliance form;
- Source code or repository reference;
- Deployment documentation;
- User manual;
- Technical documentation; and
- Other institutional requirements.

FR-55: Final Clearance Verification

The system shall verify the completeness of all final requirements before clearance.

FR-56: Final Approval

Authorized coordinators, academic administrators, and research personnel shall be able to approve final project completion.

FR-57: Institutional Archive

Cleared projects shall be stored in the secure final archive with a result of Pass or Fail only.

FR-58: Archive Immutability

Archived final documents shall not be editable by ordinary users.

FR-59: Archive Search

Authorized users shall be able to search archived projects by:

- Title;
- Academic program;
- Academic year;
- Student member;
- Adviser;
- Research area;
- Technology;
- Project status; and
- Keywords.

FR-60: Institutional Thesis and Capstone Archival

The system shall allow registered users (students and faculty) to browse the archive of completed and approved projects. The system shall limit document access by displaying only the project metadata and abstract, excluding the full paper.

FR-61: Email Notifications

The system shall send email notifications for significant workflow events.

FR-62: Notification Events

The system shall send email notifications to users involved in or affected by the current workflow for the following events:

- New assignments;
- New document submissions;
- Revision requests;
- Review completion;
- Approval or rejection;
- Defense schedule creation;
- Defense schedule modification;
- Upcoming deadlines;
- Overdue revisions;
- Final clearance; and
- Administrative announcements.

FR-63: Dashboard

Each user shall have a dashboard appropriate to the user's role.

FR-64: Student Dashboard

The student dashboard shall display:

- Active projects;
- Current workflow stage;
- Pending requirements;
- Recent comments;
- Upcoming defenses;
- Revision deadlines;
- Weekly progress status; and
- Final clearance status.

FR-65: Faculty Dashboard

The faculty dashboard shall display:

- Assigned projects;
- Pending reviews;
- Adviser assignments;
- Panel assignments;
- Upcoming defenses;

- Overdue reviews; and
- Recent student submissions.

FR-66: Coordinator Dashboard

The Coordinator dashboard shall display:

- Projects by stage;
- Projects requiring adviser assignment;
- Projects awaiting defense scheduling;
- Pending endorsements;
- Overdue projects; and
- Program-level completion rates;

FR-67: Administrative Dashboard

The Dean, Associate Dean, and System Administrator dashboards shall display:

- Total active projects;
- Projects by academic program;
- Projects by workflow stage;
- Pending approvals;
- Upcoming defenses;
- Overdue revisions;
- Completed projects;
- Archived projects; and
- System activity indicators.

Review basic compliance summaries (e.g., overdue revisions, pending approvals) drawn from existing dashboard counts; trend analysis across academic terms is deferred to Future Enhancements.

FR-68: Reports

The system shall generate reports based on authorized access.

FR-69: Report Types

Reports may include:

- Active project report;
- Project status report;
- Adviser assignment report;
- Panel assignment report;
- Defense schedule report;
- Revision compliance report;
- Weekly progress report;
- Completion report;
- Archived project report;
- User activity report; and
- Approval turnaround-time report.

FR-70: Report Export

Authorized users shall be able to export reports in PDF or spreadsheet-compatible format.

FR-71: Audit Trail

The system shall record critical activities, including:

- Login attempts;
- Account changes;
- Role assignments;
- Project registration;
- Document uploads;
- Document downloads;
- Reviewer comments;
- Approval decisions;
- Signature activities;
- Workflow transitions;
- Defense verdicts;
- Final clearance; and
- Administrative configuration changes.

FR-72: Audit Log Protection

Ordinary users shall not be able to modify or delete audit records.

FR-73: Search and Filtering

The system shall provide keyword search and multi-criteria filtering for authorized records.

11. Non-Functional Requirements

NFR-01: Usability

The interface shall be understandable and usable by students, faculty members, and administrators with minimal training.

NFR-02: Responsive Design

The system shall work properly on desktop computers, laptops, tablets, and mobile web browsers.

NFR-03: Accessibility

The system should follow accepted web accessibility practices, including:

- Keyboard-accessible controls;
- Readable contrast;
- Form labels;
- Alternative text;
- Clear error messages; and consistent navigation

NFR-04: Performance

Standard pages should load within three seconds under normal operating conditions.

NFR-05: File Upload Performance

The system shall display upload progress and confirm successful document submission.

NFR-06: Availability

The system should be available during required academic and administrative periods, subject to scheduled maintenance.

NFR-07: Reliability

The system shall prevent incomplete workflow transactions and duplicate approval actions.

NFR-08: Data Integrity

The system shall preserve the accuracy, completeness, and consistency of project records.

NFR-09: Security

The system shall use secure authentication, authorization, session management, and encrypted communication.

NFR-10: Encryption

Data transmitted between users and the system shall use HTTPS encryption.

NFR-11: Secure File Access

Uploaded files shall only be accessible to authorized users.

NFR-12: Session Security

The system shall terminate inactive sessions after a configurable period.

NFR-13: Input Validation

All user input and uploaded files shall be validated.

NFR-14: File Security

The system shall restrict file types, file sizes, and potentially harmful uploads.

NFR-15: Backup and Recovery

The system shall maintain regular backups of databases, documents, configurations, and audit records.

NFR-16: Recoverability

Authorized administrators shall be able to restore system data from valid backups.

NFR-17: Scalability

The system shall support increasing numbers of users, projects, document versions, and archived records.

NFR-18: Maintainability

The system shall use modular components and documented interfaces to support maintenance and future enhancement.

NFR-19: Compatibility

The system shall support current versions of commonly used web browsers.

NFR-20: Privacy

The system shall comply with applicable institutional privacy policies and data protection requirements.

NFR-21: AI Transparency

The system shall disclose when information was generated through artificial intelligence.

NFR-22: AI Data Protection

Confidential manuscripts shall not be transmitted to unauthorized external AI services.

NFR-23: Auditability

Academic decisions and workflow changes shall be traceable to the responsible user.

12. Business Rules

BR-01

Only registered and verified students may be included in a project group.

BR-02

A student may only access projects in which the student is an official member.

BR-03

A project shall have only one active approved title at a given time.

BR-04

Changes to an approved title shall require authorization.

BR-05

A submitted document shall not be overwritten.

BR-06

A revised manuscript shall be stored as a new version.

BR-07

Only assigned reviewers may evaluate a project.

BR-08

Only the assigned Panel Chair may record the official defense verdict.

BR-09

A project shall not proceed to defense without the required recommendation and complete documents.

BR-10

The revision countdown shall begin only after the official verdict is recorded.

BR-11

The system shall use configurable deadlines for minor and major revisions.

BR-12

Only authorized users may digitally sign institutional forms.

BR-13

A final project shall not be archived until all required clearances are completed.

BR-14

AI-generated summaries shall not be treated as official academic evaluations.

BR-15

The original submitted document shall remain accessible as part of the project history.

BR-16

Deleted or deactivated accounts shall not remove historical academic records.

BR-17

All approval and rejection decisions shall require remarks when configured by the workflow.

BR-18

Users with multiple roles shall access records according to their role within each project.

BR-19

The public gallery shall only display projects approved for public presentation.

BR-20

Confidential project documents shall not be publicly searchable.

BR-21

Individual Panel Member annotations and comments shall not be visible to other Panel Members or to the student until after the defense.

13. Major System Modules

13.1 Authentication and User Management Module

Functions:

- Institutional email registration;
- OTP verification;
- Login and logout;
- Password recovery;
- Profile management;
- Role assignment;
- Account activation and deactivation; and
- Access control.

13.2 Project Registration Module

Functions:

- Instructor-managed group creation;
- Group member assignment;
- Approved project topic registration;
- Project information entry;
- Project category selection;
- Concept paper submission;
- Title approval; and
- Project status tracking.

13.3 Conceptualization and Adviser Guidance Module

Functions:

- Concept submission;
- Instructor guidance;
- Adviser guidance;
- Reviewer comments;
- Structured recommendations;
- Revision requests;
- Concept approval; and
- Concept history.

13.4 Document Management Module

Functions:

- Document uploading;
- Document categorization;
- Version control;
- Submission history;
- Secure downloading;
- Required-document checking; and
- Final-document archiving.

13.5 AI-Assisted Review Module

Functions:

- Manuscript text extraction;
- Structured summarization;
- Summary versioning;
- AI-output labeling;
- Reviewer access control; and

- AI summary generation during major manuscript milestones.

13.6 Annotation and Evaluation Module

Functions:

- Document annotations;
- General comments;
- Reviewer recommendations;
- Student resolution notes;
- Evaluation forms;
- Approval decisions; and
- Review history.

13.7 Workflow Automation Module

Functions:

- Stage configuration;
- Automatic document routing;
- Approval sequencing;
- Return-for-revision controls;
- Stage-completion validation;
- Workflow history; and
- Escalation reminders.

13.8 Progress Monitoring Module

Functions:

- Weekly accomplishment submission;
- Attendance or consultation tracking;
- Adviser validation;
- Milestone monitoring;
- Deadline tracking; and
- Overdue requirement identification.

13.9 Defense Management Module

Functions:

- Panel assignment;
- Defense scheduling;
- Conflict checking;
- Evaluation form distribution;
- Verdict recording;
- Revision classification; and
- Revision countdown.

13.10 Digital Forms and Signature Module

Functions:

- Institutional form templates;
- Form completion;
- Digital signatures;
- Signature status;
- Signed form storage; and
- Form verification.

13.11 Notification Module

Functions:

- Email notifications;
- Deadline reminders;
- Review reminders;
- Schedule updates; and
- Approval notices;

13.12 Clearance and Archiving Module

Functions:

- Final requirement checking;
- Editor's note submission;
- Plagiarism certificate submission;
- Final approval routing;
- Clearance confirmation;
- Secure archiving; and
- Archive searching.

13.13 Reporting and Analytics Module

Functions:

- Status reports;
- Faculty assignment reports;
- Defense reports;
- Compliance reports;
- Completion reports;
- Turnaround-time reports;
- Dashboard visualizations; and
- Report exporting.

13.14 Audit and Administration Module

Functions:

- Audit log viewing;
- Academic-term configuration;
- Program configuration;
- Workflow configuration;
- Form-template management;
- Deadline configuration;
- Notification-rule management; and
- System reference-data management.

14. Data Requirements

14.1 User Data

The system shall store:

- User identifier;
- Name;
- Institutional email;
- Academic or employee identification number;
- Academic program or department;
- Global roles;
- Account status;
- Authentication records; and
- Profile information.
- Year level (for student users).

14.2 Project Data

The system shall store:

- Project identifier;
- Project title;
- Previous title versions;
- Project description;
- Project category;
- Research area;
- Academic program;
- Academic term;
- Project members;
- Adviser;
- Panel members;
- Current workflow stage;
- Project status;
- Important dates; and
- Completion information.

14.3 Document Data

The system shall store:

- Document identifier;
- Project identifier;
- Document type;
- File location;
- Version number;
- File name;
- File size;
- Uploading user;
- Submission date and time;
- Document status;
- Review status; and
- Archive status.

14.4 Review Data

The system shall store:

- Reviewer;
- Reviewer role;
- Associated document version;
- Comments;
- Annotations;
- Recommendations;
- Decision;
- Date and time; and
- Student response.

14.5 Defense Data

The system shall store:

- Defense type;
- Date;
- Time;
- Venue;
- Assigned panel;
- Evaluation forms;
- Verdict;
- Revision classification;
- Revision deadline; and
- Compliance status.

14.6 Audit Data

The system shall store:

- User;
- Activity;
- Record affected;
- Previous value;
- New value;
- Date and time;
- Network or session information; and
- Activity result.

15. User Interface Requirements

15.1 General Interface

The interface shall provide:

- Clear navigation;
- Role-specific menus;
- Breadcrumbs;
- Search controls;
- Status indicators;
- Deadline indicators;
- Confirmation messages;
- Error messages; and
- Responsive layouts.

15.2 Status Indicators

The system shall use clearly labeled status indicators, including:

- Pending;
- Submitted;
- Under Review;
- For Revision;
- Approved;
- Rejected;
- Scheduled;
- Completed;
- Overdue;
- Cleared; and
- Archived.

15.3 Project Timeline

Each project page shall include a timeline showing:

- Completed stages;
- Current stage;
- Pending stages;
- Submission dates;
- Approval dates;
- Defense dates;
- Revision deadlines; and
- Final clearance date.

15.4 Document Review Workspace

The document review workspace shall display:

- Uploaded manuscript;
- Document version;
- AI-generated summary;
- Annotations;
- Reviewer comments;
- Previous decisions;
- Evaluation form;
- Approval controls; and
- Revision history.

16. Integration Requirements

16.1 Email Service

The system shall integrate with an email service for:

- OTP delivery;
- Registration confirmation;
- Workflow notifications;
- Review reminders;
- Defense schedules; and
- Deadline alerts.

16.2 AI Service

The system may integrate with a secured AI or natural language processing service for manuscript summarization.

The AI integration shall:

- Use authorized manuscript content;
- Return structured summaries only for complete proposal-defense and final-defense manuscript submissions.
- Ignore intermediate drafts and revision submissions that require only revision notes or changelogs.
- Record the manuscript version used;
- Label AI-generated output;
- Prevent automatic academic decisions; and
- Follow institutional data-protection requirements.

16.3 File Storage

The system shall integrate with secure local or cloud-based file storage.

16.4 Institutional Authentication

Future versions may integrate with university single sign-on or directory services.

16.5 Plagiarism Detection

The system may store plagiarism certificates or references from an institutionally approved plagiarism-detection service.

The initial system shall not claim to replace specialized plagiarism-detection platforms.

17. Assumptions

1. Users have access to institutional email accounts.

2. Institutional workflows, forms, and approval authorities are formally defined.
3. Students will submit documents in supported formats.
4. Authorized faculty members will be assigned as advisers and panel members.
5. The institution will provide the required hosting and storage environment.
6. AI-assisted summaries will be used only for authorized academic support.
7. Reviewers remain responsible for reading and evaluating the submitted manuscript.
8. Revision periods will be defined by institutional policy.
9. The institution will identify documents that may be publicly displayed.
10. Administrative personnel will maintain academic terms, roles, deadlines, and workflow configurations.

---

18. Constraints

1. The system must comply with institutional data privacy and academic integrity policies.
2. Manuscripts may contain confidential or potentially patentable information.
3. Access to project records must be strictly controlled.
4. Existing institutional forms may require customization before digitization.
5. AI-generated summaries may contain errors and must be verified by reviewers.
6. AI summarization is limited to major manuscript milestones to reduce unnecessary processing and operate within available AI service usage limits.
7. Internet connectivity may affect document uploading and remote access.
8. Large document files may require storage and upload limitations.
9. Workflow changes may require administrator configuration.
10. Users may hold multiple academic roles.
11. The system shall not include an internal chat module in the initial release.

---

19. Risks and Mitigation Measures

19.1 Unauthorized Document Access

Risk: Confidential academic documents may be accessed by unauthorized users.
Mitigation: Implement role-based access control, project-specific permissions, secure file links, HTTPS, and audit logging.

19.2 Incorrect Role Assignment

Risk: A user may receive inappropriate access.

Mitigation: Restrict role assignment to authorized administrators and record all role changes.

19.3 Loss of Documents

Risk: Files may be deleted, corrupted, or lost.

Mitigation: Use version control, redundant storage, regular backups, and recovery procedures.

19.4 AI Summary Errors

Risk: The AI may misinterpret or omit important manuscript information.

Mitigation: Clearly label AI content, require human review, and link each summary to the source manuscript version.

19.5 Delayed Reviews

Risk: Reviewers may fail to complete evaluations on time.

Mitigation: Use pending-task dashboards, automated email notifications, deadlines, and escalation notifications.

19.6 Workflow Misconfiguration

Risk: Documents may be routed to the wrong user or stage.

Mitigation: Use administrator-controlled workflow configuration, testing, approval rules, and workflow logs.

19.7 Digital Signature Misuse

Risk: Signatures may be applied without proper authority.

Mitigation: Require authentication, role verification, signature confirmation, and immutable signing records.

19.8 Excessive File Storage

Risk: Multiple document versions may consume significant storage.

Mitigation: Apply configurable file-size limits, storage monitoring, file compression, and archival policies.

19.9 Resistance to Adoption

Risk: Users may continue using manual or external processes.

Mitigation: Provide training, user guides, role-specific dashboards, pilot implementation, and administrative policy support.

20. Minimum Viable Product

The Minimum Viable Product shall contain:

1. Institutional account registration and authentication;
2. User and role management;
3. Instructor-managed group creation and project registration
4. Concept paper and manuscript uploading;
5. Immutable document versioning;
6. Instructor and adviser comments;
7. Non-destructive annotations;
8. AI summarization for complete proposal and final-defense manuscripts
9. Adviser and panel assignment;
10. Automated workflow routing;
11. Proposal and final-defense scheduling;
12. Panel evaluation and verdict recording;
13. Revision classification and deadline monitoring;
14. Email notifications;
15. Digital approval records;
16. Final requirement submission;
17. Final clearance;
18. Project archiving;
19. Role-based dashboards; and
20. Audit logging.

21. Future Enhancements

Future versions may include:

- University single sign-on;
- Learning management system integration;
- Calendar synchronization;
- Similarity-detection integration;
- Research ethics review workflow;
- Intellectual property review workflow;
- Electronic certificate generation;
- Advanced project analytics;
- Adviser workload recommendation;
- Panel recommendation based on expertise;
- AI-assisted requirement checking;
- AI-assisted comment categorization;
- Full-text archive searching;
- Research-topic trend analysis;
- Student mobile application;
- Public research repository integration;
- Institutional repository integration; and
- External evaluator portal.

22. Acceptance Criteria

The proposed system shall be accepted when the following conditions are met:

AC-01

Authorized users can register and authenticate using institutional email accounts.

AC-02

Users only see records and functions permitted by their assigned roles.

AC-03

Instructor 1 can create project groups and register approved project topics.

AC-04

Students can submit concept papers, manuscripts, and required documents.

AC-05

Every revised upload creates a separate document version.

AC-06

Original submitted versions cannot be overwritten by ordinary users.

AC-07

Instructor 1 and the Adviser can provide guidance, comments, and revision instructions during conceptualization.

AC-08

The system can generate a clearly labeled AI-assisted manuscript summary.

AC-09

Reviewers can annotate documents without modifying the original uploaded files.

AC-10

Projects are routed to authorized reviewers based on the configured workflow.

AC-11

A project cannot proceed when mandatory documents or approvals are incomplete.

AC-12

The Program Coordinator can assign advisers and panel members.

AC-13

Authorized users can create proposal and final-defense schedules.

AC-14

Panel members can complete evaluation forms.

AC-15

Only the Panel Chair can record the official defense verdict.

AC-16

The system can classify revisions and calculate revision deadlines.

AC-17

Students and faculty receive email notifications for assignments, decisions, schedules, and deadlines.

AC-18

The Adviser can validate and digitally sign weekly progress records.

AC-19

Students can submit final manuscripts and clearance documents.

AC-20

Authorized personnel can approve final clearance.

AC-21

Cleared projects are transferred to the secure archive with a recorded Pass/Fail result.

AC-22

The system records document submissions, decisions, approvals, signatures, and workflow transitions in the audit log.

AC-23

Role-based dashboards display accurate pending tasks and project statuses.

AC-24

Authorized users can generate and export required reports.

AC-25

The system operates properly on supported desktop and mobile web browsers.

23. Product Success Metrics

The success of the system may be measured through:

- Reduction in average document-routing time;
- Reduction in missing or incomplete requirements;
- Reduction in duplicated or misplaced manuscript versions;
- Percentage of projects processed through the complete digital workflow;
- Percentage of reviews completed within deadlines;
- Average approval turnaround time;
- Percentage of weekly reports submitted on time;
- Percentage of post-defense revisions completed within the prescribed period;
- User satisfaction scores;
- System usability evaluation results;
- Number of successfully archived projects;
- Reduction in printed institutional forms; and
- Completeness of audit and project-history records.

24. Recommended Development Priority

Priority 1: Core Platform

- Authentication;
- Role management;
- Approved project topic registration;
- Document submission;
- Version control;
- Workflow routing;
- Comments and annotations.

Priority 2: Academic Review

- Adviser assignment;
- Panel assignment;
- AI-assisted summaries;
- Evaluation forms;
- Approval decisions;
- Email Notifications.

Priority 3: Defense and Revision

- Defense scheduling;
- Verdict recording;
- Revision classification;
- Revision countdown;
- Compliance verification.

Priority 4: Clearance and Institutional Records

- Digital forms;
- Digital signatures;
- Final requirements;
- Clearance routing;
- Archiving;
- Reports and analytics.

25. Product Completion Definition

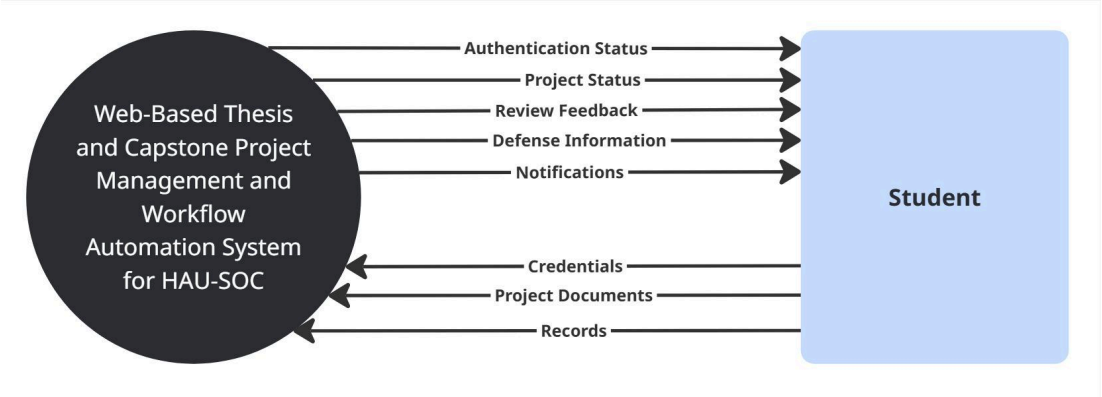
A thesis or capstone project shall be considered completed in the system when:

1. The final manuscript has been submitted.
2. All panel revisions have been verified.
3. The Adviser has approved the final version.
4. All required certifications and supporting documents have been uploaded.
5. The Program Coordinator has verified document completeness.
6. Required school-level approvals have been completed.
7. The project has received final clearance.
8. The final manuscript and related records have been transferred to the institutional archive.
9. The project workflow history and audit records are complete.
10. The system status has been changed to Completed and Archived.

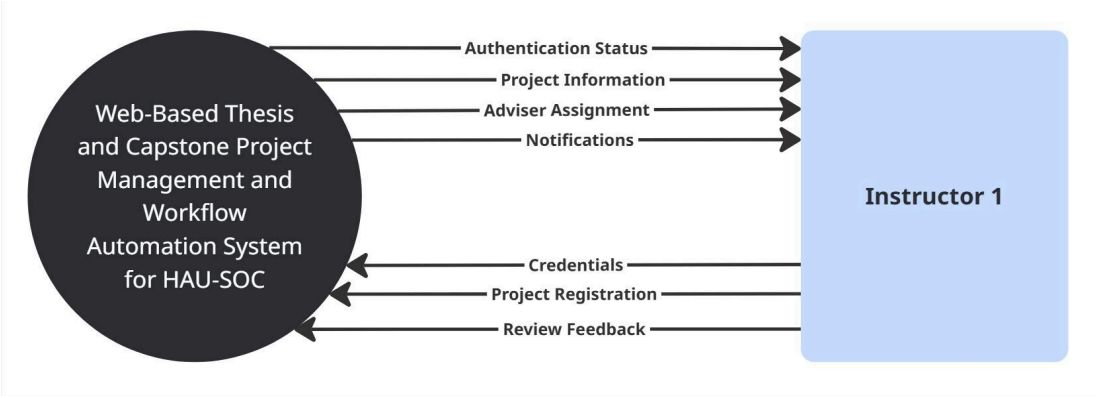
Appendix F

Context Diagram

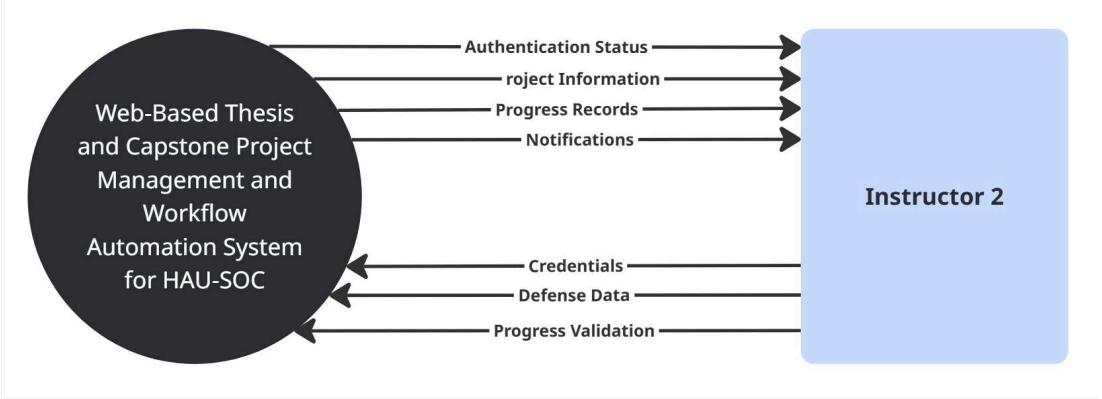
F-1: Student



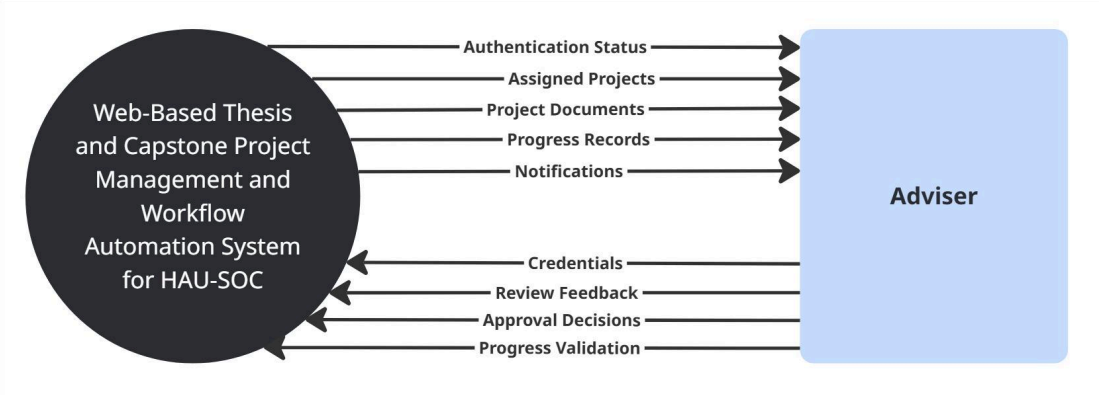
F-2: Instructor 1



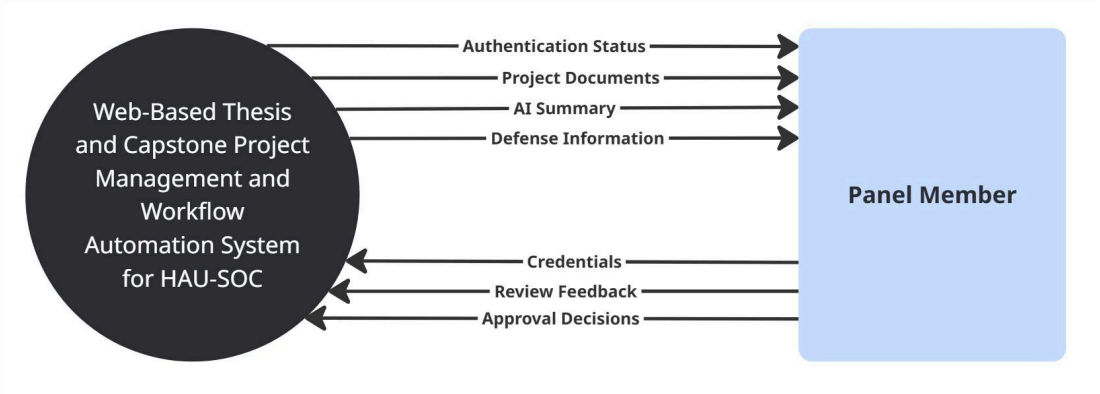
F-3: Instructor 2



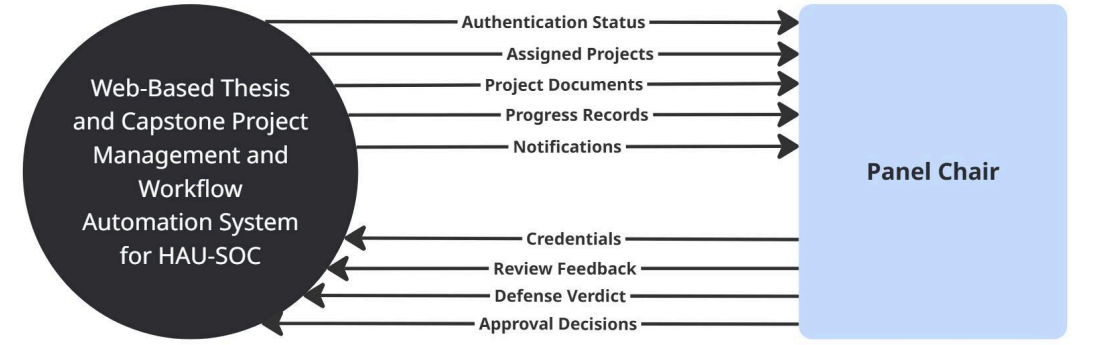
F-4: Adviser



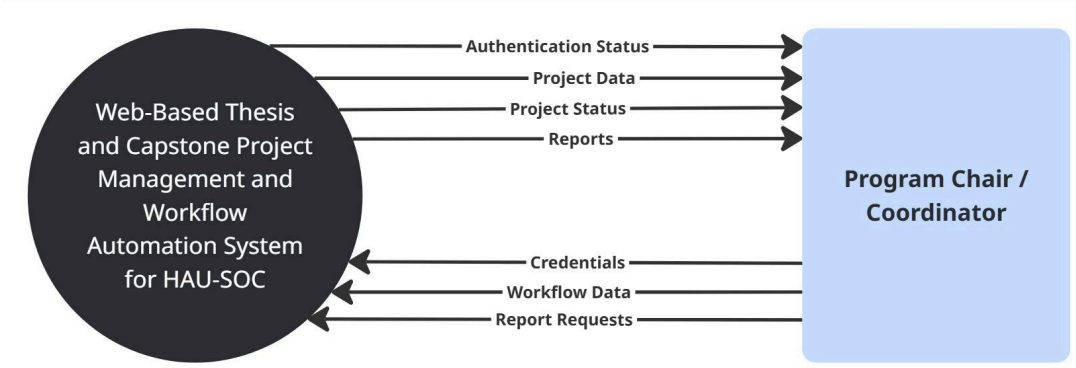
F-5: Panel Member



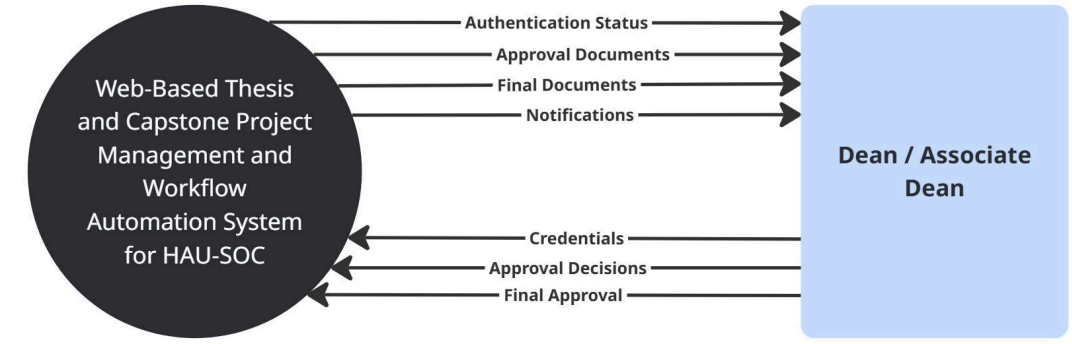
F-6: Panel Chair



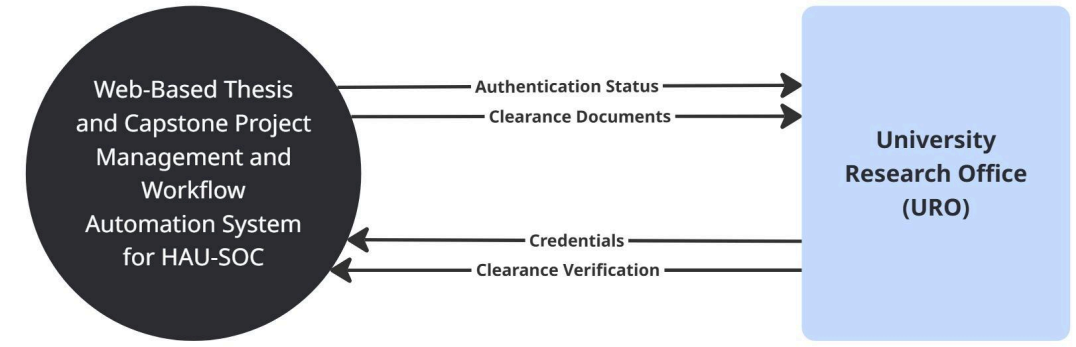
F-7: Program Chair/Coordinator



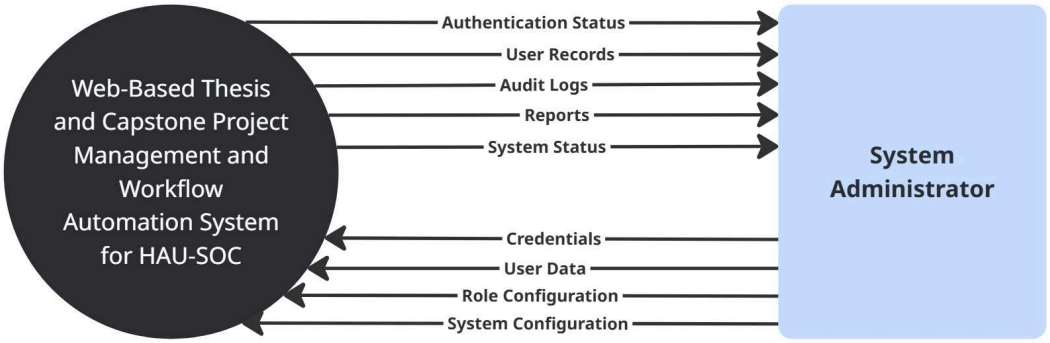
F-8: Dean/Associate Dean



F-9: University Research Office (URO)



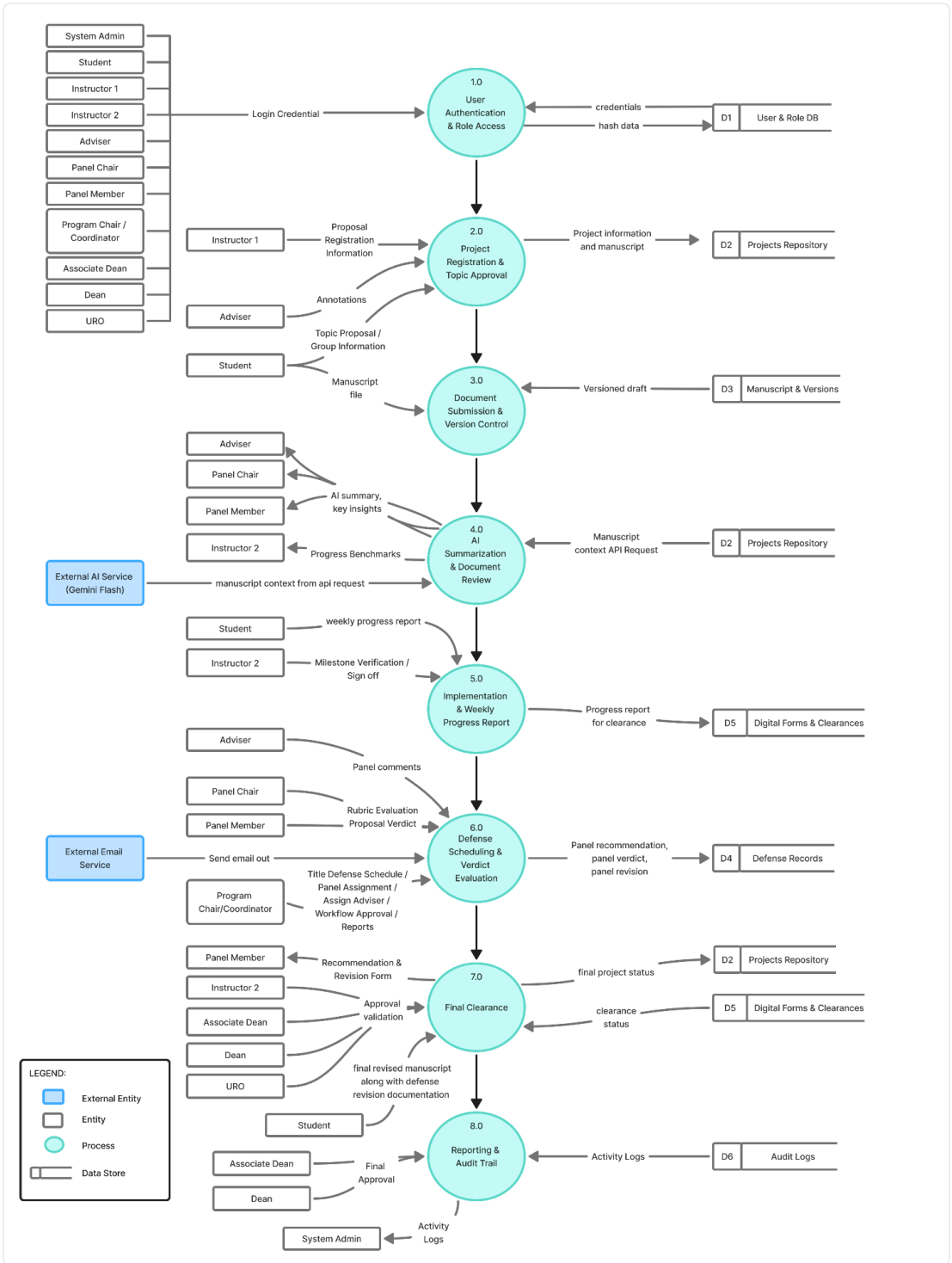
F-10: System Administrator



Appendix G

Level - 0 Data Flow Diagram

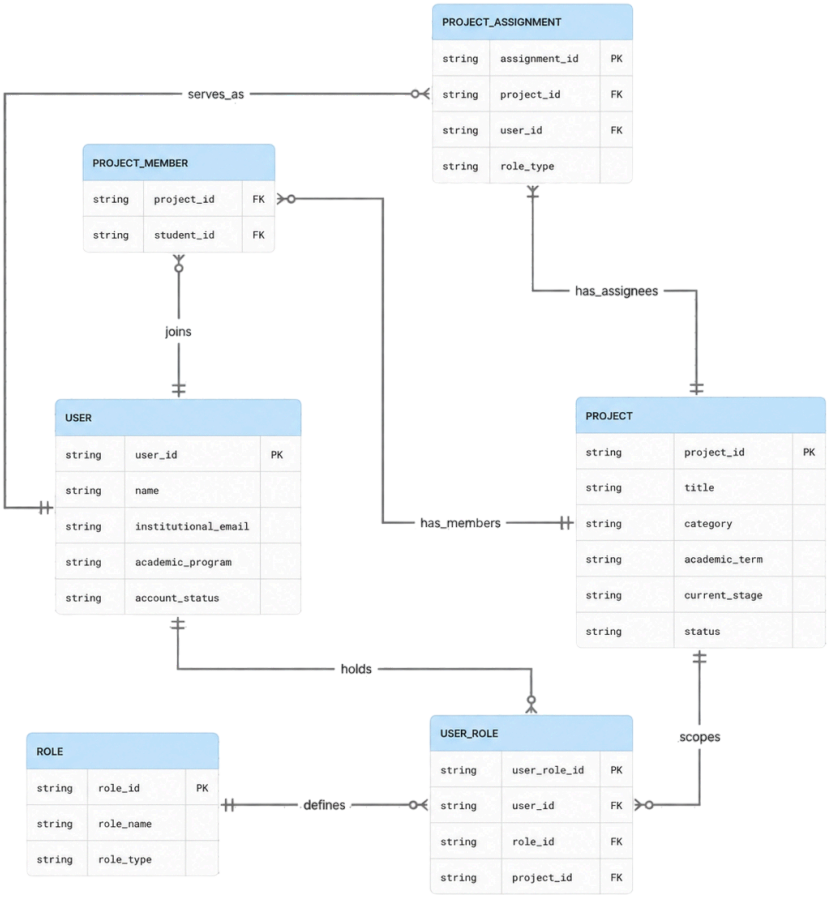
Data Flow Diagram



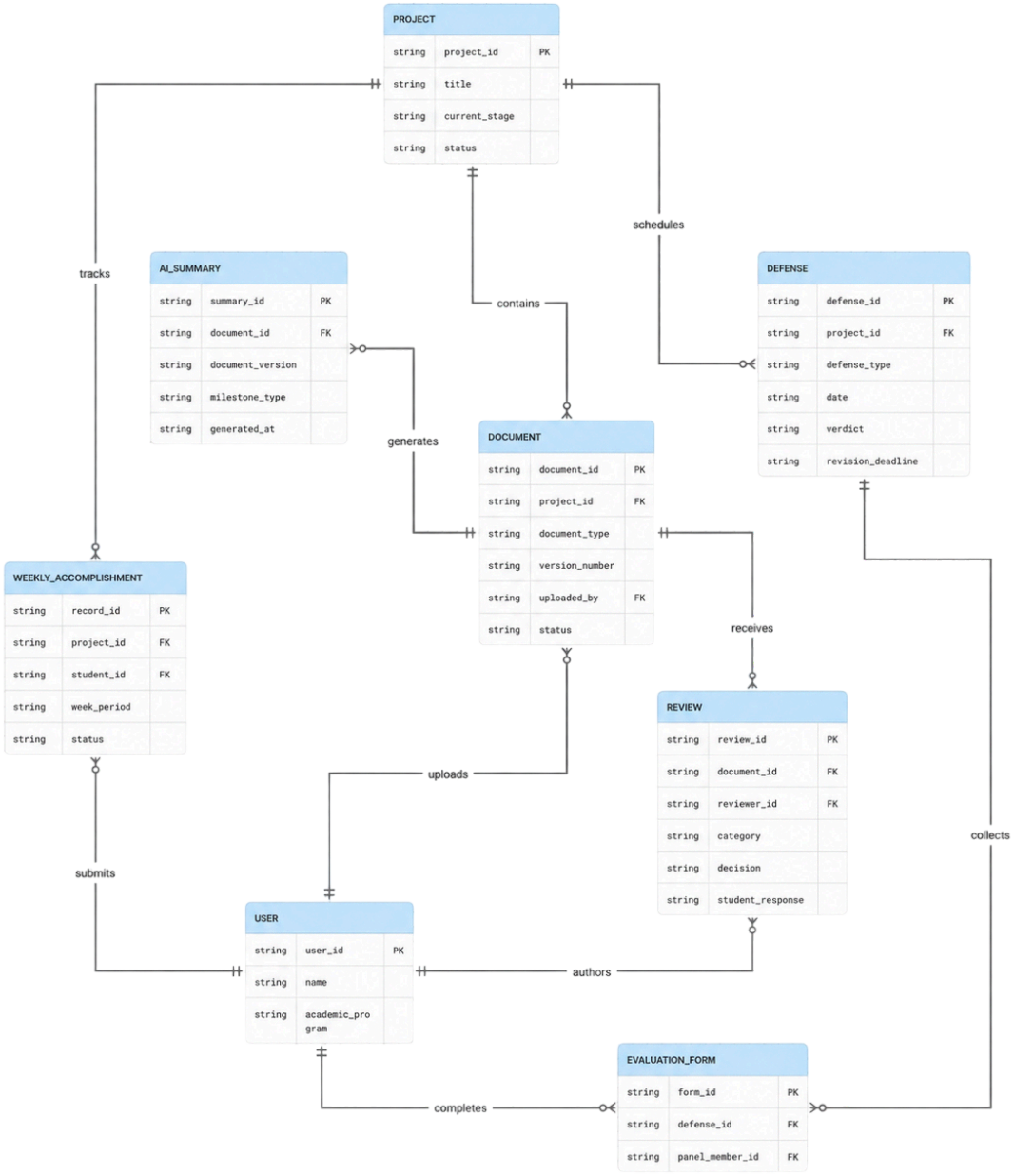
Appendix H

Entity Relationship Diagram (ERD)

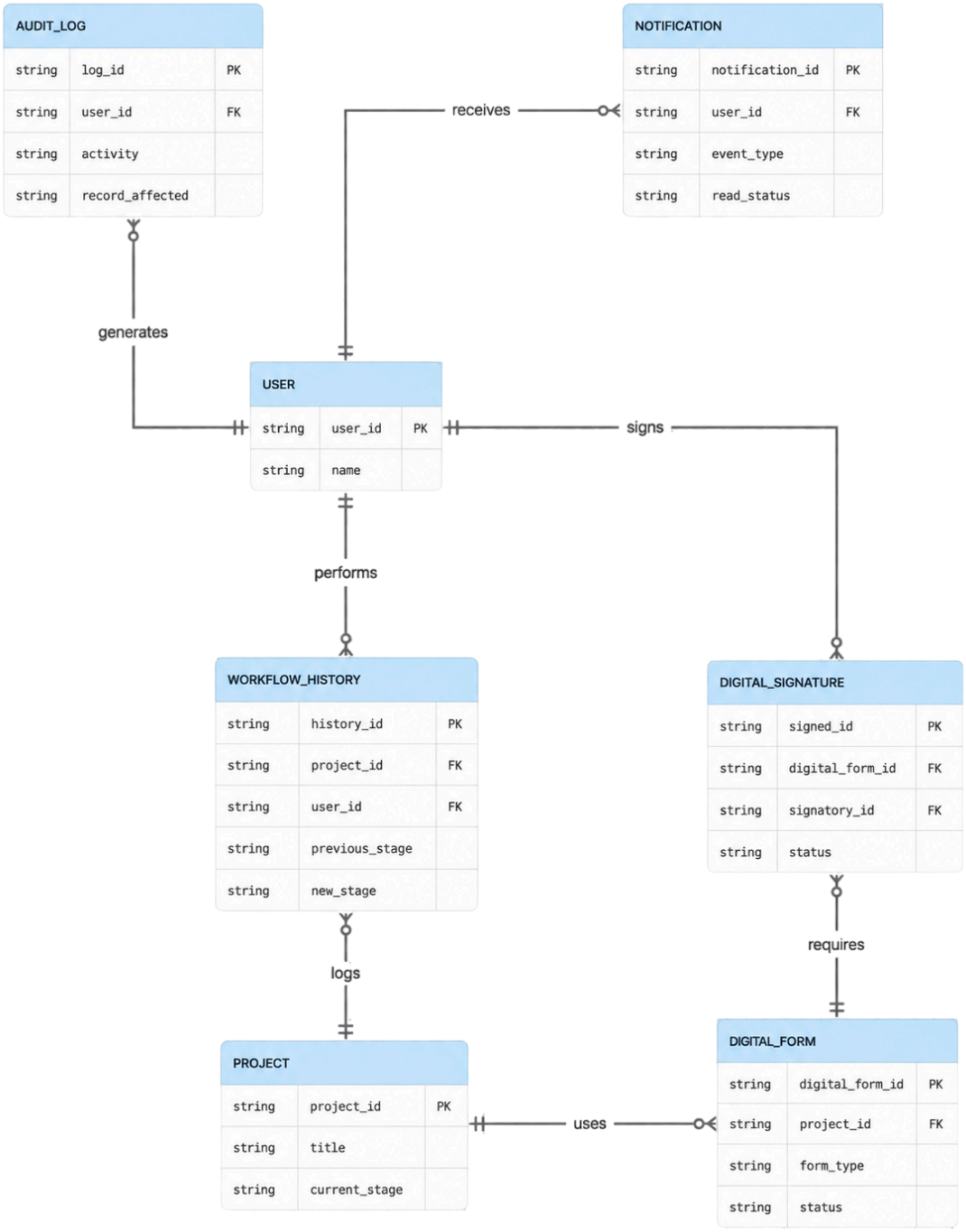
H-1: Identity, Roles, and Project Setup



H-2: Documents, Review, and Defense

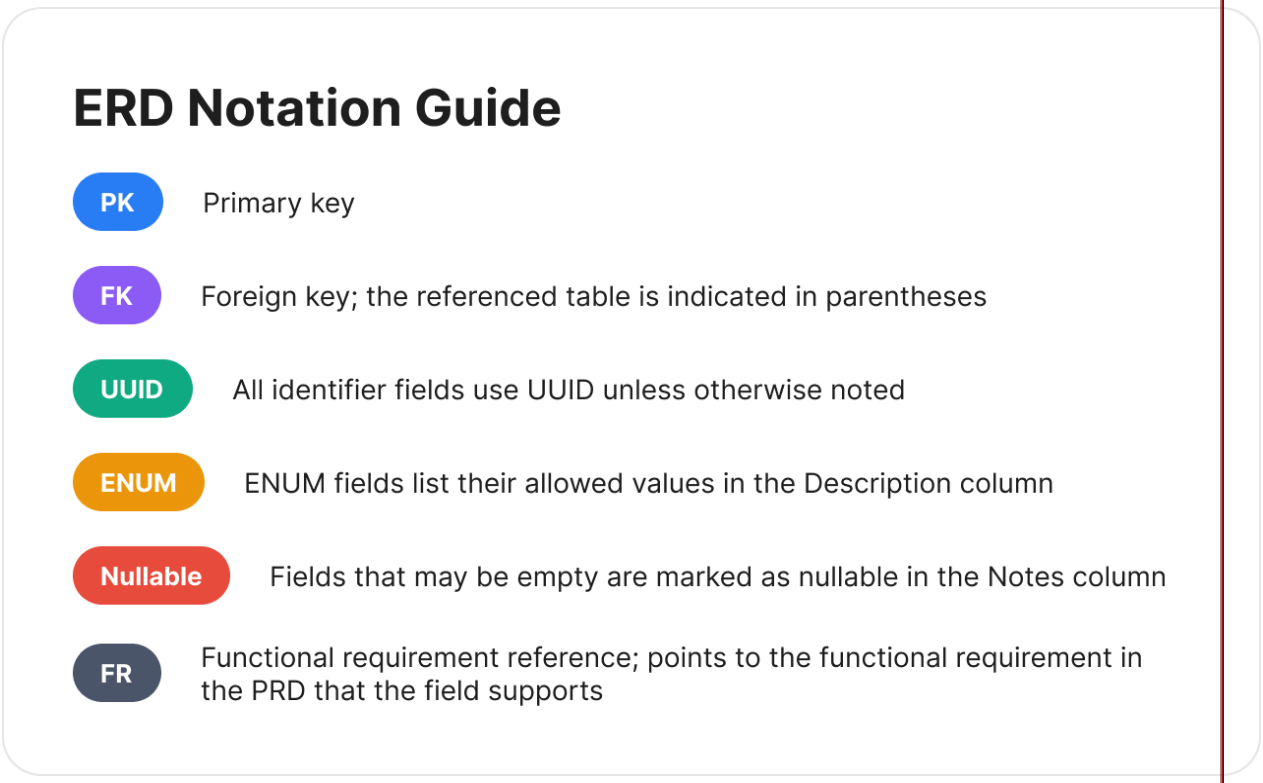


H-3: Forms, Workflow, and Records



Appendix I

Data Dictionary



I-1: USER

Field	Type	Key	Description
user_id	UUID	PK	Unique account identifier
name	VARCHAR(150)		Full name
institutional_email	VARCHAR(150)		University email used for registration, verification, and login
id_number	VARCHAR(30)		Academic or employee ID number
academic_program	VARCHAR(100)		Program or department the user belongs to
account_status	ENUM		Active, Inactive, Suspended
created_at	DATETIME		Account creation timestamp

updated_at	DATETIME		Last profile update

I-2: ROLE

Field	Type	Key	Description
role_id	UUID	PK	Unique role identifier
role_name	VARCHAR(50)		e.g. Student, Instructor 1, Instructor 2, Program Coordinator, Dean, Associate Dean, Administrator, Adviser, Panel Chair, Panel Member
role_type	ENUM		Global, Project-Specific

I-3: USER_ROLE

Field	Type	Key	Description
user_role_id	UUID	PK	Unique assignment identifier
user_id	UUID	FK (USER)	User holding the role
role_id	UUID	FK (ROLE)	Role being held
project_id	UUID	FK (PROJECT)	Project the role applies to
assigned_at	DATETIME		When the role was granted

I-4: PROJECT

Field	Type	Key	Description
project_id	UUID	PK	Unique project identifier
title	VARCHAR(255)		Current approved title
previous_titles	TEXT		JSON array of prior title versions
description	TEXT		Project description
category	VARCHAR(50)		e.g. Thesis, Capstone
research_area	VARCHAR(100)		Research or technology domain
academic_program	VARCHAR(100)		Owning academic program

academic_term	VARCHAR(20)		e.g. AY 2026-2027, 1st Sem
current_stage	ENUM		Conceptualization, Adviser Assignment, Proposal Development, Proposal Defense, Implementation, Final Defense, Final Revision, Clearance, Archived
status	ENUM		Active, On Hold, Completed, Archived
created_at	DATETIME		
updated_at	DATETIME		

I-5: PROJECT_MEMBER

Field	Type	Key	Description
project_id	UUID	PK, FK (PROJECT)	Project the student belongs to
student_id	UUID	PK, FK (USER)	Student member
joined_at	DATETIME		When the student was added to the group

I-6: PROJECT_ASSIGNMENT

Field	Type	Key	Description
assignment_id	UUID	PK	Unique assignment identifier
project_id	UUID	FK (PROJECT)	Project being served
user_id	UUID	FK (USER)	Faculty member assigned
role_type	ENUM		Instructor 1, Instructor 2, Adviser, Panel Chair, Panel Member
status	ENUM		Pending, Accepted, Declined
assigned_at	DATETIME		

I-7: DOCUMENT

Field	Type	Key	Description
document_id	UUID	PK	Unique document version identifier
project_id	UUID	FK (PROJECT)	Owning project
document_type	ENUM		Concept Paper, Proposal Manuscript, Final Manuscript, Editor's Note, Plagiarism Certificate, Revision Compliance Form, etc.
file_name	VARCHAR(255)		Original file name
file_location	VARCHAR(255)		Storage path or URL
file_size	INT		File size in KB
version_number	INT		Sequential version for this document type within the project
uploaded_by	UUID	FK (USER)	Submitting user
submission_datetime	DATETIME		
status	ENUM		Draft, Submitted, Under Review, For Revision, Approved, Rejected, Superseded, Archived
workflow_stage	VARCHAR(50)		Stage at which the document was submitted
remarks	TEXT		Submission remarks

I-8: AI_SUMMARY

Field	Type	Key	Description
summary_id	UUID	PK	Unique summary identifier

document_id	UUID	FK (DOCUMENT)	Manuscript version summarized
document_version	INT		Denormalized version number for quick reference
milestone_type	ENUM		Proposal Defense, Final Defense
summary_content	TEXT		Structured JSON: background, problem statement, objectives, methodology, expected output, scope, limitations, contributions
ai_disclosure_label	VARCHAR(100)		Fixed text confirming machine-generated, decision-support-only content
generated_at	DATETIME		

I-9: REVIEW

Field	Type	Key	Description
review_id	UUID	PK	Unique review entry identifier
document_id	UUID	FK (DOCUMENT)	Document version being reviewed
reviewer_id	UUID	FK (USER)	Author of the review
reviewer_role	VARCHAR(50)		Role held at time of review (denormalized for audit clarity)
category	ENUM		General Comment, Technical Concern, Methodology

			Concern, Documentation Issue, Formatting Issue, Required Revision, Recommendation
comment_text	TEXT		Annotation or comment content
decision	ENUM		Approve, Approve with Minor Revisions, Return for Major Revisions, Reject, Endorse to Next Stage
student_response	TEXT		Student's resolution note
created_at	DATETIME		

I-10: WEEKLY_ACCOMPLISHMENT

Field	Type	Key	Description
record_id	UUID	PK	Unique record identifier
project_id	UUID	FK (PROJECT)	
student_id	UUID	FK (USER)	Submitting student
week_period	VARCHAR(30)		e.g. "Week 5: Jun 1–7"
content	TEXT		Accomplishment narrative
attendance_status	VARCHAR(30)		Consultation/attendance record
status	ENUM		Submitted, Approved, Returned
adviser_signature_id	UUID	FK (DIGITAL_SIGNATURE)	Signature validating this record
submitted_at	DATETIME		

I-11: DEFENSE

Field	Type	Key	Description
defense_id	UUID	PK	Unique defense identifier
project_id	UUID	FK (PROJECT)	
defense_type	ENUM		Proposal Defense, Final Defense
date	DATE		
time	TIME		
venue	VARCHAR(150)		Physical venue or online meeting link
instructions	TEXT		Additional instructions
verdict	ENUM		Passed, Passed with Minor Revisions, Passed with Major Revisions, Failed
revision_classification	ENUM		Minor, Major
revision_deadline	DATE		Deadline computed from the revision countdown
revision_status	ENUM		Pending, Overdue, Completed

I-12: EVALUATION_FORM

Field	Type	Key	Description
form_id	UUID	PK	Unique form identifier
defense_id	UUID	FK (DEFENSE)	Defense being evaluated
panel_member_id	UUID	FK (USER)	Evaluator
evaluation_content	TEXT		Structured evaluation responses
submitted_at	DATETIME		

I-13: DIGITAL_FORM

Field	Type	Key	Description
digital_form_id	UUID	PK	Unique form instance identifier
project_id	UUID	FK (PROJECT)	
form_type	ENUM		Project Registration, Adviser Assignment, Capstone Recommendation, Revision Compliance, Final Clearance, Endorsement
template_ref	VARCHAR(100)		Reference to the institutional template used
status	ENUM		Draft, Completed, Signed
created_at	DATETIME		

I-14: DIGITAL_SIGNATURE

Field	Type	Key	Description
signed_id	UUID	PK	Unique signature identifier
digital_form_id	UUID	FK (DIGITAL_FORM)	Form being signed
signatory_id	UUID	FK (USER)	User who signed
role	VARCHAR(50)		Role under which the user signed
status	ENUM		Valid, Revoked
signed_at	DATETIME		

I-15: WORKFLOW_HISTORY

Field	Type	Key	Description
history_id	UUID	PK	Unique history entry identifier
project_id	UUID	FK (PROJECT)	
user_id	UUID	FK (USER)	User who triggered the transition
previous_stage	VARCHAR(50)		
new_stage	VARCHAR(50)		
decision	VARCHAR(50)		Decision associated with the transition
remarks	TEXT		
transitioned_at	DATETIME		

I-16: NOTIFICATION

Field	Type	Key	Description
notification_id	UUID	PK	Unique notification identifier
user_id	UUID	FK (USER)	Recipient
event_type	VARCHAR(50)		e.g. New Assignment, Revision Request, Defense Scheduled, Overdue Revision
message	TEXT		Notification body
read_status	BOOLEAN		Default false
sent_at	DATETIME		

I-17: AUDIT_LOG

Field	Type	Key	Description
log_id	UUID	PK	Unique log entry identifier

user_id	UUID	FK (USER)	Actor
activity	VARCHAR(100)		e.g. Login, Document Upload, Approval Decision, Signature Applied
record_affected	VARCHAR(100)		Table and record ID affected
previous_value	TEXT		
new_value	TEXT		
session_info	VARCHAR(255)		IP address or session identifier
result	ENUM		Success, Failure
created_at	DATETIME		

Appendix J

Functionality Checklist

Informed Consent

You are being invited to join the assessment of the Development of a Web-Based Thesis and Capstone Project Management and Workflow Automation System for HAU-SOC, which was designed as part of the undergraduate capstone project research done by the proponents of the School of Computing, Holy Angel University. Your participation in this survey is completely optional and you can choose not to answer the questions at your discretion. All information obtained will be kept strictly confidential, anonymity will be maintained using participant codes, all data will be for academic use only, and it will be governed by the Data Privacy Act of 2012 (Republic Act No. 10173).

By proceeding with this form, you confirm that you have read and understood this statement and voluntarily agree to participate.

☐ I have read and understood the above statement and voluntarily agree to participate.

Functionality Checklist

Web-Based System for Management and Workflow Automation of Thesis and Capstone Projects

By proceeding with this form, you confirm that you have read and understood this statement and voluntarily agree to participate.

☐ I have read and understood the above statement and voluntarily agree to participate.

User Role (Student/Faculty) _____ Date of Evaluation _____

Global Role _____

Instructions. Complete only the checklist corresponding to your role in the thesis and capstone process. Perform each listed function in the system and tick the box that reflects the observed result. Tick Pass only when the function performed exactly as described.

Response scale. Pass = the function performed as described; Fail = the function did not perform as described; N/A = the function falls outside the responsibilities of my role.

J-1 Student Functionality Checklist

	Evaluation Statement	Pass	Fail
1.	User can log in using official university email credentials (@student.hau.edu.ph) and view the Student Dashboard.	☐	☐
2.	User can submit research topics and concept papers during the ideation phase.	☐	☐
3.	User can upload required documents (Manuscripts, Editor's Note, Plagiarism Certificate) strictly in PDF format without overwriting previous entries.	☐	☐
4.	User can view previous and current versions of uploaded manuscripts.	☐	☐
5.	User can submit weekly accomplishment entries under the FM-AAC-SOC-2003 Monitoring Form.	☐	☐
6.	User can track the real-time stage of their capstone project from ideation to final archival.	☐	☐
7.	User can view reviewer annotations, comments, and AI-generated manuscript summaries.	☐	☐
8.	User receives system-triggered email alerts regarding submission updates, defense schedules, and revision requests.	☐	☐
9.	User can submit final project soft copies and repository links (GitHub/Docker) upon clearance.	☐	☐

J-2 Executive Leadership (Dean & Associate Dean) Checklist

	Evaluation Statement	Pass	Fail
1.	Executive users can review and approve adviser assignments submitted by the Program Chair.	☐	☐
2.	Executive users can review projects cleared by URO and apply final administrative digital signatures to Approval Sheets.	☐	☐

	Evaluation Statement	Pass	Fail
3.	Executive users can generate and export consolidated reports on project statuses and faculty workloads.	☐	☐
4.	Executive users can search, filter, and view completed institutional thesis/capstone records.	☐	☐

J-3 Program Chair / Coordinator Checklist

	Evaluation Statement	Pass	Fail
1.	Program Chair can assign faculty advisers to project groups using the Adviser Assignment Module.	☐	☐
2.	Program Chair can assign Panel Chairs and Panel Members to defense schedules.	☐	☐
3.	Program Chair can view program-level progress dashboards and monitor overall project completion rates.	☐	☐
4.	Program Chair can endorse cleared projects to the Dean and Associate Dean via digital sign-off.	☐	☐

J-4 Adviser Checklist

	Evaluation Statement	Pass	Fail	N/A
1.	Adviser can review student draft submissions and add non-destructive PDF annotations.	☐	☐	☐
2.	Adviser can review, approve, return, and apply digital signatures to FM-AAC-SOC-2003 weekly logs.	☐	☐	☐
3.	Adviser can complete and digitally sign the FM-AAC-SOC-2005 Recommendation Form to certify defense readiness.	☐	☐	☐
4.	Adviser can verify student revisions against panel recommendations after defenses.	☐	☐	☐

J-5 Panel Member Checklist

	Evaluation Statement	Pass	Fail	N/A
1.	Panel Member can view assigned manuscripts and add private annotations visible only to themselves.	☐	☐	☐
2.	Panel Member can view structured AI-generated summaries for proposal and final defense manuscripts.	☐	☐	☐
3.	Panel Member can verify compliance with defense conditions and sign off on the FM-AAC-SOC-2004 Revision Form.	☐	☐	☐
4.	Panel Member can apply digital signatures to the final Approval Sheet upon satisfactory revisions.	☐	☐	☐

J-6 Panel Chair Checklist

	Evaluation Statement	Pass	Fail	N/A
1.	Panel Chair can access manuscript versions, private annotations, and AI-generated summaries.	☐	☐	☐
2.	Panel Chair can officially record the defense verdict (Passed, Minor Revisions, Major Revisions, Failed).	☐	☐	☐
3.	Panel Chair can generate and submit the digital FM-AAC-SOC-2004 Capstone Revision Form.	☐	☐	☐
4.	Recording an official verdict successfully triggers the automated system revision countdown clock.	☐	☐	☐

J-7 Instructor 1 (Capstone 1) Checklist

	Evaluation Statement	Pass	Fail	N/A
1.	Instructor 1 can create project groups and assign students according to their block sections.	☐	☐	☐
2.	Instructor 1 can officially register the approved thesis or capstone title after topic evaluation.	☐	☐	☐
3.	Instructor 1 can review concept papers, add non-destructive annotations, and request revisions.	☐	☐	☐
4.	Instructor 1 can clear drafting milestones and endorse qualified project groups for proposal defense.	☐	☐	☐

J-8 Instructor 2 (Capstone 2) Checklist

	Evaluation Statement	Pass	Fail	N/A
1.	Instructor 2 can monitor continuing project implementations and track course milestone compliance.	☐	☐	☐
2.	Instructor 2 can set and publish oral defense schedules for proposal and final defenses.	☐	☐	☐
3.	Instructor 2 can review post-defense requirements before finalizing course-related requirements.	☐	☐	☐

J-9 University Research Office (URO) Checklist

	Evaluation Statement	Pass	Fail
1.	URO personnel can access and review uploaded Plagiarism Clearance Certificates and Editor's Certificates.	☐	☐
2.	URO personnel can apply digital approval verification and route cleared projects to executive approvers.	☐	☐

J-10 System Administrator Checklist

	Evaluation Statement	Pass	Fail
1.	Administrator can manage user accounts, assign roles, and verify official university emails.	☐	☐
2.	Administrator can configure Context-Aware Access Control (CAAC) tags and granular permission boundaries.	☐	☐
3.	Administrator can view protected, read-only audit logs tracking user actions and workflow state changes.	☐	☐

Appendix K

System Usability Scale Questionnaire

System Usability Scale

© Digital Equipment Corporation, 1986.

	Strongly disagree				Strongly agree
1. I think that I would like to use this system frequently	1	2	3	4	5
2. I found the system unnecessarily complex	1	2	3	4	5
3. I thought the system was easy to use	1	2	3	4	5
4. I think that I would need the support of a technical person to be able to use this system	1	2	3	4	5
5. I found the various functions in this system were well integrated	1	2	3	4	5
6. I thought there was too much inconsistency in this system	1	2	3	4	5
7. I would imagine that most people would learn to use this system very quickly	1	2	3	4	5
8. I found the system very cumbersome to use	1	2	3	4	5
9. I felt very confident using the system	1	2	3	4	5
10. I needed to learn a lot of things before I could get going with this system	1	2	3	4	5

Appendix L

Adopted System Usability Scale (SUS) Questionnaire

Informed Consent

You are being invited to participate in the evaluation of the proposed Web-based System for Management and Workflow Automation of Thesis and Capstone Projects, developed as part of an undergraduate capstone study by the proponents from the School of Computing, Holy Angel University. Your participation is entirely voluntary, and you may decline or withdraw at any point without penalty. All responses will be treated with strict confidentiality, anonymized through the use of participant codes, used solely for academic purposes, and handled in accordance with the Data Privacy Act of 2012 (Republic Act No. 10173).

By proceeding with this form, you confirm that you have read and understood this statement and voluntarily agree to participate.

☐ I have read and understood the above statement and voluntarily agree to participate.

Adopted System Usability Scale (SUS) Questionnaire

Web-Based System for Management and Workflow Automation of Thesis and Capstone Projects

By proceeding with this form, you confirm that you have read and understood this statement and voluntarily agree to participate.

☐ I have read and understood the above statement and voluntarily agree to participate.

Stakeholder Role _____ Date of Evaluation _____

Program / Department _____

Instructions. Rate your level of agreement with each statement based on your use of the system. Tick one box per statement. Please respond to every item; if you are unsure about an item, tick the midpoint rather than leaving it blank.

Response scale. 1 = Strongly Disagree · 2 = Disagree · 3 = Neutral · 4 = Agree · 5 = Strongly Agree

Statement	1	2	3	4	5
1. I think that I would like to use this system frequently.	☐	☐	☐	☐	☐
2. I found the system unnecessarily complex.	☐	☐	☐	☐	☐
3. I thought the system was easy to use.	☐	☐	☐	☐	☐
4. I think that I would need the support of a technical person to be able to use this system.	☐	☐	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☐	☐	☐	☐
6. I thought there was too much inconsistency in this system.	☐	☐	☐	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☐
8. I found the system very cumbersome to use.	☐	☐	☐	☐	☐
9. I felt very confident using the system.	☐	☐	☐	☐	☐
10. I needed to learn a lot of things before I could get going with this system.	☐	☐	☐	☐	☐

Additional comments:

Appendix M

Measurement Scales for Perceived Usefulness and Perceived Ease of Use

Perceived Usefulness

Using CHART-MASTER in my job would enable me to accomplish tasks more quickly.

likely |_____|_____|_____|_____|_____|_____|_____| unlikely
extremely quite slightly neither slightly quite extremely

Using CHART-MASTER would improve my job performance.

likely |_____|_____|_____|_____|_____|_____|_____| unlikely
extremely quite slightly neither slightly quite extremely

Using CHART-MASTER in my job would increase my productivity.

likely |_____|_____|_____|_____|_____|_____|_____| unlikely
extremely quite slightly neither slightly quite extremely

Using CHART-MASTER would enhance my effectiveness on the job.

likely |_____|_____|_____|_____|_____|_____|_____| unlikely
extremely quite slightly neither slightly quite extremely

Using CHART-MASTER would make it easier to do my job.

likely |_____|_____|_____|_____|_____|_____|_____| unlikely
extremely quite slightly neither slightly quite extremely

I would find CHART-MASTER useful in my job.

likely |_____|_____|_____|_____|_____|_____|_____| unlikely
extremely quite slightly neither slightly quite extremely

Perceived Ease of Use

Learning to operate CHART-MASTER would be easy for me.

likely |_____|_____|_____|_____|_____|_____|_____| unlikely
extremely quite slightly neither slightly quite extremely

I would find it easy to get CHART-MASTER to do what I want it to do.

likely |_____|_____|_____|_____|_____|_____|_____| unlikely
extremely quite slightly neither slightly quite extremely

My interaction with CHART-MASTER would be clear and understandable.

likely |_____|_____|_____|_____|_____|_____|_____| unlikely
extremely quite slightly neither slightly quite extremely

I would find CHART-MASTER to be flexible to interact with.

likely |_____|_____|_____|_____|_____|_____|_____| unlikely
extremely quite slightly neither slightly quite extremely

It would be easy for me to become skillful at using CHART-MASTER.

likely |_____|_____|_____|_____|_____|_____|_____| unlikely
extremely quite slightly neither slightly quite extremely

I would find CHART-MASTER easy to use.

likely |_____|_____|_____|_____|_____|_____|_____| unlikely
extremely quite slightly neither slightly quite extremely

Appendix N

Adapted Technology Acceptance Model (TAM) Questionnaire

Informed Consent

You are being invited to participate in the evaluation of the proposed Web-based System for Management and Workflow Automation of Thesis and Capstone Projects, developed as part of an undergraduate capstone study by the proponents from the School of Computing, Holy Angel University. Your participation is entirely voluntary, and you may decline or withdraw at any point without penalty. All responses will be treated with strict confidentiality, anonymized through the use of participant codes, used solely for academic purposes, and handled in accordance with the Data Privacy Act of 2012 (Republic Act No. 10173).

By proceeding with this form, you confirm that you have read and understood this statement and voluntarily agree to participate.

☐ I have read and understood the above statement and voluntarily agree to participate.

Adapted Technology Acceptance Model (TAM) Questionnaire

Web-Based System for Management and Workflow Automation of Thesis and Capstone Projects

By proceeding with this form, you confirm that you have read and understood this statement and voluntarily agree to participate.

☐ I have read and understood the above statement and voluntarily agree to participate.

Participant Code _____ Date of Evaluation _____

Program / Department _____

Instructions. Indicate how likely each statement is to be true for you, based on your use of the system. Tick one box per statement.

Response scale. 1 = Extremely Unlikely · 2 = Quite Unlikely · 3 = Slightly Unlikely · 4 = Neither · 5 = Slightly Likely · 6 = Quite Likely · 7 = Extremely Likely

Statement	1	2	3	4	5	6	7
PERCEIVED USEFULNESS							
1. Using the Web-Based Thesis and Capstone Management System in my academic role would enable me to accomplish tasks more quickly.	☐	☐	☐	☐	☐	☐	☐
2. Using the Web-Based Thesis and Capstone Management System would improve my performance in managing thesis/capstone workflows.	☐	☐	☐	☐	☐	☐	☐
3. Using the Web-Based Thesis and Capstone Management System would increase my productivity during the research project lifecycle.	☐	☐	☐	☐	☐	☐	☐
4. Using the Web-Based Thesis and Capstone Management System would enhance my effectiveness in fulfilling research requirements and approvals.	☐	☐	☐	☐	☐	☐	☐
5. Using the Web-Based Thesis and Capstone Management System would make it easier to conduct, supervise, or monitor thesis/capstone tasks.	☐	☐	☐	☐	☐	☐	☐
6. I would find the Web-Based Thesis and Capstone Management System useful in my academic responsibilities at HAU-SOC.	☐	☐	☐	☐	☐	☐	☐
PERCEIVED EASE OF USE							
7. Learning to operate the Web-Based Thesis and Capstone Management System would be easy for me.	☐	☐	☐	☐	☐	☐	☐
8. I would find it easy to get the Web-Based Thesis and Capstone Management System to do what I want it to do.	☐	☐	☐	☐	☐	☐	☐
9. My interaction with the Web-Based Thesis and Capstone Management System would be clear and understandable.	☐	☐	☐	☐	☐	☐	☐
10. I would find the Web-Based Thesis and Capstone Management System to be flexible to interact with across different devices.	☐	☐	☐	☐	☐	☐	☐

Statement	1	2	3	4	5	6	7
11. It would be easy for me to become skillful at using the Web-Based Thesis and Capstone Management System.	☐	☐	☐	☐	☐	☐	☐
12. I would find the Web-Based Thesis and Capstone Management System easy to use.	☐	☐	☐	☐	☐	☐	☐

Additional comments:

Appendix O

Expert ISO/IEC 25010 Software Quality Evaluation Form

Informed Consent

You are being invited to participate in the evaluation of the proposed Web-based System for Management and Workflow Automation of Thesis and Capstone Projects, developed as part of an undergraduate capstone study by the proponents from the School of Computing, Holy Angel University. Your participation is entirely voluntary, and you may decline or withdraw at any point without penalty. All responses will be treated with strict confidentiality, anonymized through the use of participant codes, used solely for academic purposes, and handled in accordance with the Data Privacy Act of 2012 (Republic Act No. 10173).

By proceeding with this form, you confirm that you have read and understood this statement and voluntarily agree to participate.

☐ I have read and understood the above statement and voluntarily agree to participate.

Expert ISO/IEC 25010 Software Quality Evaluation Form

Web-Based System for Management and Workflow Automation of Thesis and Capstone Projects

By proceeding with this form, you confirm that you have read and understood this statement and voluntarily agree to participate.

☐ I have read and understood the above statement and voluntarily agree to participate.

Full Name (Optional) _____ Area of Expertise _____

Years of Relevant Experience _____ Date of Evaluation _____

Instructions. Rate the system against each sub-characteristic of the ISO/IEC 25010:2011 Software Product Quality Model. Tick one box per sub-characteristic.

Response scale. 1 = Very Poor · 2 = Poor · 3 = Fair · 4 = Good · 5 = Very Good

Quality Sub-Characteristic	1	2	3	4	5
FUNCTIONAL SUITABILITY					
1. Functional Completeness — Assesses whether the system provides all functions necessary to support required tasks (e.g., submission, routing, role management, annotations) as specified in the approved requirements.	☐	☐	☐	☐	☐
2. Functional Correctness — Assesses whether the system's functions produce accurate results (e.g., correct document routing, accurate application of role-based permissions).	☐	☐	☐	☐	☐
3. Functional Appropriateness — Assesses whether the provided functions facilitate the accomplishment of the intended thesis/capstone workflow tasks.	☐	☐	☐	☐	☐
USABILITY					
4. Appropriateness Recognizability — Assesses whether users can readily recognize that the system is suitable for their needs based on its dashboards and navigation.	☐	☐	☐	☐	☐
5. Learnability — Assesses how easily users can learn to operate the system's features.	☐	☐	☐	☐	☐
6. Operability — Assesses how easily users can operate and control the system during use.	☐	☐	☐	☐	☐
7. User Error Protection — Assesses the system's ability to protect users from making errors (e.g., validation prompts, confirmation dialogs before submission or approval).	☐	☐	☐	☐	☐
8. User Interface Aesthetics — Assesses whether the interface is visually organized and satisfying to interact with.	☐	☐	☐	☐	☐
9. Accessibility — Assesses whether the system can be used effectively across standard desktop and mobile interfaces by users with diverse needs, consistent with WCAG 2.1 (World Wide Web Consortium, 2018) Level AA success criteria.	☐	☐	☐	☐	☐

Quality Sub-Characteristic	1	2	3	4	5
RELIABILITY					
10. Maturity — Assesses the system's ability to perform consistently and meet reliability needs under normal operating conditions.	☐	☐	☐	☐	☐
11. Availability — Assesses whether the system remains operational and accessible when required.	☐	☐	☐	☐	☐
12. Fault Tolerance — Assesses the system's ability to maintain stable operation and prevent duplicate transactions or approvals despite unexpected conditions.	☐	☐	☐	☐	☐
13. Recoverability — Assesses the system's ability to recover data and restore its intended state after a disruption, preserving complete data accuracy.	☐	☐	☐	☐	☐
PERFORMANCE EFFICIENCY					
14. Time Behaviour — Assesses system responsiveness and processing speed, including standard page loading within three seconds.	☐	☐	☐	☐	☐
15. Resource Utilization — Assesses the efficient use of system resources (e.g., server, storage, bandwidth) during operation.	☐	☐	☐	☐	☐
16. Capacity — Assesses the system's ability to handle document uploads and concurrent usage within defined limits.	☐	☐	☐	☐	☐
SECURITY					
17. Confidentiality — Assesses whether the system restricts access to confidential manuscripts and data to authorized users only, consistent with OWASP Top Ten (OWASP Foundation, 2025) guidance on broken access control and security misconfiguration.	☐	☐	☐	☐	☐
18. Integrity — Assesses whether the system prevents unauthorized modification of data or records, consistent with OWASP Top Ten (OWASP Foundation, 2025) guidance on injection prevention.	☐	☐	☐	☐	☐
19. Non-repudiation — Assesses whether user actions (e.g., approvals, submissions) can be reliably attributed and proven to have occurred.	☐	☐	☐	☐	☐
20. Accountability — Assesses whether user actions can be uniquely traced through logs or audit trails.	☐	☐	☐	☐	☐
21. Authenticity — Assesses whether the system reliably verifies user identity through proper authentication and access control mechanisms (e.g., HTTPS, context-aware access control).	☐	☐	☐	☐	☐

TECHNICAL STRENGTHS

What specific technical aspects or features of the developed system performed best during your evaluation?

AREAS FOR SYSTEM ENHANCEMENT

What technical improvements, optimizations, or feature modifications would you recommend for future iterations of the system?